

SIGAP — complete code listing

Repository: github.com/Alvnc/loop-tb – prototipe riset UnivaBio 2026.

Berisi: pipeline data (audit dedup), training, studi representasi SSL, evaluasi rigor, API, dan UI.

Daftar file

```
ml/evaluate.py
ml/export_onnx.py
ml/ssl_probe.py
ml/train.py
ml/data/prepare_data.py
ml/data/prepare_meta.py
ml/kaggle/hello/hello.py
ml/kaggle/make_nlm_dataset.py
ml/kaggle/out_ssl/loop-tb/api/main.py
ml/kaggle/out_ssl/loop-tb/ml/data/prepare_data.py
ml/kaggle/out_ssl/loop-tb/ml/data/prepare_meta.py
ml/kaggle/out_ssl/loop-tb/ml/evaluate.py
ml/kaggle/out_ssl/loop-tb/ml/export_onnx.py
ml/kaggle/out_ssl/loop-tb/ml/kaggle/hello/hello.py
ml/kaggle/out_ssl/loop-tb/ml/kaggle/make_nlm_dataset.py
ml/kaggle/out_ssl/loop-tb/ml/kaggle/ssl/ssl_extract.py
ml/kaggle/out_ssl/loop-tb/ml/kaggle/train/train_kernel.py
ml/kaggle/out_ssl/loop-tb/ml/ssl_probe.py
ml/kaggle/out_ssl/loop-tb/ml/train.py
ml/kaggle/out_ssl/loop-tb/tools/e2e_screen.py
ml/kaggle/out_ssl/loop-tb/tools/screenshot_ui.py
ml/kaggle/out_ssl_diag/loop-tb/api/main.py
ml/kaggle/out_ssl_diag/loop-tb/ml/data/prepare_data.py
ml/kaggle/out_ssl_diag/loop-tb/ml/data/prepare_meta.py
ml/kaggle/out_ssl_diag/loop-tb/ml/evaluate.py
ml/kaggle/out_ssl_diag/loop-tb/ml/export_onnx.py
ml/kaggle/out_ssl_diag/loop-tb/ml/kaggle/hello/hello.py
ml/kaggle/out_ssl_diag/loop-tb/ml/kaggle/make_nlm_dataset.py
ml/kaggle/out_train/loop-tb/api/main.py
ml/kaggle/out_train/loop-tb/ml/data/prepare_data.py
ml/kaggle/out_train/loop-tb/ml/data/prepare_meta.py
ml/kaggle/out_train/loop-tb/ml/evaluate.py
ml/kaggle/out_train/loop-tb/ml/export_onnx.py
ml/kaggle/out_train/loop-tb/ml/kaggle/hello/hello.py
ml/kaggle/out_train/loop-tb/ml/kaggle/train/train_kernel.py
ml/kaggle/out_train/loop-tb/ml/train.py
ml/kaggle/ssl/ssl_extract.py
ml/kaggle/train/train_kernel.py
api/main.py
api/model_config.json
api/model_config_probe.json
api/model_config_space.json
scripts/fetch_models.sh
tools/e2e_screen.py
tools/make_pdfs.py
tools/make_video.py
tools/print_check.py
tools/screenshot_ui.py
web/src/lib/api.ts
web/src/app/klinisi/page.tsx
web/src/app/layout.tsx
web/src/app/page.tsx
web/src/app/pasien/page.tsx
web/src/app/skrining/page.tsx
web/src/app/tentang/page.tsx
web/src/components/ScreeningFlow.tsx
web/src/components/SiteHeader.tsx
web/src/app/globals.css
web/package-lock.json
web/package.json
web/tsconfig.json
web/next-env.d.ts
web/next.config.ts
docs/00-charter.md
docs/01-pohon-masalah.md
docs/02-arsitektur.md
```

docs/03-eksekusi.md
docs/04-evidensi-data.md

ml/evaluate.py

```
#!/usr/bin/env python3
"""Evaluasi rigor: kalibrasi, CI bootstrap, transfer, band triase, subgroup.

Contoh (ensemble + temperature scaling):
python ml/evaluate.py --runs ml/runs/efficientnet_b0 ml/runs/convnext_tiny \
    --manifest ml/data/artifacts/manifest.json --split external --temp-scale --out ml/runs/eval_external
"""
from __future__ import annotations

import argparse
import csv
import json
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import brier_score_loss, roc_auc_score, roc_curve
from torch.utils.data import DataLoader
from tqdm import tqdm

import sys
sys.path.insert(0, str(Path(__file__).resolve().parent))
from train import CXRManifest # noqa: E402

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

def sigmoid(x: np.ndarray) -> np.ndarray:
    return 1 / (1 + np.exp(-x))

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 -> fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

def threshold_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    fpr, tpr, thr = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(thr[ok][np.argmax(tpr[ok])]) if ok.any() else 0.5

def ece(y: np.ndarray, p: np.ndarray, bins: int = 15) -> float:
    edges = np.linspace(0, 1, bins + 1)
    total = 0.0
    for i in range(bins):
        m = (p >= edges[i]) & ((p < edges[i + 1]) if i < bins - 1 else (p <= edges[i + 1]))
        if m.sum():
            total += m.mean() * abs(p[m].mean() - y[m].mean())
    return float(total)

def bootstrap_auroc(y: np.ndarray, p: np.ndarray, n: int = 1000, seed: int = 42) -> tuple[float, float]:
    rng = np.random.default_rng(seed)
    vals = []
    for _ in range(n):
        idx = rng.integers(0, len(y), len(y))
        if len(np.unique(y[idx])) > 1:
            vals.append(roc_auc_score(y[idx], p[idx]))
    return float(np.percentile(vals, 2.5)), float(np.percentile(vals, 97.5))

@torch.no_grad()
def infer(ckpt_path: Path, items: list[dict], img_size: int, device: torch.device, workers: int) -> np.ndarray:
    ckpt = torch.load(ckpt_path, map_location="cpu", weights_only=False)
    arch = ckpt["arch"]
    model = timm.create_model(arch, pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.to(device).eval()
    dl = DataLoader(CXRManifest(items, train=False, img_size=img_size), batch_size=64,
                    shuffle=False, num_workers=workers)

    out = []
    for x, _, _ in tqdm(dl, desc=f"infer {ckpt_path.parent.name}", leave=False):
        x = x.to(device)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            out.append(model(x).squeeze(1).float().cpu().numpy())
    return np.concatenate(out)
```

```

def fit_temperature(val_logits: list[np.ndarray], val_y: np.ndarray) -> float:
    """Cari T>0 yang meminimalkan NLL pada val (grid log-space lebar untuk logit besar)."""
    logits = torch.tensor(np.mean(val_logits, axis=0), dtype=torch.float32)
    y = torch.tensor(val_y, dtype=torch.float32)
    best_t, best_nll = 1.0, float("inf")
    for t in np.logspace(-1, 3.5, 500):
        nll = nn.functional.binary_cross_entropy_with_logits(logits / t, y).item()
        if nll < best_nll:
            best_t, best_nll = float(t), nll
    return best_t

def reliability_plot(y: np.ndarray, p: np.ndarray, path: Path, title: str) -> None:
    edges = np.linspace(0, 1, 11)
    mids, accs = [], []
    for i in range(10):
        m = (p >= edges[i]) & (p < edges[i + 1] if i < 9 else p <= edges[i + 1])
        if m.sum():
            mids.append(p[m].mean())
            accs.append(y[m].mean())
    plt.figure(figsize=(4.2, 4.2))
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.plot(mids, accs, "o-", color="#d62728", lw=2)
    plt.xlabel("Confidence"), plt.ylabel("Empirical TB rate"), plt.title(title)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def roc_plot(curves: dict[str, tuple[np.ndarray, np.ndarray]], path: Path) -> None:
    plt.figure(figsize=(5, 5))
    for name, (y, p) in curves.items():
        fpr, tpr, _ = roc_curve(y, p)
        plt.plot(fpr, tpr, lw=2, label=f"{name} (AUROC {roc_auc_score(y, p):.3f})")
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.xlabel("FPR"), plt.ylabel("TPR (sensitivitas)"), plt.legend(fontsize=8)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--runs", type=Path, nargs="+", required=True)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--split", default="external", choices=["val", "external"])
    ap.add_argument("--temp-scale", action="store_true")
    ap.add_argument("--meta", type=Path, default=None, help="CSV filename,sex,age untuk subgroup (opsional)")
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--workers", type=int, default=8)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    manifest = json.loads(args.manifest.read_text())
    items = manifest[args.split]
    y = np.array([it["label"] for it in items])

    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    per_run_target, per_run_val, tscales = {}, {}, {}
    for run in args.runs:
        ckpt = torch.load(run / "best.pt", map_location="cpu", weights_only=False)
        img_size = ckpt["config"]["img_size"]
        per_run_target[run.name] = infer(run / "best.pt", items, img_size, device, args.workers)
        val_items = manifest["val"]
        per_run_val[run.name] = infer(run / "best.pt", val_items, img_size, device, args.workers)
        if args.temp_scale:
            tscales[run.name] = fit_temperature([per_run_val[run.name]], np.array([it["label"] for it in val_items]))

    per_run_target = {k: sigmoid(v) for k, v in per_run_target.items()}
    per_run_val = {k: sigmoid(v) for k, v in per_run_val.items()}
    val_y = np.array([it["label"] for it in manifest["val"]])

    # kalibrasi per-run pada val, lalu terapkan T ke target (bukan val-fit pada target!)
    if args.temp_scale:
        for k in per_run_target:
            per_run_val[k] = sigmoid(np.log(np.clip(per_run_val[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_val[k], 1e-6, 1 - 1e-6)))
            / tscales[k])
            lg = np.log(np.clip(per_run_target[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_target[k], 1e-6, 1 - 1e-6)))
            per_run_target[k] = sigmoid(lg / tscales[k])

    # ensemble = rata-rata; ketidakpastian = std antar anggota
    stack_t = np.stack(list(per_run_target.values()))
    p_ens = stack_t.mean(axis=0)
    u_ens = stack_t.std(axis=0)
    stack_v = np.stack(list(per_run_val.values()))
    p_ens_v = stack_v.mean(axis=0)

    tau_high = threshold_at_spec(val_y, p_ens_v, 0.70) # spec>=0.70 (TPP WHO)
    tau_low = threshold_at_spec(val_y, p_ens_v, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    band = np.where(p_ens >= tau_high, "rujuk_prioritas", np.where(p_ens < tau_low, "negatif_skrining", "ragu"))
    band_report = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b

```

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        band_report[b] = {"n": int(m.sum()), "frac": float(m.mean()),
                          "tb_rate": float(y[m].mean()) if m.sum() else None}

lo, hi = bootstrap_auc(y, p_ens)
metrics = {
    "split": args.split, "n": int(len(y)), "pos": int(y.sum()),
    "aucroc": float(roc_auc_score(y, p_ens)), "aucroc_ci95": [lo, hi],
    "sens_at_spec_90": sens_at_spec(y, p_ens, 0.90),
    "sens_at_spec_70": sens_at_spec(y, p_ens, 0.70),
    "ece": ece(y, p_ens), "brier": float(brier_score_loss(y, p_ens)),
    "temp_scales": tscales, "tau_low": tau_low, "tau_high": tau_high,
    "bands": band_report,
    "deferral": {"rate": float((band == "ragu").mean()),
                 "tb_rate_on_deferred": float(y[band == "ragu"].mean()) if (band == "ragu").any() else None},
    "per_run_aucroc": {k: float(roc_auc_score(y, v)) for k, v in per_run_target.items()},
}

if args.meta and args.meta.exists():
    meta = {r["filename"]: r for r in csv.DictReader(args.meta.open())}
    for key in ("sex", "age"):
        groups: dict[str, list[int]] = {}
        for i, it in enumerate(items):
            row = meta.get(Path(it["path"]).name, {})
            g = row.get(key, "NA") or "NA"
            if key == "age":
                try:
                    a = float(g)
                    g = "0-30" if a < 30 else "30-50" if a < 50 else "50+"
                except ValueError:
                    g = "NA"
            groups.setdefault(g, []).append(i)
        metrics[f"aucroc_by_{key}"] = {
            g: (float(roc_auc_score(y[idx], p_ens[idx])) if len(np.unique(y[idx])) > 1 else None)
              for g, idx in groups.items() if g != "NA" and len(idx) >= 20
        }

(args.out / "metrics.json").write_text(json.dumps(metrics, indent=1))
# dump prediksi per-citra (untuk analisis lanjutan & stacking)
import csv as _csv
with (args.out / "preds.csv").open("w", newline="") as f:
    w = _csv.writer(f)
    w.writerow(["path", "label", "p_ens", "u_ens"])
    for it, p, u in zip(items, p_ens, u_ens):
        w.writerow([it["path"], it["label"], float(p), float(u)])
reliability_plot(y, p_ens, args.out / "reliability.png", f"Kalibrasi - {args.split}")
curves = {f"ens ({args.split})": (y, p_ens)}
if len(per_run_target) > 1:
    curves.update({k: (y, v) for k, v in per_run_target.items()})
roc_plot(curves, args.out / "roc.png")

print("\n=== METRICS ===")
for k, v in metrics.items():
    if isinstance(v, (int, float, str)):
        print(f"{k:18s} {v}")
print(f"bands          {band_report}")

if __name__ == "__main__":
    main()

```

ml/export_onnx.py

```
#!/usr/bin/env python3
"""Export checkpoint ke ONNX (opsi 17) + verifikasi onnxruntime.

Contoh:
python ml/export_onnx.py --ckpt ml/runs/efficientnet_b0/best.pt --out artifacts/tb_efficientnet_b0.onnx
"""
from __future__ import annotations

import argparse
from pathlib import Path

import numpy as np
import torch

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm")

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__)
    ap.add_argument("--ckpt", type=Path, required=True)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--opset", type=int, default=17)
    args = ap.parse_args()

    ckpt = torch.load(args.ckpt, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()

    img_size = ckpt["config"].get("img_size", 224)
    dummy = torch.randn(1, 3, img_size, img_size)
    args.out.parent.mkdir(parents=True, exist_ok=True)
    torch.onnx.export(
        model, dummy, args.out,
        input_names=["input"], output_names=["logit"],
        dynamic_axes={"input": {0: "batch"}, "logit": {0: "batch"}},
        opset_version=args.opset, dynamo=False, # legacy: satu file mandiri, opset stabil
    )

    import onnxruntime as ort
    sess = ort.InferenceSession(str(args.out), providers=["CPUExecutionProvider"])
    ref = model(dummy).detach().numpy()
    got = sess.run(None, {"input": dummy.numpy()})[0]
    # Verifikasi di ruang probabilitas (deployment-relevant); logit bisa sangat besar
    # pada model overkonfiden sehingga toleransi absolut di ruang logit tidak adil.
    sig = lambda z: 1 / (1 + np.exp(-z)) # noqa: E731
    prob_diff = float(np.abs(sig(ref) - sig(got)).max())
    size_mb = args.out.stat().st_size / 1e6
    print(f"ONNX ok: {args.out} ({size_mb:.1f} MB) max|Δprob|={prob_diff:.2e} opset={args.opset}")
    if prob_diff > 1e-3:
        print(f"WARN: perbedaan probabilitas {prob_diff:.4f} > 1e-3 – periksa sebelum produksi")

if __name__ == "__main__":
    main()
```

ml/ssl_probe.py

```
#!/usr/bin/env python3
"""Latih probe linear di atas embedding SSL + evaluasi eksternal rigor (ADR-007).

Contoh:
.venv/bin/python ml/ssl_probe.py --emb-dir ml/kaggle/out_ssl/ssl_embeddings \
  --out ml/runs/ssl_eval --supervised-preds ml/runs/eval_external_ens/preds.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import sys
from pathlib import Path

import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import brier_score_loss, roc_auc_score
from sklearn.preprocessing import StandardScaler

sys.path.insert(0, str(Path(__file__).resolve().parent))
from evaluate import ( # noqa: E402
    bootstrap_auroc,
    ece,
    fit_temperature,
    reliability_plot,
    roc_plot,
    sens_at_spec,
    threshold_at_spec,
)

# Statistik preprocessing piksel yang HARUS sama dengan saat ekstraksi embedding.
# (dinov2: default timm/ImageNet; rad_dino: diisi dari log kernel – processor HF)
PIXEL = {
    "dinov2": {
        "type": "timm", "id": "vit_base_patch14_dinov2.lvd142m", "size": 518,
        "mean": (0.485, 0.456, 0.406), "std": (0.229, 0.224, 0.225),
    },
    "rad_dino": {
        "type": "hf", "id": "microsoft/rad-dino", "size": 518,
        "mean": (0.5307, 0.5307, 0.5307), "std": (0.2588, 0.2588, 0.2588), # diverifikasi dari log kernel; koreksi bila beda
    },
}

def sigmoid(z: np.ndarray) -> np.ndarray:
    z = np.clip(z, -60, 60)
    return 1 / (1 + np.exp(-z))

def load_split(emb_dir: Path, enc: str, split: str) -> tuple[np.ndarray, np.ndarray, list[dict]]:
    X = np.load(emb_dir / f"emb_{enc}_{split}.npy")
    meta = json.loads((emb_dir / f"meta_{split}.json").read_text())
    y = np.array([m["label"] for m in meta])
    return X, y, meta

def pixel_of(emb_dir: Path, enc: str) -> dict:
    """Statistik preprocessing piksel: dari encmeta kernel bila ada, else fallback PIXEL."""
    f = emb_dir / f"encmeta_{enc}.json"
    if f.exists():
        d = json.loads(f.read_text())
        return {"type": d["type"], "id": d["id"], "size": d["size"], "mean": tuple(d["mean"]), "std": tuple(d["std"])}
    return PIXEL.get(enc, PIXEL["dinov2"])

def band_metrics(y: np.ndarray, p: np.ndarray, tau_low: float, tau_high: float) -> dict:
    band = np.where(p >= tau_high, "rujuk_prioritas", np.where(p < tau_low, "negatif_skrining", "ragu"))
    rep = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b
        rep[b] = {
            "n": int(m.sum()),
            "frac": float(m.mean()),
            "tb_rate": float(y[m].mean()) if m.sum() else None,
        }
    return rep

def full_metrics(y: np.ndarray, p: np.ndarray) -> dict:
    lo, hi = bootstrap_auroc(y, p)
    return {
        "n": int(len(y)),
        "pos": int(y.sum()),
        "auroc": float(roc_auc_score(y, p)),
        "auroc_ci95": [lo, hi],
        "sens_at_spec_90": sens_at_spec(y, p, 0.90),
        "sens_at_spec_70": sens_at_spec(y, p, 0.70),
    }
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        "ece": ece(y, p),
        "brier": float(brier_score_loss(y, p)),
    }

def run_encoder(enc: str, emb_dir: Path, out_dir: Path, supervised_preds: Path | None) -> dict:
    Xtr, ytr, _ = load_split(emb_dir, enc, "train")
    Xva, yva, meta_va = load_split(emb_dir, enc, "val")
    Xex, yex, meta_ex = load_split(emb_dir, enc, "external")

    scaler = StandardScaler().fit(Xtr)
    Xtr_s, Xva_s, Xex_s = scaler.transform(Xtr), scaler.transform(Xva), scaler.transform(Xex)

    best = {"auroc": -1.0, "C": None}
    for C in (0.001, 0.01, 0.1, 1.0, 10.0):
        clf = LogisticRegression(C=C, class_weight="balanced", max_iter=3000, solver="lbfgs")
        clf.fit(Xtr_s, ytr)
        z_va = clf.decision_function(Xva_s)
        try:
            auroc = roc_auc_score(yva, sigmoid(z_va))
        except ValueError:
            auroc = float("nan")
        if auroc > best["auroc"]:
            best = {"auroc": float(auroc), "C": C, "clf": clf}
        print(f" [{enc}] C={C:<6} val AUROC={auroc:.4f}")

    clf: LogisticRegression = best["clf"]
    z_va, z_ex = clf.decision_function(Xva_s), clf.decision_function(Xex_s)
    T = fit_temperature([z_va], yva)
    p_va, p_ex = sigmoid(z_va / T), sigmoid(z_ex / T)

    tau_high = threshold_at_spec(yva, p_va, 0.70)
    tau_low = threshold_at_spec(yva, p_va, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    res = {
        "encoder": enc,
        "chosen_C": best["C"],
        "val": full_metrics(yva, p_va),
        "external": full_metrics(yex, p_ex),
        "temp": T,
        "tau_low": tau_low,
        "tau_high": tau_high,
        "bands_external": band_metrics(yex, p_ex, tau_low, tau_high),
    }

    out_dir.mkdir(parents=True, exist_ok=True)
    (out_dir / f"metrics_{enc}.json").write_text(json.dumps(res, indent=1))

    # ekspor probe untuk serving API (scaler + LR + T + metadata encoder)
    pix = pixel_of(emb_dir, enc)
    np.savez(
        out_dir / f"probe_{enc}.npz",
        scaler_mean=scaler.mean_,
        scaler_scale=scaler.scale_,
        coef=clf.coef_.ravel(),
        intercept=np.array([float(clf.intercept_[0])]),
        T=np.array([T]),
        size=np.array([pix["size"]]),
        pixel_mean=np.array(pix["mean"], dtype=np.float64),
        pixel_std=np.array(pix["std"], dtype=np.float64),
        encoder_type=np.array([pix["type"]]),
        encoder_id=np.array([pix["id"]]),
    )

    reliability_plot(yex, p_ex, out_dir / f"reliability_{enc}.png", f"SSL probe {enc} - external")
    roc_plot({f"{enc} (external)": (yex, p_ex)}, out_dir / f"roc_{enc}.png")
    with (out_dir / f"preds_{enc}.csv").open("w", newline="") as f:
        w = csv.writer(f)
        w.writerow(["path", "label", "p", "z"])
        for m, p, z in zip(meta_ex, p_ex, z_ex):
            w.writerow([m["path"], m["label"], float(p), float(z)])

    # stacking dengan ensemble supervised (jika tersedia)
    if supervised_preds and supervised_preds.exists():
        sup = {}
        for row in csv.DictReader(supervised_preds.open()):
            sup[Path(row["path"]).stem] = float(row["p_ens"])
        pairs = [(Path(m["path"]).stem, m["label"], p) for m, p in zip(meta_ex, p_ex)]
        aligned = [(k, lab, p, sup[k]) for k, lab, p in pairs if k in sup]
        if len(aligned) >= 50:
            ya = np.array([a[1] for a in aligned])
            ps = (np.array([a[2] for a in aligned]) + np.array([a[3] for a in aligned])) / 2
            res["stack_with_supervised"] = {**full_metrics(ya, ps), "n_matched": len(aligned)}
            print(f" [{enc}] stack n={len(aligned)} external AUROC={res['stack_with_supervised']['auroc']:.4f}")

    (out_dir / f"metrics_{enc}.json").write_text(json.dumps(res, indent=1))
    return res

def main() -> None:

```

```

ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
ap.add_argument("--emb-dir", type=Path, required=True)
ap.add_argument("--out", type=Path, required=True)
ap.add_argument("--encoder", nargs="*", default=None, help="default: semua yang ada")
ap.add_argument("--supervised-preds", type=Path, default=None)
args = ap.parse_args()

encs = args.encoder or sorted(
    p.name[len("emb_") : -len("_train.npy")] for p in args.emb_dir.glob("emb_*_train.npy")
)
results = []
for enc in encs:
    print(f"=== {enc} ===")
    results.append(run_encoder(enc, args.emb_dir, args.out, args.supervised_preds))

print("\n=== RINGKASAN (eksternal) ===")
for r in results:
    e = r["external"]
    print(f"{r['encoder']:10s} AUROC {e['auroc']:.4f} [{e['auroc_ci95'][0]:.3f},{e['auroc_ci95'][1]:.3f}] "
          f"| sens@spec90 {e['sens_at_spec_90']:.3f} | ECE {e['ece']:.3f}"
          + (f" | stack {r['stack_with_supervised'] ['auroc']:.4f}" if "stack_with_supervised" in r else ""))

# kombinasi rata-rata probabilitas antar encoder SSL
preds_files = sorted(args.out.glob("preds_*.csv"))
if len(preds_files) > 1:
    maps = []
    for f in preds_files:
        rows = list(csv.DictReader(f.open()))
        maps.append({r["path"]: (float(r["p"]), int(r["label"])) for r in rows})
    keys = set(maps[0])
    for m in maps[1:]:
        keys |= set(m)
    ks = sorted(keys)
    y = np.array([maps[0][k][1] for k in ks])
    p = np.mean([np.array([m[k][0] for k in ks]) for m in maps], axis=0)
    comb = full_metrics(y, p)
    print(f"combined{'x'.join([]) or ''} SSL ({len(maps)} encoder): AUROC {comb['auroc']:.4f} "
          f"[{comb['auroc_ci95'][0]:.3f},{comb['auroc_ci95'][1]:.3f}] | sens@spec90 {comb['sens_at_spec_90']:.3f}")
    (args.out / "metrics_combined.json").write_text(json.dumps(comb, indent=1))

if __name__ == "__main__":
    main()

```

ml/train.py

```
#!/usr/bin/env python3
"""Train TB CXR classifier dari manifest provenance-aware.

Contoh:
python ml/train.py --manifest ml/data/artifacts/manifest.json --arch efficientnet_b0 \
    --epochs 20 --out ml/runs/efficientnet_b0

Baseline linear probe (fitur beku):
python ml/train.py ... --freeze --lr 1e-3 --out ml/runs/linear_probe
"""
from __future__ import annotations

import argparse
import csv
import json
import time
from pathlib import Path

import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import roc_auc_score, roc_curve
from torch.utils.data import DataLoader, Dataset, WeightedRandomSampler
from torchvision import transforms as T
from tqdm import tqdm

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

class CXRManifest(Dataset):
    def __init__(self, items: list[dict], train: bool, img_size: int):
        self.items = items
        if train:
            self.tx = T.Compose([
                T.RandomResizedCrop(img_size, scale=(0.75, 1.0), ratio=(0.9, 1.1)),
                T.RandomRotation(7, fill=0),
                T.ColorJitter(brightness=0.15, contrast=0.15),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])
        else:
            self.tx = T.Compose([
                T.Resize(int(img_size * 1.14)),
                T.CenterCrop(img_size),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])

    def __len__(self) -> int:
        return len(self.items)

    def __getitem__(self, i: int):
        from PIL import Image
        it = self.items[i]
        with Image.open(it["path"]) as im:
            x = self.tx(im.convert("RGB"))
            return x, torch.tensor(float(it["label"])), i

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 → fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

@torch.no_grad()
def predict(model: nn.Module, loader: DataLoader, device: torch.device) -> tuple[np.ndarray, np.ndarray]:
    model.eval()
    logits_all, labels_all = [], []
    for x, y, _ in loader:
        x = x.to(device, non_blocking=True)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            logits = model(x).squeeze(1).float()
        logits_all.append(logits.cpu().numpy())
        labels_all.append(y.numpy())
    return np.concatenate(logits_all), np.concatenate(labels_all)

def evaluate(logits: np.ndarray, y: np.ndarray) -> dict:
    p = 1 / (1 + np.exp(-logits))
    out = {"n": int(len(y)), "pos": int(y.sum())}
    if len(np.unique(y)) > 1:
```

```

        out["auroc"] = float(roc_auc_score(y, p))
        out["sens_at_spec_90"] = sens_at_spec(y, p, 0.90)
        out["sens_at_spec_70"] = sens_at_spec(y, p, 0.70)
    return out

```

```

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--arch", default="efficientnet_b0")
    ap.add_argument("--epochs", type=int, default=20)
    ap.add_argument("--batch", type=int, default=32)
    ap.add_argument("--lr", type=float, default=3e-4)
    ap.add_argument("--wd", type=float, default=1e-4)
    ap.add_argument("--img-size", type=int, default=224)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--seed", type=int, default=42)
    ap.add_argument("--workers", type=int, default=8)
    ap.add_argument("--freeze", action="store_true", help="linear probe: backbone beku")
    ap.add_argument("--balanced", action="store_true", help="WeightedRandomSampler")
    ap.add_argument("--limit", type=int, default=0, help="smoke test: batasi n sampel (0=semua)")
    args = ap.parse_args()

    torch.manual_seed(args.seed)
    np.random.seed(args.seed)
    args.out.mkdir(parents=True, exist_ok=True)

    device = torch.device("cuda" if torch.cuda.is_available() else "mps" if torch.backends.mps.is_available() else "cpu")
    manifest = json.loads(args.manifest.read_text())

    def take(items: list[dict], n: int) -> list[dict]:
        """Subset berstratifikasi untuk smoke test."""
        if not n:
            return items
        pos = [it for it in items if it["label"] == 1]
        neg = [it for it in items if it["label"] == 0]
        n = min(n, len(items))
        kp = min(len(pos), max(1, n // 4))
        return pos[:kp] + neg[: max(n - kp, 0)]

    train_items = take(manifest["train"], args.limit)
    val_items = take(manifest["val"], args.limit)

    ds_tr = CXRMManifest(train_items, train=True, img_size=args.img_size)
    ds_va = CXRMManifest(val_items, train=False, img_size=args.img_size)
    sampler = None
    if args.balanced:
        labels = np.array([it["label"] for it in train_items])
        w = np.where(labels == 1, 1.0 / labels.sum(), 1.0 / (len(labels) - labels.sum()))
        sampler = WeightedRandomSampler(torch.from_numpy(w), num_samples=len(labels), replacement=True)
    dl_tr = DataLoader(ds_tr, batch_size=args.batch, shuffle=sampler is None, sampler=sampler,
                      num_workers=args.workers, pin_memory=(device.type == "cuda"), drop_last=True)
    dl_va = DataLoader(ds_va, batch_size=args.batch, shuffle=False, num_workers=args.workers, pin_memory=(device.type == "cuda"))

    model = timm.create_model(args.arch, pretrained=True, num_classes=1).to(device)
    if args.freeze:
        for n, p in model.named_parameters():
            p.requires_grad = "head" in n or "classifier" in n or "fc" in n
    params = [p for p in model.parameters() if p.requires_grad]
    opt = torch.optim.AdamW(params, lr=args.lr, weight_decay=args.wd)
    sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=max(args.epochs, 1))
    loss_fn = nn.BCEWithLogitsLoss()
    scaler = torch.amp.GradScaler(enabled=(device.type == "cuda"))

    print(f"device={device} arch={args.arch} freeze={args.freeze} "
          f"train={len(ds_tr)} val={len(ds_va)} params={sum(p.numel() for p in params)/1e6:.2f}M")

    hist = []
    best = {"auroc": -1.0, "epoch": -1}
    for epoch in range(1, args.epochs + 1):
        model.train()
        t0, losses = time.time(), []
        for x, y, _ in tqdm(dl_tr, desc=f"epoch {epoch:02d}", leave=False):
            x, y = x.to(device, non_blocking=True), y.to(device, non_blocking=True)
            opt.zero_grad(set_to_none=True)
            with torch.autocast(device.type, enabled=(device.type == "cuda")):
                logits = model(x).squeeze(1)
                loss = loss_fn(logits.float(), y)
            scaler.scale(loss).backward()
            scaler.step(opt)
            scaler.update()
            losses.append(loss.item())
        sched.step()

        logits, y_va = predict(model, dl_va, device)
        m = evaluate(logits, y_va)
        row = {"epoch": epoch, "loss": float(np.mean(losses)), "lr": opt.param_groups[0]["lr"],
              "sec": round(time.time() - t0, 1), **m}
        hist.append(row)
    print(f"epoch {epoch:02d} | loss {row['loss']:.4f} | AUROC {m.get('auroc', float('nan')):.4f} "
          f"| sens@spec90 {m.get('sens_at_spec_90', float('nan')):.3f} | {row['sec']}s")

```

```

torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
           "epoch": epoch, "val": m}, args.out / "last.pt")
if m.get("auroc", -1) > best["auroc"]:
    best = {"auroc": m["auroc"], "epoch": epoch}
    torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
               "epoch": epoch, "val": m}, args.out / "best.pt")
    with (args.out / "val_preds.csv").open("w", newline="") as f:
        w = csv.writer(f)
        w.writerow(["path", "label", "logit"])
        for it, lg in zip(val_items, logits):
            w.writerow([it["path"], it["label"], float(lg)])

(args.out / "history.csv").write_text(
    "epoch,loss,lr,sec,n,pos,auroc,sens_at_spec_90,sens_at_spec_70\n" + "\n".join(
        ", ".join(str(r.get(k, "")) for k in ("epoch", "loss", "lr", "sec", "n", "pos", "auroc", "sens_at_spec_90",
        "sens_at_spec_70"))
        for r in hist) + "\n")
(args.out / "metrics.json").write_text(json.dumps({"best": best, "history": hist, "config": vars(args)}, indent=1, default=str))
print(f"\nBEST epoch {best['epoch']} AUROC {best['auroc']:.4f} -> {args.out/'best.pt'}")

if __name__ == "__main__":
    main()

```

ml/data/prepare_data.py

```
#!/usr/bin/env python3
"""Build a provenance-aware manifest for TB chest X-ray training.

Sumber & aturan label (lihat docs/02-arsitektur.md):
- TBX11K (Kaggle: vbookshelf/tbx11k-simplified): folder per kelas.
  positif = {tb}; negatif = {healthy, sick}; latent_tb DIKELUARKAN (ambigu).
- Rahman composite (Kaggle: tawsifurrahman/tuberculosis-tb-chest-xray-dataset):
  folder Normal/ -> 0, Tuberculosis/ -> 1.
- NLM Shenzhen/Montgomery: konvensi nama file resmi `*_0` = normal, `*_1` = TB.

Dedup (WAJIB – mitigasi class-conditional acquisition confounding):
1. pHash 64-bit; Hamming <= --dup-threshold dianggap duplikat (union-find transitif).
2. Integritas external: citra train yang sekelompok dengan citra external DIBUANG dari train.
3. Duplikat internal train dibuang (keep-first, urutan deterministik).
4. Split val dilakukan per-klaster (tidak ada kebocoran near-dup train->val).

Output (ml/data/artifacts/):
- manifest.json, dedup_report.json, train.csv, val.csv, external.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import re
from collections import Counter
from pathlib import Path

import numpy as np
from PIL import Image
from tqdm import tqdm

try:
    import imagehash
except ImportError: # pragma: no cover
    raise SystemExit("Butuh imagehash: pip install imagehash")

IMG_EXTS = {".png", ".jpg", ".jpeg", ".bmp", ".tif", ".tiff"}
POS_TOKENS = {"tb", "tuberculosis", "tuberculose"}
NEG_TOKENS = {"healthy", "normal", "sick"}
EXCLUDE_TOKENS = {"latent", "latent_tb", "latenttb"}

def _rel(p: Path, root: Path) -> str:
    """Path relatif terhadap root repo (portable lokal <-> Colab); fallback absolut."""
    try:
        return str(p.relative_to(root))
    except ValueError:
        return str(p)

def _tokens(path: Path) -> set[str]:
    parts = [p.lower() for p in path.parts]
    parts += [path.stem.lower()]
    return set(parts)

def load_tbx11k_map(raw: Path) -> dict[str, int | None]:
    """Label otoritatif dari data.csv TBX11K: 1=active_tb, 0=no_tb, None=latent (dikeluarkan)."""
    m: dict[str, int | None] = {}
    for p in raw.rglob("data.csv"):
        if "tbx11k" not in str(p).lower():
            continue
        for r in csv.DictReader(p.open()):
            f = (r.get("fname") or "").strip()
            if not f:
                continue
            t = (r.get("tb_type") or "").lower()
            tgt = (r.get("target") or "").lower()
            if "latent" in t:
                m.setdefault(f, None)
            elif tgt == "tb":
                m[f] = 1
            else:
                m.setdefault(f, 0)
    return m

def classify(path: Path, raw: Path, tbx_map: dict[str, int | None]) -> tuple[int, str] | None:
    """Return (label, source) atau None kalau tidak bisa diklasifikasi."""
    rel = path.relative_to(raw)
    low = str(rel).lower()

    # --- NLM Shenzhen / Montgomery: konvensi suffix nama file ---
    m = re.search(r"(chncxr|mcucxr)_d+_(\d+)", path.stem.lower())
    if m:
        label = int(m.group(2))
        source = "shenzhen" if m.group(1) == "chncxr" else "montgomery"
```

```

        return label, source

toks = _tokens(rel)
if toks & EXCLUDE_TOKENS or any(t.startswith("latent") for t in toks):
    return None

if "tbx11k" in low:
    if tbx_map:
        lab = tbx_map.get(path.name, None)
        return None if lab is None else (lab, "tbx11k")
    # fallback bila data.csv tidak ada: prefix nama file
    stem = path.stem.lower()
    if stem.startswith("t"):
        return 1, "tbx11k"
    if stem.startswith(("h", "s")):
        return 0, "tbx11k"
    return None

if toks & POS_TOKENS:
    return 1, "rahman"
if toks & NEG_TOKENS:
    return 0, "rahman"
return None

class DSU:
    def __init__(self, n: int):
        self.parent = list(range(n))

    def find(self, x: int) -> int:
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def union(self, a: int, b: int) -> None:
        ra, rb = self.find(a), self.find(b)
        if ra != rb:
            self.parent[rb] = ra

_POP = np.array([bin(i).count("1") for i in range(256)], dtype=np.uint8)

def hamming_pairs(hashes: np.ndarray, threshold: int, chunk: int = 512) -> list[tuple[int, int]]:
    """Semua pasangan (i<j) dengan Hamming distance <= threshold (chunked, hemat memori)."""
    n = len(hashes)
    h8 = hashes.view(np.uint8).reshape(n, 8)
    pairs: list[tuple[int, int]] = []
    for start in range(0, n, chunk):
        stop = min(start + chunk, n)
        xor = np.bitwise_xor(h8[start:stop, None, :], h8[None, :, :])
        dist = _POP[xor].sum(axis=-1) # (chunk, n)
        rows, cols = np.where(dist <= threshold)
        for r, c in zip(rows, cols):
            i, j = start + int(r), int(c)
            if i < j:
                pairs.append((i, j))
    return pairs

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--raw", type=Path, default=Path("data/raw"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts"))
    ap.add_argument("--dup-threshold", type=int, default=3, help="Hamming pHash <= ini dianggap duplikat")
    ap.add_argument("--val-frac", type=float, default=0.15)
    ap.add_argument("--seed", type=int, default=42)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    raw = args.raw.resolve()
    root = raw.parent.parent # asumsi struktur <root>/data/raw

    # ----- 1) discovery -----
    labeled: list[dict] = []
    unlabeled = Counter()
    tbx_map = load_tbx11k_map(raw)
    print(f"[discovery] label TBX11K dari data.csv: {len(tbx_map)} entri")
    for p in sorted(raw.rglob("*")):
        if p.suffix.lower() not in IMG_EXTS:
            continue
        cls = classify(p, raw, tbx_map)
        if cls is None:
            unlabeled[str(p.relative_to(raw).parent)] += 1
            continue
        label, source = cls
        labeled.append({"path": _rel(p, root), "label": label, "source": source})

    print(f"[discovery] berlabel={len(labeled)} tak-terklasifikasi={sum(unlabeled.values())}")
    for k, v in unlabeled.most_common(10):
        print(f" skip {v:6d} {k}")

```

```

if not labeled:
    raise SystemExit("Tidak ada citra berlabel – cek struktur data/raw")

# ----- 2) pHash + kluster duplikat -----
print("[dedup] menghitung pHash...")
ok: list[dict] = []
for item in tqdm(labeled, unit="img"):
    try:
        with Image.open(item["path"]) as im:
            h = imagehash.phash(im.convert("L"), hash_size=8)
            item["phash"] = int(str(h), 16)
            ok.append(item)
    except Exception as e: # gambar korup
        print(f" warn: gagal baca {item['path']}: {e}")
labeled = ok

hashes = np.array([it["phash"] for it in labeled], dtype=np.uint64)
pairs = hamming_pairs(hashes, args.dup_threshold)
dsu = DSU(len(labeled))
for i, j in pairs:
    dsu.union(i, j)
for idx, it in enumerate(labeled):
    it["cluster"] = dsu.find(idx)

by_cluster: dict[int, list[dict]] = {}
for it in labeled:
    by_cluster.setdefault(it["cluster"], []).append(it)

# ----- 3) lock external, buang kebocoran dari train -----
ext = [it for it in labeled if it["source"] in ("shenzhen", "montgomery")]
ext_clusters = {it["cluster"] for it in ext}
train_pool, dropped_cross = [], []
for it in labeled:
    if it["source"] in ("shenzhen", "montgomery"):
        continue
    (dropped_cross if it["cluster"] in ext_clusters else train_pool).append(it)

# duplikat internal train: keep-first per kluster (deterministik via path)
keep, dropped_internal = [], []
seen_clusters: set[int] = set()
for it in sorted(train_pool, key=lambda x: x["path"]):
    if it["cluster"] in seen_clusters:
        dropped_internal.append(it)
    else:
        seen_clusters.add(it["cluster"])
        keep.append(it)
train_pool = keep

# ----- 4) split val per-kluster -----
rng = np.random.default_rng(args.seed)
sizes = Counter(it["cluster"] for it in train_pool)
pool_clusters = sorted(sizes)
rng.shuffle(pool_clusters)
target_val = int(len(train_pool) * args.val_frac)
val_clusters: set[int] = set()
acc = 0
for c in pool_clusters:
    if acc >= target_val:
        break
    val_clusters.add(c)
    acc += sizes[c]
train = [it for it in train_pool if it["cluster"] not in val_clusters]
val = [it for it in train_pool if it["cluster"] in val_clusters]

def dump(items: list[dict]) -> list[dict]:
    return [
        {k: it[k] for k in ("path", "label", "source", "cluster", "phash")}
        for it in sorted(items, key=lambda x: x["path"])
    ]

manifest = {
    "meta": {
        "root": str(root),
        "raw": _rel(raw, root),
        "dup_threshold": args.dup_threshold,
        "val_frac": args.val_frac,
        "seed": args.seed,
        "counts": {
            "train": len(train), "val": len(val), "external": len(ext),
            "dropped_cross_corpus": len(dropped_cross),
            "dropped_internal_dup": len(dropped_internal),
            "unlabeled": sum(unlabeled.values()),
        },
    },
    "train": dump(train),
    "val": dump(val),
    "external": dump(ext),
}
(args.out / "manifest.json").write_text(json.dumps(manifest, indent=1))
(args.out / "dedup_report.json").write_text(json.dumps({
    "pairs_within_threshold": len(pairs),
    "clusters": len(by_cluster),
}))

```

```

        "dropped_cross_corpus": dump(dropped_cross),
        "dropped_internal_dup": dump(dropped_internal),
        "unlabeled_by_dir": dict(unlabeled),
    }, indent=1))

for split in ("train", "val", "external"):
    with (args.out / f"{split}.csv").open("w") as f:
        f.write("path,label,source,cluster\n")
        for it in manifest[split]:
            f.write(f"{it['path']},{it['label']},{it['source']},{it['cluster']}\n")

def comp(items):
    c = Counter((it["source"], it["label"]) for it in items)
    return " ".join(f"{s}/{ 'TB' if l else 'NEG' }={n}" for (s, l), n in sorted(c.items()))

print("\n=== RINGKASAN ===")
print(f"train      n={len(train):6d}  {comp(train)}")
print(f"val        n={len(val):6d}    {comp(val)}")
print(f"external  n={len(ext):6d}    {comp(ext)}")
print(f"dibuang  cross-corpus={len(dropped_cross)}  internal-dup={len(dropped_internal)}")
print(f"manifest -> {args.out/'manifest.json'}")

if __name__ == "__main__":
    main()

```

ml/data/prepare_meta.py

```
#!/usr/bin/env python3
"""Ekstrak metadata klinis NLM (Montgomery + Shenzhen) → CSV untuk analisis subgroup.

Output: ml/data/artifacts/meta_nlm.csv
Kolom: filename,sex,age,reading,scar_flag
- Montgomery: "Patient's Sex: F / Patient's Age: 027Y / <reading>"
- Shenzhen: "male 45yrs / <reading>" (kadang terkontaminasi antar-file)
- scar_flag: menyebut scar/inactive/fibrosis → untuk pembahasan mode gagal
            "healed scar false positive" (selaras audit Bilal 2026).
"""
from __future__ import annotations

import argparse
import csv
import re
from pathlib import Path

SEX_M = re.compile(r"sex[:\s]+(m|male)\b", re.I)
SEX_F = re.compile(r"sex[:\s]+(f|female)\b", re.I)
SEX_FMT2 = re.compile(r"\b(male|female)\b", re.I)
AGE_MONT = re.compile(r"age[:\s]+(\d{1,3})\s*Y?\b", re.I)
AGE_FMT2 = re.compile(r"\b(\d{1,3})\s*yrs\b", re.I)
SCAR = re.compile(r"scar|inactive|fibro", re.I)

def parse_montgomery(text: str) -> dict | None:
    age = AGE_MONT.search(text)
    sex = "F" if SEX_F.search(text) else ("M" if SEX_M.search(text) else "")
    if not age:
        return None
    # reading = baris setelah Age
    lines = [ln.strip() for ln in text.splitlines() if ln.strip()]
    reading = lines[2] if len(lines) > 2 else ""
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": int(bool(SCAR.search(reading)))}

def parse_shenzhen(text: str) -> dict | None:
    age = AGE_FMT2.search(text)
    sex_m = SEX_FMT2.findall(text)
    sex = sex_m[0][0].upper() if sex_m else ""
    if not age:
        return None
    # reading = token huruf setelah "yrs"
    tail = text[age.end():]
    reading = ""
    m = re.search(r"[A-Za-z]{2,}", tail)
    if m:
        reading = m.group(0)
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": 0}

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--nlm", type=Path, default=Path("data/raw/nlm"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts/meta_nlm.csv"))
    args = ap.parse_args()
    args.out.parent.mkdir(parents=True, exist_ok=True)

    rows, n_fail = [], 0
    for corpus, parser in (("montgomery", parse_montgomery), ("shenzhen", parse_shenzhen)):
        d = args.nlm / corpus / "ClinicalReadings"
        for p in sorted(d.glob("*.txt")):
            r = parser(p.read_text(errors="ignore"))
            if r is None:
                n_fail += 1
                continue
            r["filename"] = p.stem + ".png"
            r["corpus"] = corpus
            rows.append(r)

    with args.out.open("w", newline="") as f:
        w = csv.DictWriter(f, fieldnames=["filename", "corpus", "sex", "age", "reading", "scar_flag"])
        w.writeheader()
        w.writerows(rows)

    from collections import Counter
    print(f"meta -> {args.out} n={len(rows)} gagal-parse={n_fail}")
    print("per korpus:", Counter(r["corpus"] for r in rows))
    print("sex:", Counter(r["sex"] for r in rows))
    print("scar_flag:", Counter(r["scar_flag"] for r in rows))

if __name__ == "__main__":
    main()
```

ml/kaggle/hello/hello.py

```
"""Kernel uji: verifikasi GPU, internet, versi lib, dan struktur dataset ter-mount."""
import os
import sys

lines: list[str] = []

def log(*a) -> None:
    print(*a, flush=True)
    lines.append(" ".join(str(x) for x in a))

log("python:", sys.version.split()[0])

try:
    import torch

    log("torch:", torch.__version__, "| cuda:", torch.cuda.is_available())
    if torch.cuda.is_available():
        log("gpu:", torch.cuda.get_device_name(0))
except Exception as e: # noqa: BLE001
    log("torch error:", e)

for mod in ("timm", "sklearn", "onnx", "onnxruntime"):
    try:
        m = __import__(mod)
        log(f"{mod}: {getattr(m, '__version__', '?')}")
    except Exception as e: # noqa: BLE001
        log(f"{mod}: MISSING ({e})")

log("--- input mounts ---")
for root, dirs, _ in os.walk("/kaggle/input"):
    log(root, "dirs:", dirs[:6])
    if root.count("/") > 3:
        dirs[:] = []

log("HELLO DONE")
with open("/kaggle/working/hello_result.txt", "w") as f:
    f.write("\n".join(lines) + "\n")
```

ml/kaggle/make_nlm_dataset.py

```
#!/usr/bin/env python3
"""Siapkan dataset Kaggle privat: NLM CXR di-resize (max dim 512) + ClinicalReadings.

Dipisah dari data asli agar unggahan ringan (~150 MB) dan resolusi tetap cukup
untuk encoder SSL (DINOv2/RAD-DINO resize ke 518).
"""
from __future__ import annotations

import shutil
from pathlib import Path

from PIL import Image

SRC = Path("data/raw/nlm")
DST = Path("ml/kaggle/nlm_dataset/nlm")

def main() -> None:
    n = 0
    for corpus in ("shenzhen", "montgomery"):
        out_dir = DST / corpus / "CXR_png"
        out_dir.mkdir(parents=True, exist_ok=True)
        for p in sorted((SRC / corpus / "CXR_png").glob("*.png")):
            with Image.open(p) as im:
                im = im.convert("L")
                w, h = im.size
                scale = 512 / max(w, h)
                if scale < 1:
                    im = im.resize((max(1, round(w * scale)), max(1, round(h * scale))), Image.LANCZOS)
                im.save(out_dir / (p.stem + ".jpg"), quality=92)
            n += 1
        readings = SRC / corpus / "ClinicalReadings"
        if readings.exists():
            out_r = DST / corpus / "ClinicalReadings"
            out_r.mkdir(parents=True, exist_ok=True)
            for f in readings.glob("*.txt"):
                shutil.copy(f, out_r / f.name)
        print(f"selesai: {n} citra -> {DST}")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl/loop-tb/api/main.py

```
"""SIGAP API – inference ensemble (supervised ckpt &/atau probe SSL) + Grad-CAM + band triase.

Jalankan lokal:
    uvicorn api.main:app --host 0.0.0.0 --port 8000

Konfigurasi (default `api/model_config.json`, override env LOOPTB_CONFIG):
{
    "score_members": [ {"path": "ml/runs/ssl_eval/probe_dinov2.npz"} ],
    "uncertainty_members": [ {"path": "ml/runs/eb0/best.pt", "T": 1.2336},
                             {"path": "ml/runs/convnext/best.pt", "T": 1.3405} ],
    "bands": {"tau_low": ..., "tau_high": ..., "tau_uncertainty": 0.2}
}

Skor p = rata-rata prob anggota `score_members` (model terbaik untuk AUROC eksternal).
Ketidakpastian u = std antar SEMUA anggota unik (disagreement lintas-keluarga SSL-supervised).
Jika hanya ada satu anggota total, u = 0.
"""
from __future__ import annotations

import base64
import io
import json
import os
from pathlib import Path

import matplotlib

matplotlib.use("Agg")
import matplotlib.cm as cm
import numpy as np
import timm
import torch
import torch.nn as nn
from fastapi import FastAPI, File, HTTPException, UploadFile
from fastapi.middleware.cors import CORSMiddleware
from PIL import Image
from torchvision import transforms as T

CONFIG_PATH = Path(os.environ.get("LOOPTB_CONFIG", "api/model_config.json"))
MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

app = FastAPI(title="loop-tb (SIGAP)", version="0.4.0")
app.add_middleware(
    CORSMiddleware, allow_origins=["*"], allow_credentials=False,
    allow_methods=["*"], allow_headers=["*"],
)

_state: dict = {}

def sigmoid(z: float) -> float:
    return float(1.0 / (1.0 + np.exp(-np.clip(z, -60, 60))))

class HFViT(nn.Module):
    """Pembungkus model HF (mis. RAD-DINO) → CLS token sebagai fitur."""
    def __init__(self, model_id: str):
        super().__init__()
        from transformers import AutoModel

        self.m = AutoModel.from_pretrained(model_id)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        out = self.m(pixel_values=x)
        return out.last_hidden_state[:, 0] if hasattr(out, "last_hidden_state") else out.pooler_output

def _load_member(spec: dict) -> dict:
    path = spec["path"]
    if path.endswith(".npz"):
        d = np.load(path, allow_pickle=False)
        enc_type, enc_id = str(d["encoder_type"][0]), str(d["encoder_id"][0])
        model: nn.Module = HFViT(enc_id) if enc_type == "hf" else timm.create_model(
            enc_id, pretrained=True, num_classes=0, dynamic_img_size=True
        )
    return {
        "kind": "probe",
        "model": model.eval(),
        "T": float(d["T"][0]),
        "size": int(d["size"][0]),
        "pixel_mean": tuple(float(v) for v in d["pixel_mean"]),
        "pixel_std": tuple(float(v) for v in d["pixel_std"]),
        "scaler_mean": d["scaler_mean"].astype(np.float32),
        "scaler_scale": d["scaler_scale"].astype(np.float32),
        "coef": d["coef"].astype(np.float32),
        "intercept": float(d["intercept"][0]),
        "path": path,
    }
```

```

    }
    ckpt = torch.load(path, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    return {
        "kind": "torch",
        "model": model.eval(),
        "T": float(spec.get("T", 1.0)),
        "size": int(ckpt["config"].get("img_size", 224)),
        "path": path,
    }

def _load() -> None:
    cfg = json.loads(CONFIG_PATH.read_text())
    score_specs = cfg.get("score_members") or cfg.get("members") or []
    unc_specs = cfg.get("uncertainty_members") or score_specs
    cache: dict[str, dict] = {}

    def get(spec: dict) -> dict:
        p = spec["path"]
        if p not in cache:
            cache[p] = _load_member(spec)
        return cache[p]

    _state.update(
        score_members=[get(s) for s in score_specs],
        unc_members=[get(s) for s in unc_specs],
        bands={"tau_high": 0.5, "tau_low": 0.5, **cfg.get("bands", {})},
    )
    print(f"[SIGAP] score={len(_state['score_members'])} unc={len(_state['unc_members'])} bands={_state['bands']}")

def _names(members: list[dict]) -> list[str]:
    return [f"{Path(m['path']).parent.name}/{Path(m['path']).name}" for m in members]

def preprocess(img: Image.Image, size: int, mean=MEAN, std=STD) -> torch.Tensor:
    tx = T.Compose([T.Resize(int(size * 1.14)), T.CenterCrop(size), T.ToTensor(), T.Normalize(mean, std)])
    return tx(img.convert("RGB")).unsqueeze(0)

def band_of(p: float, u: float, bands: dict) -> str:
    if u > float(bands.get("tau_uncertainty", 1.0)):
        return "ragu"
    if p >= bands["tau_high"]:
        return "rujuk_prioritas"
    if p < bands["tau_low"]:
        return "negatif_skrining"
    return "ragu"

def infer_prob(member: dict, img: Image.Image) -> float:
    if member["kind"] == "probe":
        x = preprocess(img, member["size"], member.get("pixel_mean", MEAN), member.get("pixel_std", STD))
        with torch.no_grad():
            feats = member["model"](x).float().numpy()[0]
            z = float(((feats - member["scaler_mean"]) / member["scaler_scale"]) @ member["coef"] + member["intercept"])
    else:
        x = preprocess(img, member["size"])
        with torch.no_grad():
            z = float(member["model"](x).squeeze(1).item())
    return sigmoid(z / member["T"])

def _cam_from_tokens(grads: torch.Tensor, acts: torch.Tensor, out_hw: tuple[int, int]) -> torch.Tensor | None:
    g, a = grads[1:], acts[1:]
    n = g.shape[0]
    side = int(round(n ** 0.5))
    if side * side != n:
        return None
    g = g.reshape(side, side, -1)
    a = a.reshape(side, side, -1)
    cam = (g * a).sum(-1).clamp(min=0)[None, None]
    return nn.functional.interpolate(cam, size=out_hw, mode="bilinear", align_corners=False)[0, 0]

def gradcam(member: dict, img: Image.Image) -> np.ndarray | None:
    model = member["model"]
    if member["kind"] == "torch":
        x = preprocess(img, member["size"])
        model.zero_grad(set_to_none=True)
        feats = model.forward_features(x)
        feats.retain_grad()
        model.forward_head(feats).squeeze(1).backward()
        grads, acts = feats.grad, feats
        if grads.dim() == 4:
            w = grads.mean(dim=(2, 3), keepdim=True)
            cam = (w * acts).sum(1, keepdim=True).clamp(min=0)
            cam = nn.functional.interpolate(cam, size=x.shape[-2:], mode="bilinear", align_corners=False)[0, 0]
        else:
            cam = _cam_from_tokens(grads[0], acts[0], tuple(x.shape[-2:]))
    
```

```

else:
    mean, std = member.get("pixel_mean", MEAN), member.get("pixel_std", STD)
    x = preprocess(img, member["size"], mean, std)
    model.zero_grad(set_to_none=True)
    if hasattr(model, "forward_features"):
        feats = model.forward_features(x)
        pooled = model.forward_head(feats)
    else:
        feats = model.m(pixel_values=x).last_hidden_state
        pooled = feats[:, 0]
    feats = feats if feats.dim() == 3 else feats.flatten(2).transpose(1, 2)
    feats.retain_grad()
    mt = torch.tensor(member["scaler_mean"])
    st = torch.tensor(member["scaler_scale"])
    cf = torch.tensor(member["coef"])
    z = ((pooled[0].float() - mt) / st) @ cf + member["intercept"]
    z.backward()
    cam = _cam_from_tokens(feats.grad[0], feats[0], tuple(x.shape[-2:]))
if cam is None:
    return None
cam = cam.detach()
return (cam / (cam.amax() + 1e-8)).numpy()

def overlay_png(img: Image.Image, cam: np.ndarray) -> str:
    img_np = np.asarray(img.convert("RGB").resize((cam.shape[1], cam.shape[0])), dtype=np.float32) / 255.0
    heat = cm.jet(cam)[..., :3].astype(np.float32)
    alpha = cam[..., None] * 0.45
    blend = img_np * (1 - alpha) + heat * alpha
    out = Image.fromarray((np.clip(blend, 0, 1) * 255).astype(np.uint8))
    buf = io.BytesIO()
    out.save(buf, format="PNG")
    return base64.b64encode(buf.getvalue()).decode()

@app.on_event("startup")
def startup() -> None:
    _load()

@app.get("/health")
def health() -> dict:
    return {
        "status": "ok" if _state.get("score_members") else "no-model",
        "score_members": _names(_state.get("score_members", [])),
        "uncertainty_members": _names(_state.get("unc_members", [])),
        "bands": _state.get("bands"),
    }

@app.post("/predict")
async def predict(file: UploadFile = File(...)) -> dict:
    if not _state.get("score_members"):
        raise HTTPException(503, "model belum siap")
    raw = await file.read()
    try:
        img = Image.open(io.BytesIO(raw))
        img.load()
    except Exception as e:
        raise HTTPException(400, f"bukan citra valid: {e}")

    score_probs = [infer_prob(m, img) for m in _state["score_members"]]
    extra = [m for m in _state["unc_members"] if m not in _state["score_members"]]
    all_probs = score_probs + [infer_prob(m, img) for m in extra]

    p = float(np.mean(score_probs))
    u = float(np.std(all_probs)) if len(all_probs) > 1 else 0.0
    band = band_of(p, u, _state["bands"])

    cam = gradcam(_state["score_members"][0], img)
    heatmap_b64 = overlay_png(img, cam) if cam is not None else None

    return {
        "p_tb": round(p, 4),
        "uncertainty": round(u, 4),
        "band": band,
        "bands": _state["bands"],
        "members": _names(_state["score_members"]),
        "uncertainty_members": _names(_state["unc_members"]),
        "disclaimer": "Alat triase skrining – bukan diagnosis. Wajib konfirmasi (GeneXpert) oleh tenaga kesehatan.",
        "heatmap_png_b64": heatmap_b64,
    }

```

ml/kaggle/out_ssl/loop-tb/ml/data/prepare_data.py

```
#!/usr/bin/env python3
"""Build a provenance-aware manifest for TB chest X-ray training.

Sumber & aturan label (lihat docs/02-arsitektur.md):
- TBX11K (Kaggle: vbookshelf/tbx11k-simplified): folder per kelas.
  positif = {tb}; negatif = {healthy, sick}; latent_tb DIKELUARKAN (ambigu).
- Rahman composite (Kaggle: tawsifurrahman/tuberculosis-tb-chest-xray-dataset):
  folder Normal/ -> 0, Tuberculosis/ -> 1.
- NLM Shenzhen/Montgomery: konvensi nama file resmi `*_0` = normal, `*_1` = TB.

Dedup (WAJIB – mitigasi class-conditional acquisition confounding):
1. pHash 64-bit; Hamming <= --dup-threshold dianggap duplikat (union-find transitif).
2. Integritas external: citra train yang sekelompok dengan citra external DIBUANG dari train.
3. Duplikat internal train dibuang (keep-first, urutan deterministik).
4. Split val dilakukan per-klaster (tidak ada kebocoran near-dup train->val).

Output (ml/data/artifacts/):
- manifest.json, dedup_report.json, train.csv, val.csv, external.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import re
from collections import Counter
from pathlib import Path

import numpy as np
from PIL import Image
from tqdm import tqdm

try:
    import imagehash
except ImportError: # pragma: no cover
    raise SystemExit("Butuh imagehash: pip install imagehash")

IMG_EXTS = {".png", ".jpg", ".jpeg", ".bmp", ".tif", ".tiff"}
POS_TOKENS = {"tb", "tuberculosis", "tuberculose"}
NEG_TOKENS = {"healthy", "normal", "sick"}
EXCLUDE_TOKENS = {"latent", "latent_tb", "latenttb"}

def _rel(p: Path, root: Path) -> str:
    """Path relatif terhadap root repo (portable lokal <-> Colab); fallback absolut."""
    try:
        return str(p.relative_to(root))
    except ValueError:
        return str(p)

def _tokens(path: Path) -> set[str]:
    parts = [p.lower() for p in path.parts]
    parts += [path.stem.lower()]
    return set(parts)

def load_tbx11k_map(raw: Path) -> dict[str, int | None]:
    """Label otoritatif dari data.csv TBX11K: 1=active_tb, 0=no_tb, None=latent (dikeluarkan)."""
    m: dict[str, int | None] = {}
    for p in raw.rglob("data.csv"):
        if "tbx11k" not in str(p).lower():
            continue
        for r in csv.DictReader(p.open()):
            f = (r.get("fname") or "").strip()
            if not f:
                continue
            t = (r.get("tb_type") or "").lower()
            tgt = (r.get("target") or "").lower()
            if "latent" in t:
                m.setdefault(f, None)
            elif tgt == "tb":
                m[f] = 1
            else:
                m.setdefault(f, 0)
    return m

def classify(path: Path, raw: Path, tbx_map: dict[str, int | None]) -> tuple[int, str] | None:
    """Return (label, source) atau None kalau tidak bisa diklasifikasi."""
    rel = path.relative_to(raw)
    low = str(rel).lower()

    # --- NLM Shenzhen / Montgomery: konvensi suffix nama file ---
    m = re.search(r"(chncxr|mcucxr)_d+_(\d+)", path.stem.lower())
    if m:
        label = int(m.group(2))
        source = "shenzhen" if m.group(1) == "chncxr" else "montgomery"
```

```

        return label, source

toks = _tokens(rel)
if toks & EXCLUDE_TOKENS or any(t.startswith("latent") for t in toks):
    return None

if "tbx11k" in low:
    if tbx_map:
        lab = tbx_map.get(path.name, None)
        return None if lab is None else (lab, "tbx11k")
    # fallback bila data.csv tidak ada: prefix nama file
    stem = path.stem.lower()
    if stem.startswith("t"):
        return 1, "tbx11k"
    if stem.startswith(("h", "s")):
        return 0, "tbx11k"
    return None

if toks & POS_TOKENS:
    return 1, "rahman"
if toks & NEG_TOKENS:
    return 0, "rahman"
return None

class DSU:
    def __init__(self, n: int):
        self.parent = list(range(n))

    def find(self, x: int) -> int:
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def union(self, a: int, b: int) -> None:
        ra, rb = self.find(a), self.find(b)
        if ra != rb:
            self.parent[rb] = ra

_POP = np.array([bin(i).count("1") for i in range(256)], dtype=np.uint8)

def hamming_pairs(hashes: np.ndarray, threshold: int, chunk: int = 512) -> list[tuple[int, int]]:
    """Semua pasangan (i<j) dengan Hamming distance <= threshold (chunked, hemat memori)."""
    n = len(hashes)
    h8 = hashes.view(np.uint8).reshape(n, 8)
    pairs: list[tuple[int, int]] = []
    for start in range(0, n, chunk):
        stop = min(start + chunk, n)
        xor = np.bitwise_xor(h8[start:stop, None, :], h8[None, :, :])
        dist = _POP[xor].sum(axis=-1) # (chunk, n)
        rows, cols = np.where(dist <= threshold)
        for r, c in zip(rows, cols):
            i, j = start + int(r), int(c)
            if i < j:
                pairs.append((i, j))
    return pairs

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--raw", type=Path, default=Path("data/raw"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts"))
    ap.add_argument("--dup-threshold", type=int, default=3, help="Hamming pHash <= ini dianggap duplikat")
    ap.add_argument("--val-frac", type=float, default=0.15)
    ap.add_argument("--seed", type=int, default=42)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    raw = args.raw.resolve()
    root = raw.parent.parent # asumsi struktur <root>/data/raw

    # ----- 1) discovery -----
    labeled: list[dict] = []
    unlabeled = Counter()
    tbx_map = load_tbx11k_map(raw)
    print(f"[discovery] label TBX11K dari data.csv: {len(tbx_map)} entri")
    for p in sorted(raw.rglob("*")):
        if p.suffix.lower() not in IMG_EXTS:
            continue
        cls = classify(p, raw, tbx_map)
        if cls is None:
            unlabeled[str(p.relative_to(raw).parent)] += 1
            continue
        label, source = cls
        labeled.append({"path": _rel(p, root), "label": label, "source": source})

    print(f"[discovery] berlabel={len(labeled)} tak-terklasifikasi={sum(unlabeled.values())}")
    for k, v in unlabeled.most_common(10):
        print(f" skip {v:6d} {k}")

```

```

if not labeled:
    raise SystemExit("Tidak ada citra berlabel – cek struktur data/raw")

# ----- 2) pHash + kluster duplikat -----
print("[dedup] menghitung pHash...")
ok: list[dict] = []
for item in tqdm(labeled, unit="img"):
    try:
        with Image.open(item["path"]) as im:
            h = imagehash.phash(im.convert("L"), hash_size=8)
            item["phash"] = int(str(h), 16)
            ok.append(item)
    except Exception as e: # gambar korup
        print(f"warn: gagal baca {item['path']}: {e}")
labeled = ok

hashes = np.array([it["phash"] for it in labeled], dtype=np.uint64)
pairs = hamming_pairs(hashes, args.dup_threshold)
dsu = DSU(len(labeled))
for i, j in pairs:
    dsu.union(i, j)
for idx, it in enumerate(labeled):
    it["cluster"] = dsu.find(idx)

by_cluster: dict[int, list[dict]] = {}
for it in labeled:
    by_cluster.setdefault(it["cluster"], []).append(it)

# ----- 3) lock external, buang kebocoran dari train -----
ext = [it for it in labeled if it["source"] in ("shenzhen", "montgomery")]
ext_clusters = {it["cluster"] for it in ext}
train_pool, dropped_cross = [], []
for it in labeled:
    if it["source"] in ("shenzhen", "montgomery"):
        continue
    (dropped_cross if it["cluster"] in ext_clusters else train_pool).append(it)

# duplikat internal train: keep-first per kluster (deterministik via path)
keep, dropped_internal = [], []
seen_clusters: set[int] = set()
for it in sorted(train_pool, key=lambda x: x["path"]):
    if it["cluster"] in seen_clusters:
        dropped_internal.append(it)
    else:
        seen_clusters.add(it["cluster"])
        keep.append(it)
train_pool = keep

# ----- 4) split val per-kluster -----
rng = np.random.default_rng(args.seed)
sizes = Counter(it["cluster"] for it in train_pool)
pool_clusters = sorted(sizes)
rng.shuffle(pool_clusters)
target_val = int(len(train_pool) * args.val_frac)
val_clusters: set[int] = set()
acc = 0
for c in pool_clusters:
    if acc >= target_val:
        break
    val_clusters.add(c)
    acc += sizes[c]
train = [it for it in train_pool if it["cluster"] not in val_clusters]
val = [it for it in train_pool if it["cluster"] in val_clusters]

def dump(items: list[dict]) -> list[dict]:
    return [
        {k: it[k] for k in ("path", "label", "source", "cluster", "phash")}
        for it in sorted(items, key=lambda x: x["path"])
    ]

manifest = {
    "meta": {
        "root": str(root),
        "raw": _rel(raw, root),
        "dup_threshold": args.dup_threshold,
        "val_frac": args.val_frac,
        "seed": args.seed,
        "counts": {
            "train": len(train), "val": len(val), "external": len(ext),
            "dropped_cross_corpus": len(dropped_cross),
            "dropped_internal_dup": len(dropped_internal),
            "unlabeled": sum(unlabeled.values()),
        },
    },
    "train": dump(train),
    "val": dump(val),
    "external": dump(ext),
}
(args.out / "manifest.json").write_text(json.dumps(manifest, indent=1))
(args.out / "dedup_report.json").write_text(json.dumps({
    "pairs_within_threshold": len(pairs),
    "clusters": len(by_cluster),
}))

```

```

        "dropped_cross_corpus": dump(dropped_cross),
        "dropped_internal_dup": dump(dropped_internal),
        "unlabeled_by_dir": dict(unlabeled),
    }, indent=1))

for split in ("train", "val", "external"):
    with (args.out / f"{split}.csv").open("w") as f:
        f.write("path,label,source,cluster\n")
        for it in manifest[split]:
            f.write(f"{it['path']},{it['label']},{it['source']},{it['cluster']}\n")

def comp(items):
    c = Counter((it["source"], it["label"]) for it in items)
    return " ".join(f"{s}/{ 'TB' if l else 'NEG' }={n}" for (s, l), n in sorted(c.items()))

print("\n=== RINGKASAN ===")
print(f"train      n={len(train):6d}  {comp(train)}")
print(f"val        n={len(val):6d}    {comp(val)}")
print(f"external  n={len(ext):6d}    {comp(ext)}")
print(f"dibuang  cross-corpus={len(dropped_cross)}  internal-dup={len(dropped_internal)}")
print(f"manifest -> {args.out/'manifest.json'}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl/loop-tb/ml/data/prepare_meta.py

```
#!/usr/bin/env python3
"""Ekstrak metadata klinis NLM (Montgomery + Shenzhen) → CSV untuk analisis subgroup.

Output: ml/data/artifacts/meta_nlm.csv
Kolom: filename,sex,age,reading,scar_flag
- Montgomery: "Patient's Sex: F / Patient's Age: 027Y / <reading>"
- Shenzhen: "male 45yrs / <reading>" (kadang terkontaminasi antar-file)
- scar_flag: menyebut scar/inactive/fibrosis → untuk pembahasan mode gagal
            "healed scar false positive" (selaras audit Bilal 2026).
"""
from __future__ import annotations

import argparse
import csv
import re
from pathlib import Path

SEX_M = re.compile(r"sex[:\s]+(m|male)\b", re.I)
SEX_F = re.compile(r"sex[:\s]+(f|female)\b", re.I)
SEX_FMT2 = re.compile(r"\b(male|female)\b", re.I)
AGE_MONT = re.compile(r"age[:\s]+(\d{1,3})\s*Y?\b", re.I)
AGE_FMT2 = re.compile(r"\b(\d{1,3})\s*yrs\b", re.I)
SCAR = re.compile(r"scar|inactive|fibro", re.I)

def parse_montgomery(text: str) -> dict | None:
    age = AGE_MONT.search(text)
    sex = "F" if SEX_F.search(text) else ("M" if SEX_M.search(text) else "")
    if not age:
        return None
    # reading = baris setelah Age
    lines = [ln.strip() for ln in text.splitlines() if ln.strip()]
    reading = lines[2] if len(lines) > 2 else ""
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": int(bool(SCAR.search(reading)))}

def parse_shenzhen(text: str) -> dict | None:
    age = AGE_FMT2.search(text)
    sex_m = SEX_FMT2.findall(text)
    sex = sex_m[0][0].upper() if sex_m else ""
    if not age:
        return None
    # reading = token huruf setelah "yrs"
    tail = text[age.end():]
    reading = ""
    m = re.search(r"[A-Za-z]{2,}", tail)
    if m:
        reading = m.group(0)
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": 0}

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--nlm", type=Path, default=Path("data/raw/nlm"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts/meta_nlm.csv"))
    args = ap.parse_args()
    args.out.parent.mkdir(parents=True, exist_ok=True)

    rows, n_fail = [], 0
    for corpus, parser in (("montgomery", parse_montgomery), ("shenzhen", parse_shenzhen)):
        d = args.nlm / corpus / "ClinicalReadings"
        for p in sorted(d.glob("*.txt")):
            r = parser(p.read_text(errors="ignore"))
            if r is None:
                n_fail += 1
                continue
            r["filename"] = p.stem + ".png"
            r["corpus"] = corpus
            rows.append(r)

    with args.out.open("w", newline="") as f:
        w = csv.DictWriter(f, fieldnames=["filename", "corpus", "sex", "age", "reading", "scar_flag"])
        w.writeheader()
        w.writerows(rows)

    from collections import Counter
    print(f"meta -> {args.out} n={len(rows)} gagal-parse={n_fail}")
    print("per korpus:", Counter(r["corpus"] for r in rows))
    print("sex:", Counter(r["sex"] for r in rows))
    print("scar_flag:", Counter(r["scar_flag"] for r in rows))

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl/loop-tb/ml/evaluate.py

```
#!/usr/bin/env python3
"""Evaluasi rigor: kalibrasi, CI bootstrap, transfer, band triase, subgroup.

Contoh (ensemble + temperature scaling):
python ml/evaluate.py --runs ml/runs/efficientnet_b0 ml/runs/convnext_tiny \
    --manifest ml/data/artifacts/manifest.json --split external --temp-scale --out ml/runs/eval_external
"""
from __future__ import annotations

import argparse
import csv
import json
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import brier_score_loss, roc_auc_score, roc_curve
from torch.utils.data import DataLoader
from tqdm import tqdm

import sys
sys.path.insert(0, str(Path(__file__).resolve().parent))
from train import CXRManifest # noqa: E402

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

def sigmoid(x: np.ndarray) -> np.ndarray:
    return 1 / (1 + np.exp(-x))

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 -> fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

def threshold_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    fpr, tpr, thr = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(thr[ok][np.argmax(tpr[ok])]) if ok.any() else 0.5

def ece(y: np.ndarray, p: np.ndarray, bins: int = 15) -> float:
    edges = np.linspace(0, 1, bins + 1)
    total = 0.0
    for i in range(bins):
        m = (p >= edges[i]) & ((p < edges[i + 1]) if i < bins - 1 else (p <= edges[i + 1]))
        if m.sum():
            total += m.mean() * abs(p[m].mean() - y[m].mean())
    return float(total)

def bootstrap_auroc(y: np.ndarray, p: np.ndarray, n: int = 1000, seed: int = 42) -> tuple[float, float]:
    rng = np.random.default_rng(seed)
    vals = []
    for _ in range(n):
        idx = rng.integers(0, len(y), len(y))
        if len(np.unique(y[idx])) > 1:
            vals.append(roc_auc_score(y[idx], p[idx]))
    return float(np.percentile(vals, 2.5)), float(np.percentile(vals, 97.5))

@torch.no_grad()
def infer(ckpt_path: Path, items: list[dict], img_size: int, device: torch.device, workers: int) -> np.ndarray:
    ckpt = torch.load(ckpt_path, map_location="cpu", weights_only=False)
    arch = ckpt["arch"]
    model = timm.create_model(arch, pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.to(device).eval()
    dl = DataLoader(CXRManifest(items, train=False, img_size=img_size), batch_size=64,
                    shuffle=False, num_workers=workers)

    out = []
    for x, _, _ in tqdm(dl, desc=f"infer {ckpt_path.parent.name}", leave=False):
        x = x.to(device)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            out.append(model(x).squeeze(1).float().cpu().numpy())
    return np.concatenate(out)
```

```

def fit_temperature(val_logits: list[np.ndarray], val_y: np.ndarray) -> float:
    """Cari T>0 yang meminimalkan NLL pada val (grid log-space lebar untuk logit besar)."""
    logits = torch.tensor(np.mean(val_logits, axis=0), dtype=torch.float32)
    y = torch.tensor(val_y, dtype=torch.float32)
    best_t, best_nll = 1.0, float("inf")
    for t in np.logspace(-1, 3.5, 500):
        nll = nn.functional.binary_cross_entropy_with_logits(logits / t, y).item()
        if nll < best_nll:
            best_t, best_nll = float(t), nll
    return best_t

def reliability_plot(y: np.ndarray, p: np.ndarray, path: Path, title: str) -> None:
    edges = np.linspace(0, 1, 11)
    mids, accs = [], []
    for i in range(10):
        m = (p >= edges[i]) & (p < edges[i + 1] if i < 9 else p <= edges[i + 1])
        if m.sum():
            mids.append(p[m].mean())
            accs.append(y[m].mean())
    plt.figure(figsize=(4.2, 4.2))
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.plot(mids, accs, "o-", color="#d62728", lw=2)
    plt.xlabel("Confidence"), plt.ylabel("Empirical TB rate"), plt.title(title)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def roc_plot(curves: dict[str, tuple[np.ndarray, np.ndarray]], path: Path) -> None:
    plt.figure(figsize=(5, 5))
    for name, (y, p) in curves.items():
        fpr, tpr, _ = roc_curve(y, p)
        plt.plot(fpr, tpr, lw=2, label=f"{name} (AUROC {roc_auc_score(y, p):.3f})")
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.xlabel("FPR"), plt.ylabel("TPR (sensitivitas)"), plt.legend(fontsize=8)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--runs", type=Path, nargs="+", required=True)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--split", default="external", choices=["val", "external"])
    ap.add_argument("--temp-scale", action="store_true")
    ap.add_argument("--meta", type=Path, default=None, help="CSV filename,sex,age untuk subgroup (opsional)")
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--workers", type=int, default=8)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    manifest = json.loads(args.manifest.read_text())
    items = manifest[args.split]
    y = np.array([it["label"] for it in items])

    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    per_run_target, per_run_val, tscales = {}, {}, {}
    for run in args.runs:
        ckpt = torch.load(run / "best.pt", map_location="cpu", weights_only=False)
        img_size = ckpt["config"]["img_size"]
        per_run_target[run.name] = infer(run / "best.pt", items, img_size, device, args.workers)
        val_items = manifest["val"]
        per_run_val[run.name] = infer(run / "best.pt", val_items, img_size, device, args.workers)
        if args.temp_scale:
            tscales[run.name] = fit_temperature([per_run_val[run.name]], np.array([it["label"] for it in val_items]))

    per_run_target = {k: sigmoid(v) for k, v in per_run_target.items()}
    per_run_val = {k: sigmoid(v) for k, v in per_run_val.items()}
    val_y = np.array([it["label"] for it in manifest["val"]])

    # kalibrasi per-run pada val, lalu terapkan T ke target (bukan val-fit pada target!)
    if args.temp_scale:
        for k in per_run_target:
            per_run_val[k] = sigmoid(np.log(np.clip(per_run_val[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_val[k], 1e-6, 1 - 1e-6)))
            / tscales[k])
            lg = np.log(np.clip(per_run_target[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_target[k], 1e-6, 1 - 1e-6)))
            per_run_target[k] = sigmoid(lg / tscales[k])

    # ensemble = rata-rata; ketidakpastian = std antar anggota
    stack_t = np.stack(list(per_run_target.values()))
    p_ens = stack_t.mean(axis=0)
    u_ens = stack_t.std(axis=0)
    stack_v = np.stack(list(per_run_val.values()))
    p_ens_v = stack_v.mean(axis=0)

    tau_high = threshold_at_spec(val_y, p_ens_v, 0.70) # spec>=0.70 (TPP WHO)
    tau_low = threshold_at_spec(val_y, p_ens_v, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    band = np.where(p_ens >= tau_high, "rujuk_prioritas", np.where(p_ens < tau_low, "negatif_skrining", "ragu"))
    band_report = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b

```

```

        band_report[b] = {"n": int(m.sum()), "frac": float(m.mean()),
                          "tb_rate": float(y[m].mean()) if m.sum() else None}

lo, hi = bootstrap_aucroc(y, p_ens)
metrics = {
    "split": args.split, "n": int(len(y)), "pos": int(y.sum()),
    "aucroc": float(roc_auc_score(y, p_ens)), "aucroc_ci95": [lo, hi],
    "sens_at_spec_90": sens_at_spec(y, p_ens, 0.90),
    "sens_at_spec_70": sens_at_spec(y, p_ens, 0.70),
    "ece": ece(y, p_ens), "brier": float(brier_score_loss(y, p_ens)),
    "temp_scales": tscales, "tau_low": tau_low, "tau_high": tau_high,
    "bands": band_report,
    "deferral": {"rate": float((band == "ragu").mean()),
                 "tb_rate_on_deferred": float(y[band == "ragu"].mean()) if (band == "ragu").any() else None},
    "per_run_aucroc": {k: float(roc_auc_score(y, v)) for k, v in per_run_target.items()},
}

if args.meta and args.meta.exists():
    meta = {r["filename"]: r for r in csv.DictReader(args.meta.open())}
    for key in ("sex", "age"):
        groups: dict[str, list[int]] = {}
        for i, it in enumerate(items):
            row = meta.get(Path(it["path"]).name, {})
            g = row.get(key, "NA") or "NA"
            if key == "age":
                try:
                    a = float(g)
                    g = "0-30" if a < 30 else "30-50" if a < 50 else "50+"
                except ValueError:
                    g = "NA"
            groups.setdefault(g, []).append(i)
        metrics[f"aucroc_by_{key}"] = {
            g: (float(roc_auc_score(y[idx], p_ens[idx])) if len(np.unique(y[idx])) > 1 else None)
              for g, idx in groups.items() if g != "NA" and len(idx) >= 20
        }

(args.out / "metrics.json").write_text(json.dumps(metrics, indent=1))
# dump prediksi per-citra (untuk analisis lanjutan & stacking)
import csv as _csv
with (args.out / "preds.csv").open("w", newline="") as f:
    w = _csv.writer(f)
    w.writerow(["path", "label", "p_ens", "u_ens"])
    for it, p, u in zip(items, p_ens, u_ens):
        w.writerow([it["path"], it["label"], float(p), float(u)])
reliability_plot(y, p_ens, args.out / "reliability.png", f"Kalibrasi - {args.split}")
curves = {f"ens ({args.split})": (y, p_ens)}
if len(per_run_target) > 1:
    curves.update({k: (y, v) for k, v in per_run_target.items()})
roc_plot(curves, args.out / "roc.png")

print("\n=== METRICS ===")
for k, v in metrics.items():
    if isinstance(v, (int, float, str)):
        print(f"{k:18s} {v}")
print(f"bands          {band_report}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl/loop-tb/ml/export_onnx.py

```
#!/usr/bin/env python3
"""Export checkpoint ke ONNX (opsi 17) + verifikasi onnxruntime.

Contoh:
python ml/export_onnx.py --ckpt ml/runs/efficientnet_b0/best.pt --out artifacts/tb_efficientnet_b0.onnx
"""
from __future__ import annotations

import argparse
from pathlib import Path

import numpy as np
import torch

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm")

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__)
    ap.add_argument("--ckpt", type=Path, required=True)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--opset", type=int, default=17)
    args = ap.parse_args()

    ckpt = torch.load(args.ckpt, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()

    img_size = ckpt["config"].get("img_size", 224)
    dummy = torch.randn(1, 3, img_size, img_size)
    args.out.parent.mkdir(parents=True, exist_ok=True)
    torch.onnx.export(
        model, dummy, args.out,
        input_names=["input"], output_names=["logit"],
        dynamic_axes={"input": {0: "batch"}, "logit": {0: "batch"}},
        opset_version=args.opset, dynamo=False, # legacy: satu file mandiri, opset stabil
    )

    import onnxruntime as ort
    sess = ort.InferenceSession(str(args.out), providers=["CPUExecutionProvider"])
    ref = model(dummy).detach().numpy()
    got = sess.run(None, {"input": dummy.numpy()})[0]
    # Verifikasi di ruang probabilitas (deployment-relevant); logit bisa sangat besar
    # pada model overkonfiden sehingga toleransi absolut di ruang logit tidak adil.
    sig = lambda z: 1 / (1 + np.exp(-z)) # noqa: E731
    prob_diff = float(np.abs(sig(ref) - sig(got)).max())
    size_mb = args.out.stat().st_size / 1e6
    print(f"ONNX ok: {args.out} ({size_mb:.1f} MB) max|Δprob|={prob_diff:.2e} opset={args.opset}")
    if prob_diff > 1e-3:
        print(f"WARN: perbedaan probabilitas {prob_diff:.4f} > 1e-3 – periksa sebelum produksi")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl/loop-tb/ml/kaggle/hello/hello.py

```
"""Kernel uji: verifikasi GPU, internet, versi lib, dan struktur dataset ter-mount."""
import os
import sys

lines: list[str] = []

def log(*a) -> None:
    print(*a, flush=True)
    lines.append(" ".join(str(x) for x in a))

log("python:", sys.version.split()[0])

try:
    import torch

    log("torch:", torch.__version__, "| cuda:", torch.cuda.is_available())
    if torch.cuda.is_available():
        log("gpu:", torch.cuda.get_device_name(0))
except Exception as e: # noqa: BLE001
    log("torch error:", e)

for mod in ("timm", "sklearn", "onnx", "onnxruntime"):
    try:
        m = __import__(mod)
        log(f"{mod}: {getattr(m, '__version__', '?')}")
    except Exception as e: # noqa: BLE001
        log(f"{mod}: MISSING ({e})")

log("--- input mounts ---")
for root, dirs, _ in os.walk("/kaggle/input"):
    log(root, "dirs:", dirs[:6])
    if root.count("/") > 3:
        dirs[:] = []

log("HELLO DONE")
with open("/kaggle/working/hello_result.txt", "w") as f:
    f.write("\n".join(lines) + "\n")
```

ml/kaggle/out_ssl/loop-tb/ml/kaggle/make_nlm_dataset.py

```
#!/usr/bin/env python3
"""Siapkan dataset Kaggle privat: NLM CXR di-resize (max dim 512) + ClinicalReadings.

Dipisah dari data asli agar unggahan ringan (~150 MB) dan resolusi tetap cukup
untuk encoder SSL (DINOv2/RAD-DINO resize ke 518).
"""
from __future__ import annotations

import shutil
from pathlib import Path

from PIL import Image

SRC = Path("data/raw/nlm")
DST = Path("ml/kaggle/nlm_dataset/nlm")

def main() -> None:
    n = 0
    for corpus in ("shenzhen", "montgomery"):
        out_dir = DST / corpus / "CXR_png"
        out_dir.mkdir(parents=True, exist_ok=True)
        for p in sorted((SRC / corpus / "CXR_png").glob("*.png")):
            with Image.open(p) as im:
                im = im.convert("L")
                w, h = im.size
                scale = 512 / max(w, h)
                if scale < 1:
                    im = im.resize((max(1, round(w * scale)), max(1, round(h * scale))), Image.LANCZOS)
                im.save(out_dir / (p.stem + ".jpg"), quality=92)
            n += 1
        readings = SRC / corpus / "ClinicalReadings"
        if readings.exists():
            out_r = DST / corpus / "ClinicalReadings"
            out_r.mkdir(parents=True, exist_ok=True)
            for f in readings.glob("*.txt"):
                shutil.copy(f, out_r / f.name)
        print(f"selesai: {n} citra -> {DST}")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl/loop-tb/ml/kaggle/ssl/ssl_extract.py

```
"""Ekstraksi embedding SSL untuk studi representasi (ADR-007).

Encoder:
- dinov2 : timm vit_base_patch14_dinov2.lvd142m (SSL umum, 518 px)
- rad_dino: hf-hub:microsoft/rad-dino (DINOv2 pada X-ray data, 518 px)

Output di /kaggle/working/loop-tb/ssl_embeddings/:
meta_{train,val,external}.json + emb_{encoder}_{split}.npy
Dipakai oleh ml/ssl_probe.py secara lokal (sklearn, tanpa GPU).
"""
from __future__ import annotations

import json
import os
import subprocess
import sys

import numpy as np

WORK = "/kaggle/working"
REPO = f"{WORK}/loop-tb"

def sh(cmd: list[str]) -> None:
    print("+", " ".join(map(str, cmd)), flush=True)
    subprocess.run(cmd, check=True)

if not os.path.exists(REPO):
    sh(["git", "clone", "--depth", "1", "https://github.com/Alvnc/loop-tb.git", REPO])
os.chdir(REPO)
sh([sys.executable, "-m", "pip", "install", "-q", "timm", "huggingface_hub", "transformers"])

# --- mount datasets (self-healing) ---
os.makedirs("data/raw", exist_ok=True)

def pick(dst: str, cands: list[str], markers: tuple[str, ...]) -> None:
    if os.path.exists(dst):
        return
    for c in cands:
        if any(os.path.isdir(os.path.join(c, m)) for m in markers):
            os.symlink(c, dst)
            print("symlink", dst, "->", c, flush=True)
            return
    for c in cands:
        if os.path.exists(c):
            os.symlink(c, dst)
            print("symlink (fallback)", dst, "->", c, flush=True)
            return
    print("WARNING: tidak ada mount untuk", dst, flush=True)

pick("data/raw/tbx11k-simplified",
     ["/kaggle/input/datasets/vbookshelf/tbx11k-simplified/tbx11k-simplified",
      "/kaggle/input/datasets/vbookshelf/tbx11k-simplified", "/kaggle/input/tbx11k-simplified"],
     ("images",))
pick("data/raw/TB_Chest_Radiography_Database",
     ["/kaggle/input/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
      "/kaggle/input/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database"],
     ("Normal", "Tuberculosis"))
pick("data/raw/nlm",
     ["/kaggle/input/datasets/alvinreba/nlm-tb-cxr-resized/nlm",
      "/kaggle/input/datasets/alvinreba/nlm-tb-cxr-resized",
      "/kaggle/input/nlm-tb-cxr-resized/nlm",
      "/kaggle/input/nlm-tb-cxr-resized"],
     ("shenzhen", "montgomery"))

# --- siapkan daftar item (external: .png -> .jpg hasil resize) ---
manifest = json.load(open("ml/data/artifacts/manifest.json"))
train_items = manifest["train"]
val_items = manifest["val"]
ext_items = []
for it in manifest["external"]:
    p = it["path"]
    jp = p[:-4] + ".jpg" if p.lower().endswith(".png") else p
    ext_items.append({**it, "path": jp})

for name, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
    missing = [it["path"] for it in items if not os.path.exists(it["path"])]
    print(f"{name}: n={len(items)} missing={len(missing)}", missing[:2], flush=True)
    assert not missing, f"data {name} tidak lengkap"

out_dir = "ssl_embeddings"
os.makedirs(out_dir, exist_ok=True)
for split, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
    with open(f"{out_dir}/meta_{split}.json", "w") as f:
        json.dump([{"path": it["path"], "label": it["label"], "source": it["source"]} for it in items], f)
```

```

# --- ekstraksi ---
import torch # noqa: E402
from PIL import Image # noqa: E402
from torch.utils.data import DataLoader, Dataset # noqa: E402
from torchvision import transforms as T # noqa: E402

MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

class Items(Dataset):
    def __init__(self, items: list[dict], size: int, mean: tuple = MEAN, std: tuple = STD):
        self.items = items
        self.tx = T.Compose([T.Resize(size), T.CenterCrop(size), T.ToTensor(), T.Normalize(mean, std)])

    def __len__(self) -> int:
        return len(self.items)

    def __getitem__(self, i: int):
        with Image.open(self.items[i]["path"]) as im:
            x = self.tx(im.convert("RGB"))
        return x, i

@torch.no_grad()
def extract(model, items: list[dict], size: int, device: torch.device, tag: str,
            mean: tuple = MEAN, std: tuple = STD) -> np.ndarray:
    dl = DataLoader(Items(items, size, mean, std), batch_size=32, shuffle=False, num_workers=4, pin_memory=True)
    out = np.zeros((len(items), 0), dtype=np.float32)
    embs = []
    for bi, (x, _) in enumerate(dl):
        x = x.to(device, non_blocking=True)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            feats = model(x)
            feats = feats.float().cpu().numpy()
            embs.append(feats)
        if bi % 20 == 0:
            print(f" {tag} batch {bi}/{len(dl)}", flush=True)
    out = np.concatenate(embs, axis=0)
    return out

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("device:", device, flush=True)

ENCODERS = [
    ("dinov2", "vit_base_patch14_dinov2.lvd142m", 518),
    ("rad_dino", "hf-hub:microsoft/rad-dino", 518),
]

import timm # noqa: E402

for enc_name, model_id, size in ENCODERS:
    if all(os.path.exists(f"{out_dir}/emb_{enc_name}_{s}.npz") for s in ("train", "val", "external")):
        print(f"skip {enc_name} (sudah ada)", flush=True)
        continue
    try:
        print(f"=== encoder {enc_name} ({model_id}) ===", flush=True)
        mean, std = MEAN, STD
        if enc_name == "rad_dino":
            from transformers import AutoImageProcessor, AutoModel

            hf_id = model_id.replace("hf-hub:", "")

            class HFWrap(torch.nn.Module):
                def __init__(self, m):
                    super().__init__()
                    self.m = m

                def forward(self, x):
                    out = self.m(pixel_values=x)
                    return out.last_hidden_state[:, 0] if hasattr(out, "last_hidden_state") else out.pooler_output

            proc = AutoImageProcessor.from_pretrained(hf_id)
            print("rad_dino processor:", proc.image_mean, proc.image_std, proc.size, flush=True)
            if proc.image_mean:
                mean, std = tuple(proc.image_mean), tuple(proc.image_std)
            model = HFWrap(AutoModel.from_pretrained(hf_id))
        else:
            model = timm.create_model(model_id, pretrained=True, num_classes=0, dynamic_img_size=True)
        model.to(device).eval()
        for split, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
            emb = extract(model, items, size, device, f"{enc_name}_{split}", mean, std)
            np.save(f"{out_dir}/emb_{enc_name}_{split}.npz", emb)
            print(f"{enc_name} {split}: {emb.shape}", flush=True)
        with open(f"{out_dir}/encmeta_{enc_name}.json", "w") as f:
            json.dump({
                "type": "hf" if enc_name == "rad_dino" else "timm",
                "id": model_id.replace("hf-hub:", ""),
                "size": size,
                "mean": list(mean),
                "std": list(std),
            }, f)
    except Exception as e:
        print(f"Error processing {enc_name} {model_id}: {e}")

```

```
    }, f)
    del model
    torch.cuda.empty_cache()
except Exception as e: # noqa: BLE001
    print(f"ERROR encoder {enc_name}: {e}", flush=True)

print("SSL EXTRACT DONE", flush=True)
```

ml/kaggle/out_ssl/loop-tb/ml/kaggle/train/train_kernel.py

```
"""Kernel training loop-tb: clone repo → symlink dataset → train 2 arsitektur + baseline → export ONNX.

Output di /kaggle/working/loop-tb/{runs,artifacts}; diunduh via `kaggle kernels output`.
Manifest yang dipakai = split v1 frozen dari repo (deterministik; NLM external tidak dibutuhkan di sini).
"""
import json
import os
import subprocess
import sys

WORK = "/kaggle/working"
REPO = f"{WORK}/loop-tb"

def sh(cmd: list[str], **kw) -> None:
    print("+", " ".join(map(str, cmd)), flush=True)
    subprocess.run(cmd, check=True, **kw)

# 1) repo + deps
if not os.path.exists(REPO):
    sh(["git", "clone", "--depth", "1", "https://github.com/Alvnc/loop-tb.git", REPO])
os.chdir(REPO)
sh([sys.executable, "-m", "pip", "install", "-q", "timm", "onnxruntime"])

# 2) symlink dataset ke struktur yang diharapkan manifest (tahan variasi struktur mount)
os.makedirs("data/raw", exist_ok=True)

def pick(dst: str, cands: list[str], markers: tuple[str, ...]) -> None:
    """Symlink kandidat mount pertama yang punya 'marker' (images/Normal/Tuberculosis)."""
    if os.path.exists(dst):
        return
    for c in cands:
        if any(os.path.isdir(os.path.join(c, m)) for m in markers):
            os.symlink(c, dst)
            print("symlink", dst, "->", c, flush=True)
            return
    for c in cands:
        if os.path.exists(c):
            os.symlink(c, dst)
            print("symlink (fallback)", dst, "->", c, flush=True)
            return
    print("WARNING: tidak ada kandidat mount untuk", dst, flush=True)

TBX_CANDS = [
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified",
]
RAH_CANDS = [
    "/kaggle/input/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
    "/kaggle/input/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
]
pick("data/raw/tbx11k-simplified", TBX_CANDS, ("images",))
pick("data/raw/TB_Chest_Radiography_Database", RAH_CANDS, ("Normal", "Tuberculosis"))

for root, dirs, files in os.walk("/kaggle/input"):
    print("mount:", root, "dirs:", dirs[:6], "files:", len(files), flush=True)
    if root.count("/") > 4:
        dirs[:] = []

# 3) sanity: semua path train/val harus ada
manifest = json.load(open("ml/data/artifacts/manifest.json"))
missing = [it["path"] for s in ("train", "val") for it in manifest[s] if not os.path.exists(it["path"])]
print(f"train={len(manifest['train'])} val={len(manifest['val'])} missing={len(missing)}", flush=True)
print("missing contoh:", missing[:3], flush=True)
assert not missing, "data train/val tidak lengkap – cek mount dataset"

# 4) training (dua arsitektur + baseline fitur beku)
JOBS = [
    ("efficientnet_b0", "runs/eb0", ["--epochs", "14", "--batch", "64"]),
    ("convnext_tiny", "runs/convnext", ["--epochs", "12", "--batch", "48"]),
]
for arch, out, extra in JOBS:
    sh(
        [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
         "--arch", arch, "--balanced", "--out", out, *extra]
    )

sh(
    [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
     "--arch", "efficientnet_b0", "--freeze", "--lr", "1e-3", "--epochs", "8", "--batch", "64",
     "--out", "runs/linear_probe"]
)
```

```
# 5) export ONNX (satu file mandiri per model)
os.makedirs("artifacts", exist_ok=True)
for run, name in (("runs/eb0", "eb0"), ("runs/convnext", "convnext")):
    sh([sys.executable, "ml/export_onnx.py", "--ckpt", f"{run}/best.pt", "--out", f"artifacts/{name}.onnx"])

print("ALL DONE", flush=True)
```

ml/kaggle/out_ssl/loop-tb/ml/ssl_probe.py

```
#!/usr/bin/env python3
"""Latih probe linear di atas embedding SSL + evaluasi eksternal rigor (ADR-007).

Contoh:
.venv/bin/python ml/ssl_probe.py --emb-dir ml/kaggle/out_ssl/ssl_embeddings \
  --out ml/runs/ssl_eval --supervised-preds ml/runs/eval_external_ens/preds.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import sys
from pathlib import Path

import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import brier_score_loss, roc_auc_score
from sklearn.preprocessing import StandardScaler

sys.path.insert(0, str(Path(__file__).resolve().parent))
from evaluate import ( # noqa: E402
    bootstrap_auroc,
    ece,
    fit_temperature,
    reliability_plot,
    roc_plot,
    sens_at_spec,
    threshold_at_spec,
)

# Statistik preprocessing piksel yang HARUS sama dengan saat ekstraksi embedding.
# (dinov2: default timm/ImageNet; rad_dino: diisi dari log kernel – processor HF)
PIXEL = {
    "dinov2": {
        "type": "timm", "id": "vit_base_patch14_dinov2.lvd142m", "size": 518,
        "mean": (0.485, 0.456, 0.406), "std": (0.229, 0.224, 0.225),
    },
    "rad_dino": {
        "type": "hf", "id": "microsoft/rad-dino", "size": 518,
        "mean": (0.5307, 0.5307, 0.5307), "std": (0.2588, 0.2588, 0.2588), # diverifikasi dari log kernel; koreksi bila beda
    },
}

def sigmoid(z: np.ndarray) -> np.ndarray:
    z = np.clip(z, -60, 60)
    return 1 / (1 + np.exp(-z))

def load_split(emb_dir: Path, enc: str, split: str) -> tuple[np.ndarray, np.ndarray, list[dict]]:
    X = np.load(emb_dir / f"emb_{enc}_{split}.npy")
    meta = json.loads((emb_dir / f"meta_{split}.json").read_text())
    y = np.array([m["label"] for m in meta])
    return X, y, meta

def pixel_of(emb_dir: Path, enc: str) -> dict:
    """Statistik preprocessing piksel: dari encmeta kernel bila ada, else fallback PIXEL."""
    f = emb_dir / f"encmeta_{enc}.json"
    if f.exists():
        d = json.loads(f.read_text())
        return {"type": d["type"], "id": d["id"], "size": d["size"], "mean": tuple(d["mean"]), "std": tuple(d["std"])}
    return PIXEL.get(enc, PIXEL["dinov2"])

def band_metrics(y: np.ndarray, p: np.ndarray, tau_low: float, tau_high: float) -> dict:
    band = np.where(p >= tau_high, "rujuk_prioritas", np.where(p < tau_low, "negatif_skrining", "ragu"))
    rep = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b
        rep[b] = {
            "n": int(m.sum()),
            "frac": float(m.mean()),
            "tb_rate": float(y[m].mean()) if m.sum() else None,
        }
    return rep

def full_metrics(y: np.ndarray, p: np.ndarray) -> dict:
    lo, hi = bootstrap_auroc(y, p)
    return {
        "n": int(len(y)),
        "pos": int(y.sum()),
        "auroc": float(roc_auc_score(y, p)),
        "auroc_ci95": [lo, hi],
        "sens_at_spec_90": sens_at_spec(y, p, 0.90),
        "sens_at_spec_70": sens_at_spec(y, p, 0.70),
    }
```

```

        "ece": ece(y, p),
        "brier": float(brier_score_loss(y, p)),
    }

def run_encoder(enc: str, emb_dir: Path, out_dir: Path, supervised_preds: Path | None) -> dict:
    Xtr, ytr, _ = load_split(emb_dir, enc, "train")
    Xva, yva, meta_va = load_split(emb_dir, enc, "val")
    Xex, yex, meta_ex = load_split(emb_dir, enc, "external")

    scaler = StandardScaler().fit(Xtr)
    Xtr_s, Xva_s, Xex_s = scaler.transform(Xtr), scaler.transform(Xva), scaler.transform(Xex)

    best = {"auroc": -1.0, "C": None}
    for C in (0.001, 0.01, 0.1, 1.0, 10.0):
        clf = LogisticRegression(C=C, class_weight="balanced", max_iter=3000, solver="lbfgs")
        clf.fit(Xtr_s, ytr)
        z_va = clf.decision_function(Xva_s)
        try:
            auroc = roc_auc_score(yva, sigmoid(z_va))
        except ValueError:
            auroc = float("nan")
        if auroc > best["auroc"]:
            best = {"auroc": float(auroc), "C": C, "clf": clf}
        print(f" [{enc}] C={C:<6} val AUROC={auroc:.4f}")

    clf: LogisticRegression = best["clf"]
    z_va, z_ex = clf.decision_function(Xva_s), clf.decision_function(Xex_s)
    T = fit_temperature([z_va], yva)
    p_va, p_ex = sigmoid(z_va / T), sigmoid(z_ex / T)

    tau_high = threshold_at_spec(yva, p_va, 0.70)
    tau_low = threshold_at_spec(yva, p_va, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    res = {
        "encoder": enc,
        "chosen_C": best["C"],
        "val": full_metrics(yva, p_va),
        "external": full_metrics(yex, p_ex),
        "temp": T,
        "tau_low": tau_low,
        "tau_high": tau_high,
        "bands_external": band_metrics(yex, p_ex, tau_low, tau_high),
    }

    out_dir.mkdir(parents=True, exist_ok=True)
    (out_dir / f"metrics_{enc}.json").write_text(json.dumps(res, indent=1))

    # ekspor probe untuk serving API (scaler + LR + T + metadata encoder)
    pix = pixel_of(emb_dir, enc)
    np.savez(
        out_dir / f"probe_{enc}.npz",
        scaler_mean=scaler.mean_,
        scaler_scale=scaler.scale_,
        coef=clf.coef_.ravel(),
        intercept=np.array([float(clf.intercept_[0])]),
        T=np.array([T]),
        size=np.array([pix["size"]]),
        pixel_mean=np.array(pix["mean"], dtype=np.float64),
        pixel_std=np.array(pix["std"], dtype=np.float64),
        encoder_type=np.array([pix["type"]]),
        encoder_id=np.array([pix["id"]]),
    )

    reliability_plot(yex, p_ex, out_dir / f"reliability_{enc}.png", f"SSL probe {enc} - external")
    roc_plot({f"{enc} (external)": (yex, p_ex)}, out_dir / f"roc_{enc}.png")
    with (out_dir / f"preds_{enc}.csv").open("w", newline="") as f:
        w = csv.writer(f)
        w.writerow(["path", "label", "p", "z"])
        for m, p, z in zip(meta_ex, p_ex, z_ex):
            w.writerow([m["path"], m["label"], float(p), float(z)])

    # stacking dengan ensemble supervised (jika tersedia)
    if supervised_preds and supervised_preds.exists():
        sup = {}
        for row in csv.DictReader(supervised_preds.open()):
            sup[Path(row["path"]).stem] = float(row["p_ens"])
        pairs = [(Path(m["path"]).stem, m["label"], p) for m, p in zip(meta_ex, p_ex)]
        aligned = [(k, lab, p, sup[k]) for k, lab, p in pairs if k in sup]
        if len(aligned) >= 50:
            ya = np.array([a[1] for a in aligned])
            ps = (np.array([a[2] for a in aligned]) + np.array([a[3] for a in aligned])) / 2
            res["stack_with_supervised"] = {**full_metrics(ya, ps), "n_matched": len(aligned)}
            print(f" [{enc}] stack n={len(aligned)} external AUROC={res['stack_with_supervised']['auroc']:.4f}")

    (out_dir / f"metrics_{enc}.json").write_text(json.dumps(res, indent=1))
    return res

def main() -> None:

```

```

ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
ap.add_argument("--emb-dir", type=Path, required=True)
ap.add_argument("--out", type=Path, required=True)
ap.add_argument("--encoder", nargs="*", default=None, help="default: semua yang ada")
ap.add_argument("--supervised-preds", type=Path, default=None)
args = ap.parse_args()

encs = args.encoder or sorted(
    p.name[len("emb_") : -len("_train.npy")] for p in args.emb_dir.glob("emb_*_train.npy")
)
results = []
for enc in encs:
    print(f"=== {enc} ===")
    results.append(run_encoder(enc, args.emb_dir, args.out, args.supervised_preds))

print("\n=== RINGKASAN (eksternal) ===")
for r in results:
    e = r["external"]
    print(f"{r['encoder']:10s} AUROC {e['auroc']:.4f} [{e['auroc_ci95'][0]:.3f},{e['auroc_ci95'][1]:.3f}] "
          f"| sens@spec90 {e['sens_at_spec_90']:.3f} | ECE {e['ece']:.3f}"
          + (f" | stack {r['stack_with_supervised']['auroc']:.4f}" if "stack_with_supervised" in r else ""))

# kombinasi rata-rata probabilitas antar encoder SSL
preds_files = sorted(args.out.glob("preds_*.csv"))
if len(preds_files) > 1:
    maps = []
    for f in preds_files:
        rows = list(csv.DictReader(f.open()))
        maps.append({r["path"]: (float(r["p"]), int(r["label"])) for r in rows})
    keys = set(maps[0])
    for m in maps[1:]:
        keys |= set(m)
    ks = sorted(keys)
    y = np.array([maps[0][k][1] for k in ks])
    p = np.mean([np.array([m[k][0] for k in ks]) for m in maps], axis=0)
    comb = full_metrics(y, p)
    print(f"combined{'x'.join([]) or ''} SSL ({len(maps)} encoder): AUROC {comb['auroc']:.4f} "
          f"[{comb['auroc_ci95'][0]:.3f},{comb['auroc_ci95'][1]:.3f}] | sens@spec90 {comb['sens_at_spec_90']:.3f}")
    (args.out / "metrics_combined.json").write_text(json.dumps(comb, indent=1))

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl/loop-tb/ml/train.py

```
#!/usr/bin/env python3
"""Train TB CXR classifier dari manifest provenance-aware.

Contoh:
python ml/train.py --manifest ml/data/artifacts/manifest.json --arch efficientnet_b0 \
    --epochs 20 --out ml/runs/efficientnet_b0

Baseline linear probe (fitur beku):
python ml/train.py ... --freeze --lr 1e-3 --out ml/runs/linear_probe
"""
from __future__ import annotations

import argparse
import csv
import json
import time
from pathlib import Path

import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import roc_auc_score, roc_curve
from torch.utils.data import DataLoader, Dataset, WeightedRandomSampler
from torchvision import transforms as T
from tqdm import tqdm

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

class CXRManifest(Dataset):
    def __init__(self, items: list[dict], train: bool, img_size: int):
        self.items = items
        if train:
            self.tx = T.Compose([
                T.RandomResizedCrop(img_size, scale=(0.75, 1.0), ratio=(0.9, 1.1)),
                T.RandomRotation(7, fill=0),
                T.ColorJitter(brightness=0.15, contrast=0.15),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])
        else:
            self.tx = T.Compose([
                T.Resize(int(img_size * 1.14)),
                T.CenterCrop(img_size),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])

    def __len__(self) -> int:
        return len(self.items)

    def __getitem__(self, i: int):
        from PIL import Image
        it = self.items[i]
        with Image.open(it["path"]) as im:
            x = self.tx(im.convert("RGB"))
        return x, torch.tensor(float(it["label"])), i

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 → fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

@torch.no_grad()
def predict(model: nn.Module, loader: DataLoader, device: torch.device) -> tuple[np.ndarray, np.ndarray]:
    model.eval()
    logits_all, labels_all = [], []
    for x, y, _ in loader:
        x = x.to(device, non_blocking=True)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            logits = model(x).squeeze(1).float()
        logits_all.append(logits.cpu().numpy())
        labels_all.append(y.numpy())
    return np.concatenate(logits_all), np.concatenate(labels_all)

def evaluate(logits: np.ndarray, y: np.ndarray) -> dict:
    p = 1 / (1 + np.exp(-logits))
    out = {"n": int(len(y)), "pos": int(y.sum())}
    if len(np.unique(y)) > 1:
```

```

        out["auroc"] = float(roc_auc_score(y, p))
        out["sens_at_spec_90"] = sens_at_spec(y, p, 0.90)
        out["sens_at_spec_70"] = sens_at_spec(y, p, 0.70)
    return out

```

```

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--arch", default="efficientnet_b0")
    ap.add_argument("--epochs", type=int, default=20)
    ap.add_argument("--batch", type=int, default=32)
    ap.add_argument("--lr", type=float, default=3e-4)
    ap.add_argument("--wd", type=float, default=1e-4)
    ap.add_argument("--img-size", type=int, default=224)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--seed", type=int, default=42)
    ap.add_argument("--workers", type=int, default=8)
    ap.add_argument("--freeze", action="store_true", help="linear probe: backbone beku")
    ap.add_argument("--balanced", action="store_true", help="WeightedRandomSampler")
    ap.add_argument("--limit", type=int, default=0, help="smoke test: batasi n sampel (0=semua)")
    args = ap.parse_args()

    torch.manual_seed(args.seed)
    np.random.seed(args.seed)
    args.out.mkdir(parents=True, exist_ok=True)

    device = torch.device("cuda" if torch.cuda.is_available() else "mps" if torch.backends.mps.is_available() else "cpu")
    manifest = json.loads(args.manifest.read_text())

    def take(items: list[dict], n: int) -> list[dict]:
        """Subset berstratifikasi untuk smoke test."""
        if not n:
            return items
        pos = [it for it in items if it["label"] == 1]
        neg = [it for it in items if it["label"] == 0]
        n = min(n, len(items))
        kp = min(len(pos), max(1, n // 4))
        return pos[:kp] + neg[: max(n - kp, 0)]

    train_items = take(manifest["train"], args.limit)
    val_items = take(manifest["val"], args.limit)

    ds_tr = CXRMManifest(train_items, train=True, img_size=args.img_size)
    ds_va = CXRMManifest(val_items, train=False, img_size=args.img_size)
    sampler = None
    if args.balanced:
        labels = np.array([it["label"] for it in train_items])
        w = np.where(labels == 1, 1.0 / labels.sum(), 1.0 / (len(labels) - labels.sum()))
        sampler = WeightedRandomSampler(torch.from_numpy(w), num_samples=len(labels), replacement=True)
    dl_tr = DataLoader(ds_tr, batch_size=args.batch, shuffle=sampler is None, sampler=sampler,
                      num_workers=args.workers, pin_memory=(device.type == "cuda"), drop_last=True)
    dl_va = DataLoader(ds_va, batch_size=args.batch, shuffle=False, num_workers=args.workers, pin_memory=(device.type == "cuda"))

    model = timm.create_model(args.arch, pretrained=True, num_classes=1).to(device)
    if args.freeze:
        for n, p in model.named_parameters():
            p.requires_grad = "head" in n or "classifier" in n or "fc" in n
    params = [p for p in model.parameters() if p.requires_grad]
    opt = torch.optim.AdamW(params, lr=args.lr, weight_decay=args.wd)
    sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=max(args.epochs, 1))
    loss_fn = nn.BCEWithLogitsLoss()
    scaler = torch.amp.GradScaler(enabled=(device.type == "cuda"))

    print(f"device={device} arch={args.arch} freeze={args.freeze} "
          f"train={len(ds_tr)} val={len(ds_va)} params={sum(p.numel() for p in params)/1e6:.2f}M")

    hist = []
    best = {"auroc": -1.0, "epoch": -1}
    for epoch in range(1, args.epochs + 1):
        model.train()
        t0, losses = time.time(), []
        for x, y, _ in tqdm(dl_tr, desc=f"epoch {epoch:02d}", leave=False):
            x, y = x.to(device, non_blocking=True), y.to(device, non_blocking=True)
            opt.zero_grad(set_to_none=True)
            with torch.autocast(device.type, enabled=(device.type == "cuda")):
                logits = model(x).squeeze(1)
                loss = loss_fn(logits.float(), y)
            scaler.scale(loss).backward()
            scaler.step(opt)
            scaler.update()
            losses.append(loss.item())
        sched.step()

        logits, y_va = predict(model, dl_va, device)
        m = evaluate(logits, y_va)
        row = {"epoch": epoch, "loss": float(np.mean(losses)), "lr": opt.param_groups[0]["lr"],
              "sec": round(time.time() - t0, 1), **m}
        hist.append(row)
    print(f"epoch {epoch:02d} | loss {row['loss']:.4f} | AUROC {m.get('auroc', float('nan')):.4f} "
          f"| sens@spec90 {m.get('sens_at_spec_90', float('nan')):.3f} | {row['sec']}s")

```

```

torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
           "epoch": epoch, "val": m}, args.out / "last.pt")
if m.get("auroc", -1) > best["auroc"]:
    best = {"auroc": m["auroc"], "epoch": epoch}
    torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
               "epoch": epoch, "val": m}, args.out / "best.pt")
    with (args.out / "val_preds.csv").open("w", newline="") as f:
        w = csv.writer(f)
        w.writerow(["path", "label", "logit"])
        for it, lg in zip(val_items, logits):
            w.writerow([it["path"], it["label"], float(lg)])

(args.out / "history.csv").write_text(
    "epoch,loss,lr,sec,n,pos,auroc,sens_at_spec_90,sens_at_spec_70\n" + "\n".join(
        ", ".join(str(r.get(k, "")) for k in ("epoch", "loss", "lr", "sec", "n", "pos", "auroc", "sens_at_spec_90",
        "sens_at_spec_70"))
        for r in hist) + "\n")
(args.out / "metrics.json").write_text(json.dumps({"best": best, "history": hist, "config": vars(args)}, indent=1, default=str))
print(f"\nBEST epoch {best['epoch']} AUROC {best['auroc']:.4f} -> {args.out/'best.pt'}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl/loop-tb/tools/e2e_screen.py

```
#!/usr/bin/env python3
"""Uji integrasi UI+API: unggah citra X-ray lewat halaman /skrining, ambil screenshot hasil.

Jalankan dengan helper server (web 3210 + api 8000):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
--server "cd web && npm start -- --port 3210" --port 3210 \
--server ".venv/bin/python -m uvicorn api.main:app --port 8000" --port 8000 \
-- .venv/bin/python tools/e2e_screen.py
"""
from __future__ import annotations

import sys
from pathlib import Path

from playwright.sync_api import sync_playwright

OUT = Path("submission/screenshots")
OUT.mkdir(parents=True, exist_ok=True)

CASES = [
    ("data/raw/nlm/shenzhen/CXR_png/CHNCXR_0327_1.png", "tb", "S-014"),
    ("data/raw/nlm/shenzhen/CXR_png/CHNCXR_0001_0.png", "normal", "S-015"),
]

def run_case(page, img: str, tag: str, pid: str) -> None:
    page.goto("http://localhost:3210/skrining", wait_until="networkidle")
    page.set_input_files('input[type="file"]', img)
    page.fill('input[placeholder="mis. S-014"]', pid)
    page.fill('input[placeholder="34"]', "42")
    page.click("text=Analisis citra")
    page.wait_for_selector("text=Hasil analisis", timeout=120_000)
    page.wait_for_timeout(600)
    page.screenshot(path=str(OUT / f"skrining_hasil_{tag}_mobile.png"), full_page=True)
    verdict = page.locator("section >> text=Hasil analisis").first
    print(f"[ok] {tag}: {img}")

def main() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        ctx = browser.new_context(viewport={"width": 390, "height": 844}, device_scale_factor=2)
        page = ctx.new_page()
        console: list[str] = []
        page.on("console", lambda m: console.append(f"{m.type}: {m.text}") if m.type == "error" else None)
        for img, tag, pid in CASES:
            if not Path(img).exists():
                print("[skip]", img)
                continue
            run_case(page, img, tag, pid)
        if console:
            print("[console errors]", console[:6])
        ctx.close()
        browser.close()
    print("E2E SCREENSHOTS DONE")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl/loop-tb/tools/screenshot_ui.py

```
#!/usr/bin/env python3
"""QA visual + screenshot semua halaman SIGAP (mobile & desktop).

Jalankan lewat helper server (lihat ml/kaggle/README atau riwayat commit):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
--server "cd web && npm run dev -- --port 3210" --port 3210 \
-- .env/bin/python tools/screenshot_ui.py http://localhost:3210
"""
from __future__ import annotations

import sys
from pathlib import Path

from playwright.sync_api import sync_playwright

BASE = sys.argv[1] if len(sys.argv) > 1 else "http://localhost:3210"
OUT = Path("submission/screenshots")
OUT.mkdir(parents=True, exist_ok=True)

ROUTES = [
    ("", "landing"),
    ("skrining", "skrining"),
    ("pasien", "pasien"),
    ("klinisi", "klinisi"),
    ("tentang", "tentang"),
]
VIEWPORTS = [("mobile", 390, 844), ("desktop", 1280, 900)]

def main() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        for vp_name, w, h in VIEWPORTS:
            ctx = browser.new_context(viewport={"width": w, "height": h}, device_scale_factor=2)
            page = ctx.new_page()
            console_errors: list[str] = []
            page.on("console", lambda m: console_errors.append(m.text) if m.type == "error" else None)
            for route, name in ROUTES:
                page.goto(f"{BASE}/{route}", wait_until="networkidle")
                page.wait_for_timeout(400)
                page.screenshot(path=str(OUT / f"{name}_{vp_name}.png"), full_page=True)
                print(f"[ok] {vp_name:7s} {name}")
            if console_errors:
                print(f"[warn] console errors ({vp_name}): {console_errors[:5]}")
            ctx.close()
        browser.close()
    print(f"SCREENSHOTS -> {OUT}")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl_diag/loop-tb/api/main.py

```
"""Loop-TB API – inference + Grad-CAM + band triase.

Jalankan lokal:
    uvicorn api.main:app --host 0.0.0.0 --port 8000

Env:
    LOOPTB_CKPT path ke checkpoint best.pt (default ml/runs/convnext_tiny/best.pt)
    LOOPTB_BANDS path ke bands.json hasil evaluate.py (opsional; default  $\tau=0.5/0.5$ )
"""
from __future__ import annotations

import base64
import io
import json
import os
from pathlib import Path

import matplotlib

matplotlib.use("Agg")
import matplotlib.cm as cm
import numpy as np
import timm
import torch
import torch.nn as nn
from fastapi import FastAPI, File, HTTPException, UploadFile
from fastapi.middleware.cors import CORSMiddleware
from PIL import Image
from torchvision import transforms as T

CKPT_PATH = Path(os.environ.get("LOOPTB_CKPT", "ml/runs/convnext_tiny/best.pt"))
BANDS_PATH = Path(os.environ.get("LOOPTB_BANDS", "")) if os.environ.get("LOOPTB_BANDS") else None
MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

app = FastAPI(title="loop-tb", version="0.1.0")
app.add_middleware(
    CORSMiddleware, allow_origins=["*"], allow_credentials=False,
    allow_methods=["*"], allow_headers=["*"],
)

_state: dict = {}

def _load() -> None:
    if not CKPT_PATH.exists():
        raise RuntimeError(f"Checkpoint tidak ditemukan: {CKPT_PATH}")
    ckpt = torch.load(CKPT_PATH, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()
    bands = {"tau_high": 0.5, "tau_low": 0.5}
    if BANDS_PATH and BANDS_PATH.exists():
        b = json.loads(BANDS_PATH.read_text())
        bands = {"tau_high": float(b.get("tau_high", 0.5)), "tau_low": float(b.get("tau_low", 0.5))}
    size = int(ckpt["config"].get("img_size", 224))
    _state.update(model=model, arch=ckpt["arch"], img_size=size, bands=bands)
    print(f"[loop-tb] loaded {ckpt['arch']} @ {size}px bands={bands}")

_tx = None

def preprocess(img: Image.Image) -> torch.Tensor:
    global _tx
    if _tx is None:
        s = _state["img_size"]
        _tx = T.Compose([T.Resize(int(s * 1.14)), T.CenterCrop(s), T.ToTensor(), T.Normalize(MEAN, STD)])
    return _tx(img.convert("RGB")).unsqueeze(0)

def band_of(p: float, bands: dict) -> str:
    if p >= bands["tau_high"]:
        return "rujuk_prioritas"
    if p < bands["tau_low"]:
        return "negatif_skrining"
    return "ragu"

def gradcam(model: nn.Module, x: torch.Tensor) -> np.ndarray:
    """Grad-CAM generik via forward_features (timm), tanpa asumsi arsitektur."""
    model.zero_grad(set_to_none=True)
    feats = model.forward_features(x)
    feats.retain_grad()
    logit = model.forward_head(feats).squeeze(1)
    logit.backward()
    w = feats.grad.mean(dim=(2, 3), keepdim=True)
    cam = (w * feats).sum(dim=1, keepdim=True).clamp(min=0)
    cam = nn.functional.interpolate(cam, size=x.shape[-2:], mode="bilinear", align_corners=False)
```

```

cam = cam / (cam.amax(dim=(2, 3), keepdim=True) + 1e-8)
return cam[0, 0].detach().numpy()

def overlay_png(img: Image.Image, cam: np.ndarray) -> str:
    img_np = np.asarray(img.convert("RGB").resize((cam.shape[1], cam.shape[0])), dtype=np.float32) / 255.0
    heat = cm.jet(cam)[..., :3]
    alpha = (cam[..., None] * 0.45)
    blend = img_np * (1 - alpha) + heat.astype(np.float32) * alpha
    out = Image.fromarray((np.clip(blend, 0, 1) * 255).astype(np.uint8))
    buf = io.BytesIO()
    out.save(buf, format="PNG")
    return base64.b64encode(buf.getvalue()).decode()

@app.on_event("startup")
def startup() -> None:
    _load()

@app.get("/health")
def health() -> dict:
    return {
        "status": "ok",
        "arch": _state.get("arch"),
        "img_size": _state.get("img_size"),
        "bands": _state.get("bands"),
    }

@app.post("/predict")
async def predict(file: UploadFile = File(...)) -> dict:
    if "model" not in _state:
        raise HTTPException(503, "model belum siap")
    raw = await file.read()
    try:
        img = Image.open(io.BytesIO(raw))
        img.load()
    except Exception as e:
        raise HTTPException(400, f"bukan citra valid: {e}")

    x = preprocess(img)
    with torch.no_grad():
        logit = _state["model"](x).squeeze(1)
    p = float(torch.sigmoid(logit).item())
    band = band_of(p, _state["bands"])

    cam = gradcam(_state["model"], x)
    heatmap_b64 = overlay_png(img, cam)

    return {
        "p_tb": round(p, 4),
        "band": band,
        "bands": _state["bands"],
        "arch": _state.get("arch"),
        "disclaimer": "Alat triase skrining – bukan diagnosis. Wajib konfirmasi (GeneXpert) oleh tenaga kesehatan.",
        "heatmap_png_b64": heatmap_b64,
    }

```

ml/kaggle/out_ssl_diag/loop-tb/ml/data/prepare_data.py

```
#!/usr/bin/env python3
"""Build a provenance-aware manifest for TB chest X-ray training.

Sumber & aturan label (lihat docs/02-arsitektur.md):
- TBX11K (Kaggle: vbookshelf/tbx11k-simplified): folder per kelas.
    positif = {tb}; negatif = {healthy, sick}; latent_tb DIKELUARKAN (ambigu).
- Rahman composite (Kaggle: tawsifurrahman/tuberculosis-tb-chest-xray-dataset):
    folder Normal/ -> 0, Tuberculosis/ -> 1.
- NLM Shenzhen/Montgomery: konvensi nama file resmi `*_0` = normal, `*_1` = TB.

Dedup (WAJIB – mitigasi class-conditional acquisition confounding):
1. pHash 64-bit; Hamming <= --dup-threshold dianggap duplikat (union-find transitif).
2. Integritas external: citra train yang sekelompok dengan citra external DIBUANG dari train.
3. Duplikat internal train dibuang (keep-first, urutan deterministik).
4. Split val dilakukan per-klaster (tidak ada kebocoran near-dup train->val).

Output (ml/data/artifacts/):
- manifest.json, dedup_report.json, train.csv, val.csv, external.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import re
from collections import Counter
from pathlib import Path

import numpy as np
from PIL import Image
from tqdm import tqdm

try:
    import imagehash
except ImportError: # pragma: no cover
    raise SystemExit("Butuh imagehash: pip install imagehash")

IMG_EXTS = {".png", ".jpg", ".jpeg", ".bmp", ".tif", ".tiff"}
POS_TOKENS = {"tb", "tuberculosis", "tuberculose"}
NEG_TOKENS = {"healthy", "normal", "sick"}
EXCLUDE_TOKENS = {"latent", "latent_tb", "latenttb"}

def _rel(p: Path, root: Path) -> str:
    """Path relatif terhadap root repo (portable lokal <-> Colab); fallback absolut."""
    try:
        return str(p.relative_to(root))
    except ValueError:
        return str(p)

def _tokens(path: Path) -> set[str]:
    parts = [p.lower() for p in path.parts]
    parts += [path.stem.lower()]
    return set(parts)

def load_tbx11k_map(raw: Path) -> dict[str, int | None]:
    """Label otoritatif dari data.csv TBX11K: 1=active_tb, 0=no_tb, None=latent (dikeluarkan)."""
    m: dict[str, int | None] = {}
    for p in raw.rglob("data.csv"):
        if "tbx11k" not in str(p).lower():
            continue
        for r in csv.DictReader(p.open()):
            f = (r.get("fname") or "").strip()
            if not f:
                continue
            t = (r.get("tb_type") or "").lower()
            tgt = (r.get("target") or "").lower()
            if "latent" in t:
                m.setdefault(f, None)
            elif tgt == "tb":
                m[f] = 1
            else:
                m.setdefault(f, 0)
    return m

def classify(path: Path, raw: Path, tbx_map: dict[str, int | None]) -> tuple[int, str] | None:
    """Return (label, source) atau None kalau tidak bisa diklasifikasi."""
    rel = path.relative_to(raw)
    low = str(rel).lower()

    # --- NLM Shenzhen / Montgomery: konvensi suffix nama file ---
    m = re.search(r"(chncxr|mcucxr)_d+_(\d+)", path.stem.lower())
    if m:
        label = int(m.group(2))
        source = "shenzhen" if m.group(1) == "chncxr" else "montgomery"
```

```

        return label, source

toks = _tokens(rel)
if toks & EXCLUDE_TOKENS or any(t.startswith("latent") for t in toks):
    return None

if "tbx11k" in low:
    if tbx_map:
        lab = tbx_map.get(path.name, None)
        return None if lab is None else (lab, "tbx11k")
    # fallback bila data.csv tidak ada: prefix nama file
    stem = path.stem.lower()
    if stem.startswith("t"):
        return 1, "tbx11k"
    if stem.startswith(("h", "s")):
        return 0, "tbx11k"
    return None

if toks & POS_TOKENS:
    return 1, "rahman"
if toks & NEG_TOKENS:
    return 0, "rahman"
return None

class DSU:
    def __init__(self, n: int):
        self.parent = list(range(n))

    def find(self, x: int) -> int:
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def union(self, a: int, b: int) -> None:
        ra, rb = self.find(a), self.find(b)
        if ra != rb:
            self.parent[rb] = ra

_POP = np.array([bin(i).count("1") for i in range(256)], dtype=np.uint8)

def hamming_pairs(hashes: np.ndarray, threshold: int, chunk: int = 512) -> list[tuple[int, int]]:
    """Semua pasangan (i<j) dengan Hamming distance <= threshold (chunked, hemat memori)."""
    n = len(hashes)
    h8 = hashes.view(np.uint8).reshape(n, 8)
    pairs: list[tuple[int, int]] = []
    for start in range(0, n, chunk):
        stop = min(start + chunk, n)
        xor = np.bitwise_xor(h8[start:stop, None, :], h8[None, :, :])
        dist = _POP[xor].sum(axis=-1) # (chunk, n)
        rows, cols = np.where(dist <= threshold)
        for r, c in zip(rows, cols):
            i, j = start + int(r), int(c)
            if i < j:
                pairs.append((i, j))
    return pairs

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--raw", type=Path, default=Path("data/raw"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts"))
    ap.add_argument("--dup-threshold", type=int, default=3, help="Hamming pHash <= ini dianggap duplikat")
    ap.add_argument("--val-frac", type=float, default=0.15)
    ap.add_argument("--seed", type=int, default=42)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    raw = args.raw.resolve()
    root = raw.parent.parent # asumsi struktur <root>/data/raw

    # ----- 1) discovery -----
    labeled: list[dict] = []
    unlabeled = Counter()
    tbx_map = load_tbx11k_map(raw)
    print(f"[discovery] label TBX11K dari data.csv: {len(tbx_map)} entri")
    for p in sorted(raw.rglob("*")):
        if p.suffix.lower() not in IMG_EXTS:
            continue
        cls = classify(p, raw, tbx_map)
        if cls is None:
            unlabeled[str(p.relative_to(raw).parent)] += 1
            continue
        label, source = cls
        labeled.append({"path": _rel(p, root), "label": label, "source": source})

    print(f"[discovery] berlabel={len(labeled)} tak-terklasifikasi={sum(unlabeled.values())}")
    for k, v in unlabeled.most_common(10):
        print(f" skip {v:6d} {k}")

```

```

if not labeled:
    raise SystemExit("Tidak ada citra berlabel – cek struktur data/raw")

# ----- 2) pHash + kluster duplikat -----
print("[dedup] menghitung pHash...")
ok: list[dict] = []
for item in tqdm(labeled, unit="img"):
    try:
        with Image.open(item["path"]) as im:
            h = imagehash.phash(im.convert("L"), hash_size=8)
            item["phash"] = int(str(h), 16)
            ok.append(item)
    except Exception as e: # gambar korup
        print(f"warn: gagal baca {item['path']}: {e}")
labeled = ok

hashes = np.array([it["phash"] for it in labeled], dtype=np.uint64)
pairs = hamming_pairs(hashes, args.dup_threshold)
dsu = DSU(len(labeled))
for i, j in pairs:
    dsu.union(i, j)
for idx, it in enumerate(labeled):
    it["cluster"] = dsu.find(idx)

by_cluster: dict[int, list[dict]] = {}
for it in labeled:
    by_cluster.setdefault(it["cluster"], []).append(it)

# ----- 3) lock external, buang kebocoran dari train -----
ext = [it for it in labeled if it["source"] in ("shenzhen", "montgomery")]
ext_clusters = {it["cluster"] for it in ext}
train_pool, dropped_cross = [], []
for it in labeled:
    if it["source"] in ("shenzhen", "montgomery"):
        continue
    (dropped_cross if it["cluster"] in ext_clusters else train_pool).append(it)

# duplikat internal train: keep-first per kluster (deterministik via path)
keep, dropped_internal = [], []
seen_clusters: set[int] = set()
for it in sorted(train_pool, key=lambda x: x["path"]):
    if it["cluster"] in seen_clusters:
        dropped_internal.append(it)
    else:
        seen_clusters.add(it["cluster"])
        keep.append(it)
train_pool = keep

# ----- 4) split val per-kluster -----
rng = np.random.default_rng(args.seed)
sizes = Counter(it["cluster"] for it in train_pool)
pool_clusters = sorted(sizes)
rng.shuffle(pool_clusters)
target_val = int(len(train_pool) * args.val_frac)
val_clusters: set[int] = set()
acc = 0
for c in pool_clusters:
    if acc >= target_val:
        break
    val_clusters.add(c)
    acc += sizes[c]
train = [it for it in train_pool if it["cluster"] not in val_clusters]
val = [it for it in train_pool if it["cluster"] in val_clusters]

def dump(items: list[dict]) -> list[dict]:
    return [
        {k: it[k] for k in ("path", "label", "source", "cluster", "phash")}
        for it in sorted(items, key=lambda x: x["path"])
    ]

manifest = {
    "meta": {
        "root": str(root),
        "raw": _rel(raw, root),
        "dup_threshold": args.dup_threshold,
        "val_frac": args.val_frac,
        "seed": args.seed,
        "counts": {
            "train": len(train), "val": len(val), "external": len(ext),
            "dropped_cross_corpus": len(dropped_cross),
            "dropped_internal_dup": len(dropped_internal),
            "unlabeled": sum(unlabeled.values()),
        },
    },
    "train": dump(train),
    "val": dump(val),
    "external": dump(ext),
}
(args.out / "manifest.json").write_text(json.dumps(manifest, indent=1))
(args.out / "dedup_report.json").write_text(json.dumps({
    "pairs_within_threshold": len(pairs),
    "clusters": len(by_cluster),
}))

```

```

        "dropped_cross_corpus": dump(dropped_cross),
        "dropped_internal_dup": dump(dropped_internal),
        "unlabeled_by_dir": dict(unlabeled),
    }, indent=1))

for split in ("train", "val", "external"):
    with (args.out / f"{split}.csv").open("w") as f:
        f.write("path,label,source,cluster\n")
        for it in manifest[split]:
            f.write(f"{it['path']},{it['label']},{it['source']},{it['cluster']}\n")

def comp(items):
    c = Counter((it["source"], it["label"]) for it in items)
    return " ".join(f"{s}/{ 'TB' if l else 'NEG' }={n}" for (s, l), n in sorted(c.items()))

print("\n=== RINGKASAN ===")
print(f"train      n={len(train):6d}  {comp(train)}")
print(f"val        n={len(val):6d}    {comp(val)}")
print(f"external  n={len(ext):6d}    {comp(ext)}")
print(f"dibuang  cross-corpus={len(dropped_cross)}  internal-dup={len(dropped_internal)}")
print(f"manifest -> {args.out/'manifest.json'}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl_diag/loop-tb/ml/data/prepare_meta.py

```
#!/usr/bin/env python3
"""Ekstrak metadata klinis NLM (Montgomery + Shenzhen) → CSV untuk analisis subgrup.

Output: ml/data/artifacts/meta_nlm.csv
Kolom: filename,sex,age,reading,scar_flag
- Montgomery: "Patient's Sex: F / Patient's Age: 027Y / <reading>"
- Shenzhen: "male 45yrs / <reading>" (kadang terkontaminasi antar-file)
- scar_flag: menyebut scar/inactive/fibrosis → untuk pembahasan mode gagal
            "healed scar false positive" (selaras audit Bilal 2026).
"""
from __future__ import annotations

import argparse
import csv
import re
from pathlib import Path

SEX_M = re.compile(r"sex[:\s]+(m|male)\b", re.I)
SEX_F = re.compile(r"sex[:\s]+(f|female)\b", re.I)
SEX_FMT2 = re.compile(r"\b(male|female)\b", re.I)
AGE_MONT = re.compile(r"age[:\s]+(\d{1,3})\s*Y?\b", re.I)
AGE_FMT2 = re.compile(r"\b(\d{1,3})\s*yrs\b", re.I)
SCAR = re.compile(r"scar|inactive|fibro", re.I)

def parse_montgomery(text: str) -> dict | None:
    age = AGE_MONT.search(text)
    sex = "F" if SEX_F.search(text) else ("M" if SEX_M.search(text) else "")
    if not age:
        return None
    # reading = baris setelah Age
    lines = [ln.strip() for ln in text.splitlines() if ln.strip()]
    reading = lines[2] if len(lines) > 2 else ""
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": int(bool(SCAR.search(reading)))}

def parse_shenzhen(text: str) -> dict | None:
    age = AGE_FMT2.search(text)
    sex_m = SEX_FMT2.findall(text)
    sex = sex_m[0][0].upper() if sex_m else ""
    if not age:
        return None
    # reading = token huruf setelah "yrs"
    tail = text[age.end():]
    reading = ""
    m = re.search(r"[A-Za-z]{2,}", tail)
    if m:
        reading = m.group(0)
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": 0}

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--nlm", type=Path, default=Path("data/raw/nlm"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts/meta_nlm.csv"))
    args = ap.parse_args()
    args.out.parent.mkdir(parents=True, exist_ok=True)

    rows, n_fail = [], 0
    for corpus, parser in (("montgomery", parse_montgomery), ("shenzhen", parse_shenzhen)):
        d = args.nlm / corpus / "ClinicalReadings"
        for p in sorted(d.glob("*.txt")):
            r = parser(p.read_text(errors="ignore"))
            if r is None:
                n_fail += 1
                continue
            r["filename"] = p.stem + ".png"
            r["corpus"] = corpus
            rows.append(r)

    with args.out.open("w", newline="") as f:
        w = csv.DictWriter(f, fieldnames=["filename", "corpus", "sex", "age", "reading", "scar_flag"])
        w.writeheader()
        w.writerows(rows)

    from collections import Counter
    print(f"meta -> {args.out} n={len(rows)} gagal-parse={n_fail}")
    print("per korpus:", Counter(r["corpus"] for r in rows))
    print("sex:", Counter(r["sex"] for r in rows))
    print("scar_flag:", Counter(r["scar_flag"] for r in rows))

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl_diag/loop-tb/ml/evaluate.py

```
#!/usr/bin/env python3
"""Evaluasi rigor: kalibrasi, CI bootstrap, transfer, band triase, subgroup.

Contoh (ensemble + temperature scaling):
python ml/evaluate.py --runs ml/runs/efficientnet_b0 ml/runs/convnext_tiny \
    --manifest ml/data/artifacts/manifest.json --split external --temp-scale --out ml/runs/eval_external
"""
from __future__ import annotations

import argparse
import csv
import json
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import brier_score_loss, roc_auc_score, roc_curve
from torch.utils.data import DataLoader
from tqdm import tqdm

import sys
sys.path.insert(0, str(Path(__file__).resolve().parent))
from train import CXRManifest # noqa: E402

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

def sigmoid(x: np.ndarray) -> np.ndarray:
    return 1 / (1 + np.exp(-x))

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 -> fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

def threshold_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    fpr, tpr, thr = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(thr[ok][np.argmax(tpr[ok])]) if ok.any() else 0.5

def ece(y: np.ndarray, p: np.ndarray, bins: int = 15) -> float:
    edges = np.linspace(0, 1, bins + 1)
    total = 0.0
    for i in range(bins):
        m = (p >= edges[i]) & ((p < edges[i + 1]) if i < bins - 1 else (p <= edges[i + 1]))
        if m.sum():
            total += m.mean() * abs(p[m].mean() - y[m].mean())
    return float(total)

def bootstrap_auroc(y: np.ndarray, p: np.ndarray, n: int = 1000, seed: int = 42) -> tuple[float, float]:
    rng = np.random.default_rng(seed)
    vals = []
    for _ in range(n):
        idx = rng.integers(0, len(y), len(y))
        if len(np.unique(y[idx])) > 1:
            vals.append(roc_auc_score(y[idx], p[idx]))
    return float(np.percentile(vals, 2.5)), float(np.percentile(vals, 97.5))

@torch.no_grad()
def infer(ckpt_path: Path, items: list[dict], img_size: int, device: torch.device, workers: int) -> np.ndarray:
    ckpt = torch.load(ckpt_path, map_location="cpu", weights_only=False)
    arch = ckpt["arch"]
    model = timm.create_model(arch, pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.to(device).eval()
    dl = DataLoader(CXRManifest(items, train=False, img_size=img_size), batch_size=64,
                    shuffle=False, num_workers=workers)

    out = []
    for x, _, _ in tqdm(dl, desc=f"infer {ckpt_path.parent.name}", leave=False):
        x = x.to(device)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            out.append(model(x).squeeze(1).float().cpu().numpy())
    return np.concatenate(out)
```

```

def fit_temperature(val_logits: list[np.ndarray], val_y: np.ndarray) -> float:
    """Cari T>0 yang meminimalkan NLL pada val (grid log-space lebar untuk logit besar)."""
    logits = torch.tensor(np.mean(val_logits, axis=0), dtype=torch.float32)
    y = torch.tensor(val_y, dtype=torch.float32)
    best_t, best_nll = 1.0, float("inf")
    for t in np.logspace(-1, 3.5, 500):
        nll = nn.functional.binary_cross_entropy_with_logits(logits / t, y).item()
        if nll < best_nll:
            best_t, best_nll = float(t), nll
    return best_t

def reliability_plot(y: np.ndarray, p: np.ndarray, path: Path, title: str) -> None:
    edges = np.linspace(0, 1, 11)
    mids, accs = [], []
    for i in range(10):
        m = (p >= edges[i]) & (p < edges[i + 1] if i < 9 else p <= edges[i + 1])
        if m.sum():
            mids.append(p[m].mean())
            accs.append(y[m].mean())
    plt.figure(figsize=(4.2, 4.2))
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.plot(mids, accs, "o-", color="#d62728", lw=2)
    plt.xlabel("Confidence"), plt.ylabel("Empirical TB rate"), plt.title(title)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def roc_plot(curves: dict[str, tuple[np.ndarray, np.ndarray]], path: Path) -> None:
    plt.figure(figsize=(5, 5))
    for name, (y, p) in curves.items():
        fpr, tpr, _ = roc_curve(y, p)
        plt.plot(fpr, tpr, lw=2, label=f"{name} (AUROC {roc_auc_score(y, p):.3f})")
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.xlabel("FPR"), plt.ylabel("TPR (sensitivitas)"), plt.legend(fontsize=8)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--runs", type=Path, nargs="+", required=True)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--split", default="external", choices=["val", "external"])
    ap.add_argument("--temp-scale", action="store_true")
    ap.add_argument("--meta", type=Path, default=None, help="CSV filename,sex,age untuk subgroup (opsional)")
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--workers", type=int, default=8)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    manifest = json.loads(args.manifest.read_text())
    items = manifest[args.split]
    y = np.array([it["label"] for it in items])

    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    per_run_target, per_run_val, tscales = {}, {}, {}
    for run in args.runs:
        ckpt = torch.load(run / "best.pt", map_location="cpu", weights_only=False)
        img_size = ckpt["config"]["img_size"]
        per_run_target[run.name] = infer(run / "best.pt", items, img_size, device, args.workers)
        val_items = manifest["val"]
        per_run_val[run.name] = infer(run / "best.pt", val_items, img_size, device, args.workers)
        if args.temp_scale:
            tscales[run.name] = fit_temperature([per_run_val[run.name]], np.array([it["label"] for it in val_items]))

    per_run_target = {k: sigmoid(v) for k, v in per_run_target.items()}
    per_run_val = {k: sigmoid(v) for k, v in per_run_val.items()}
    val_y = np.array([it["label"] for it in manifest["val"]])

    # kalibrasi per-run pada val, lalu terapkan T ke target (bukan val-fit pada target!)
    if args.temp_scale:
        for k in per_run_target:
            per_run_val[k] = sigmoid(np.log(np.clip(per_run_val[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_val[k], 1e-6, 1 - 1e-6)))
            / tscales[k])
            lg = np.log(np.clip(per_run_target[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_target[k], 1e-6, 1 - 1e-6)))
            per_run_target[k] = sigmoid(lg / tscales[k])

    # ensemble = rata-rata; ketidakpastian = std antar anggota
    stack_t = np.stack(list(per_run_target.values()))
    p_ens = stack_t.mean(axis=0)
    u_ens = stack_t.std(axis=0)
    stack_v = np.stack(list(per_run_val.values()))
    p_ens_v = stack_v.mean(axis=0)

    tau_high = threshold_at_spec(val_y, p_ens_v, 0.70) # spec>=0.70 (TPP WHO)
    tau_low = threshold_at_spec(val_y, p_ens_v, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    band = np.where(p_ens >= tau_high, "rujuk_prioritas", np.where(p_ens < tau_low, "negatif_skrining", "ragu"))
    band_report = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b

```

```

        band_report[b] = {"n": int(m.sum()), "frac": float(m.mean()),
                          "tb_rate": float(y[m].mean()) if m.sum() else None}

lo, hi = bootstrap_auc(y, p_ens)
metrics = {
    "split": args.split, "n": int(len(y)), "pos": int(y.sum()),
    "aucroc": float(roc_auc_score(y, p_ens)), "aucroc_ci95": [lo, hi],
    "sens_at_spec_90": sens_at_spec(y, p_ens, 0.90),
    "sens_at_spec_70": sens_at_spec(y, p_ens, 0.70),
    "ece": ece(y, p_ens), "brier": float(brier_score_loss(y, p_ens)),
    "temp_scales": tscales, "tau_low": tau_low, "tau_high": tau_high,
    "bands": band_report,
    "deferral": {"rate": float((band == "ragu").mean()),
                 "tb_rate_on_deferred": float(y[band == "ragu"].mean()) if (band == "ragu").any() else None},
    "per_run_aucroc": {k: float(roc_auc_score(y, v)) for k, v in per_run_target.items()},
}

if args.meta and args.meta.exists():
    meta = {r["filename"]: r for r in csv.DictReader(args.meta.open())}
    for key in ("sex", "age"):
        groups: dict[str, list[int]] = {}
        for i, it in enumerate(items):
            row = meta.get(Path(it["path"]).name, {})
            g = row.get(key, "NA") or "NA"
            if key == "age":
                try:
                    a = float(g)
                    g = "0-30" if a < 30 else "30-50" if a < 50 else "50+"
                except ValueError:
                    g = "NA"
            groups.setdefault(g, []).append(i)
        metrics[f"aucroc_by_{key}"] = {
            g: (float(roc_auc_score(y[idx], p_ens[idx])) if len(np.unique(y[idx])) > 1 else None)
              for g, idx in groups.items() if g != "NA" and len(idx) >= 20
        }

(args.out / "metrics.json").write_text(json.dumps(metrics, indent=1))
# dump prediksi per-citra (untuk analisis lanjutan & stacking)
import csv as _csv
with (args.out / "preds.csv").open("w", newline="") as f:
    w = _csv.writer(f)
    w.writerow(["path", "label", "p_ens", "u_ens"])
    for it, p, u in zip(items, p_ens, u_ens):
        w.writerow([it["path"], it["label"], float(p), float(u)])
reliability_plot(y, p_ens, args.out / "reliability.png", f"Kalibrasi - {args.split}")
curves = {f"ens ({args.split})": (y, p_ens)}
if len(per_run_target) > 1:
    curves.update({k: (y, v) for k, v in per_run_target.items()})
roc_plot(curves, args.out / "roc.png")

print("\n=== METRICS ===")
for k, v in metrics.items():
    if isinstance(v, (int, float, str)):
        print(f"{k:18s} {v}")
print(f"bands          {band_report}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_ssl_diag/loop-tb/ml/export_onnx.py

```
#!/usr/bin/env python3
"""Export checkpoint ke ONNX (opsi 17) + verifikasi onnxruntime.

Contoh:
python ml/export_onnx.py --ckpt ml/runs/efficientnet_b0/best.pt --out artifacts/tb_efficientnet_b0.onnx
"""
from __future__ import annotations

import argparse
from pathlib import Path

import numpy as np
import torch

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm")

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__)
    ap.add_argument("--ckpt", type=Path, required=True)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--opset", type=int, default=17)
    args = ap.parse_args()

    ckpt = torch.load(args.ckpt, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()

    img_size = ckpt["config"].get("img_size", 224)
    dummy = torch.randn(1, 3, img_size, img_size)
    args.out.parent.mkdir(parents=True, exist_ok=True)
    torch.onnx.export(
        model, dummy, args.out,
        input_names=["input"], output_names=["logit"],
        dynamic_axes={"input": {0: "batch"}, "logit": {0: "batch"}},
        opset_version=args.opset, dynamo=False, # legacy: satu file mandiri, opset stabil
    )

    import onnxruntime as ort
    sess = ort.InferenceSession(str(args.out), providers=["CPUExecutionProvider"])
    ref = model(dummy).detach().numpy()
    got = sess.run(None, {"input": dummy.numpy()})[0]
    # Verifikasi di ruang probabilitas (deployment-relevant); logit bisa sangat besar
    # pada model overkonfiden sehingga toleransi absolut di ruang logit tidak adil.
    sig = lambda z: 1 / (1 + np.exp(-z)) # noqa: E731
    prob_diff = float(np.abs(sig(ref) - sig(got)).max())
    size_mb = args.out.stat().st_size / 1e6
    print(f"ONNX ok: {args.out} ({size_mb:.1f} MB) max|Δprob|={prob_diff:.2e} opset={args.opset}")
    if prob_diff > 1e-3:
        print(f"WARN: perbedaan probabilitas {prob_diff:.4f} > 1e-3 – periksa sebelum produksi")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_ssl_diag/loop-tb/ml/kaggle/hello/hello.py

```
"""Kernel uji: verifikasi GPU, internet, versi lib, dan struktur dataset ter-mount."""
import os
import sys

lines: list[str] = []

def log(*a) -> None:
    print(*a, flush=True)
    lines.append(" ".join(str(x) for x in a))

log("python:", sys.version.split()[0])

try:
    import torch

    log("torch:", torch.__version__, "| cuda:", torch.cuda.is_available())
    if torch.cuda.is_available():
        log("gpu:", torch.cuda.get_device_name(0))
except Exception as e: # noqa: BLE001
    log("torch error:", e)

for mod in ("timm", "sklearn", "onnx", "onnxruntime"):
    try:
        m = __import__(mod)
        log(f"{mod}: {getattr(m, '__version__', '?')}")
    except Exception as e: # noqa: BLE001
        log(f"{mod}: MISSING ({e})")

log("--- input mounts ---")
for root, dirs, _ in os.walk("/kaggle/input"):
    log(root, "dirs:", dirs[:6])
    if root.count("/") > 3:
        dirs[:] = []

log("HELLO DONE")
with open("/kaggle/working/hello_result.txt", "w") as f:
    f.write("\n".join(lines) + "\n")
```

ml/kaggle/out_ssl_diag/loop-tb/ml/kaggle/make_nlm_dataset.py

```
#!/usr/bin/env python3
"""Siapkan dataset Kaggle privat: NLM CXR di-resize (max dim 512) + ClinicalReadings.

Dipisah dari data asli agar unggahan ringan (~150 MB) dan resolusi tetap cukup
untuk encoder SSL (DINOv2/RAD-DINO resize ke 518).
"""
from __future__ import annotations

import shutil
from pathlib import Path

from PIL import Image

SRC = Path("data/raw/nlm")
DST = Path("ml/kaggle/nlm_dataset/nlm")

def main() -> None:
    n = 0
    for corpus in ("shenzhen", "montgomery"):
        out_dir = DST / corpus / "CXR_png"
        out_dir.mkdir(parents=True, exist_ok=True)
        for p in sorted((SRC / corpus / "CXR_png").glob("*.png")):
            with Image.open(p) as im:
                im = im.convert("L")
                w, h = im.size
                scale = 512 / max(w, h)
                if scale < 1:
                    im = im.resize((max(1, round(w * scale)), max(1, round(h * scale))), Image.LANCZOS)
                im.save(out_dir / (p.stem + ".jpg"), quality=92)
            n += 1
        readings = SRC / corpus / "ClinicalReadings"
        if readings.exists():
            out_r = DST / corpus / "ClinicalReadings"
            out_r.mkdir(parents=True, exist_ok=True)
            for f in readings.glob("*.txt"):
                shutil.copy(f, out_r / f.name)
        print(f"selesai: {n} citra -> {DST}")

if __name__ == "__main__":
    main()
```

ml/kaggle/out_train/loop-tb/api/main.py

```
"""Loop-TB API – inference + Grad-CAM + band triase.

Jalankan lokal:
    uvicorn api.main:app --host 0.0.0.0 --port 8000

Env:
    LOOPTB_CKPT path ke checkpoint best.pt (default ml/runs/convnext_tiny/best.pt)
    LOOPTB_BANDS path ke bands.json hasil evaluate.py (opsional; default  $\tau=0.5/0.5$ )
"""
from __future__ import annotations

import base64
import io
import json
import os
from pathlib import Path

import matplotlib

matplotlib.use("Agg")
import matplotlib.cm as cm
import numpy as np
import timm
import torch
import torch.nn as nn
from fastapi import FastAPI, File, HTTPException, UploadFile
from fastapi.middleware.cors import CORSMiddleware
from PIL import Image
from torchvision import transforms as T

CKPT_PATH = Path(os.environ.get("LOOPTB_CKPT", "ml/runs/convnext_tiny/best.pt"))
BANDS_PATH = Path(os.environ.get("LOOPTB_BANDS", "")) if os.environ.get("LOOPTB_BANDS") else None
MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

app = FastAPI(title="loop-tb", version="0.1.0")
app.add_middleware(
    CORSMiddleware, allow_origins=["*"], allow_credentials=False,
    allow_methods=["*"], allow_headers=["*"],
)

_state: dict = {}

def _load() -> None:
    if not CKPT_PATH.exists():
        raise RuntimeError(f"Checkpoint tidak ditemukan: {CKPT_PATH}")
    ckpt = torch.load(CKPT_PATH, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()
    bands = {"tau_high": 0.5, "tau_low": 0.5}
    if BANDS_PATH and BANDS_PATH.exists():
        b = json.loads(BANDS_PATH.read_text())
        bands = {"tau_high": float(b.get("tau_high", 0.5)), "tau_low": float(b.get("tau_low", 0.5))}
    size = int(ckpt["config"].get("img_size", 224))
    _state.update(model=model, arch=ckpt["arch"], img_size=size, bands=bands)
    print(f"[loop-tb] loaded {ckpt['arch']} @ {size}px bands={bands}")

_tx = None

def preprocess(img: Image.Image) -> torch.Tensor:
    global _tx
    if _tx is None:
        s = _state["img_size"]
        _tx = T.Compose([T.Resize(int(s * 1.14)), T.CenterCrop(s), T.ToTensor(), T.Normalize(MEAN, STD)])
    return _tx(img.convert("RGB")).unsqueeze(0)

def band_of(p: float, bands: dict) -> str:
    if p >= bands["tau_high"]:
        return "rujuk_prioritas"
    if p < bands["tau_low"]:
        return "negatif_skrining"
    return "ragu"

def gradcam(model: nn.Module, x: torch.Tensor) -> np.ndarray:
    """Grad-CAM generik via forward_features (timm), tanpa asumsi arsitektur."""
    model.zero_grad(set_to_none=True)
    feats = model.forward_features(x)
    feats.retain_grad()
    logit = model.forward_head(feats).squeeze(1)
    logit.backward()
    w = feats.grad.mean(dim=(2, 3), keepdim=True)
    cam = (w * feats).sum(dim=1, keepdim=True).clamp(min=0)
    cam = nn.functional.interpolate(cam, size=x.shape[-2:], mode="bilinear", align_corners=False)
```

```

cam = cam / (cam.amax(dim=(2, 3), keepdim=True) + 1e-8)
return cam[0, 0].detach().numpy()

def overlay_png(img: Image.Image, cam: np.ndarray) -> str:
    img_np = np.asarray(img.convert("RGB").resize((cam.shape[1], cam.shape[0])), dtype=np.float32) / 255.0
    heat = cm.jet(cam)[..., :3]
    alpha = (cam[..., None] * 0.45)
    blend = img_np * (1 - alpha) + heat.astype(np.float32) * alpha
    out = Image.fromarray((np.clip(blend, 0, 1) * 255).astype(np.uint8))
    buf = io.BytesIO()
    out.save(buf, format="PNG")
    return base64.b64encode(buf.getvalue()).decode()

@app.on_event("startup")
def startup() -> None:
    _load()

@app.get("/health")
def health() -> dict:
    return {
        "status": "ok",
        "arch": _state.get("arch"),
        "img_size": _state.get("img_size"),
        "bands": _state.get("bands"),
    }

@app.post("/predict")
async def predict(file: UploadFile = File(...)) -> dict:
    if "model" not in _state:
        raise HTTPException(503, "model belum siap")
    raw = await file.read()
    try:
        img = Image.open(io.BytesIO(raw))
        img.load()
    except Exception as e:
        raise HTTPException(400, f"bukan citra valid: {e}")

    x = preprocess(img)
    with torch.no_grad():
        logit = _state["model"](x).squeeze(1)
    p = float(torch.sigmoid(logit).item())
    band = band_of(p, _state["bands"])

    cam = gradcam(_state["model"], x)
    heatmap_b64 = overlay_png(img, cam)

    return {
        "p_tb": round(p, 4),
        "band": band,
        "bands": _state["bands"],
        "arch": _state.get("arch"),
        "disclaimer": "Alat triase skrining – bukan diagnosis. Wajib konfirmasi (GeneXpert) oleh tenaga kesehatan.",
        "heatmap_png_b64": heatmap_b64,
    }

```

ml/kaggle/out_train/loop-tb/ml/data/prepare_data.py

```
#!/usr/bin/env python3
"""Build a provenance-aware manifest for TB chest X-ray training.

Sumber & aturan label (lihat docs/02-arsitektur.md):
- TBX11K (Kaggle: vbookshelf/tbx11k-simplified): folder per kelas.
  positif = {tb}; negatif = {healthy, sick}; latent_tb DIKELUARKAN (ambigu).
- Rahman composite (Kaggle: tawsifurrahman/tuberculosis-tb-chest-xray-dataset):
  folder Normal/ -> 0, Tuberculosis/ -> 1.
- NLM Shenzhen/Montgomery: konvensi nama file resmi `*_0` = normal, `*_1` = TB.

Dedup (WAJIB – mitigasi class-conditional acquisition confounding):
1. pHash 64-bit; Hamming <= --dup-threshold dianggap duplikat (union-find transitif).
2. Integritas external: citra train yang sekelompok dengan citra external DIBUANG dari train.
3. Duplikat internal train dibuang (keep-first, urutan deterministik).
4. Split val dilakukan per-klaster (tidak ada kebocoran near-dup train->val).

Output (ml/data/artifacts/):
- manifest.json, dedup_report.json, train.csv, val.csv, external.csv
"""
from __future__ import annotations

import argparse
import csv
import json
import re
from collections import Counter
from pathlib import Path

import numpy as np
from PIL import Image
from tqdm import tqdm

try:
    import imagehash
except ImportError: # pragma: no cover
    raise SystemExit("Butuh imagehash: pip install imagehash")

IMG_EXTS = {".png", ".jpg", ".jpeg", ".bmp", ".tif", ".tiff"}
POS_TOKENS = {"tb", "tuberculosis", "tuberculose"}
NEG_TOKENS = {"healthy", "normal", "sick"}
EXCLUDE_TOKENS = {"latent", "latent_tb", "latenttb"}

def _rel(p: Path, root: Path) -> str:
    """Path relatif terhadap root repo (portable lokal <-> Colab); fallback absolut."""
    try:
        return str(p.relative_to(root))
    except ValueError:
        return str(p)

def _tokens(path: Path) -> set[str]:
    parts = [p.lower() for p in path.parts]
    parts += [path.stem.lower()]
    return set(parts)

def load_tbx11k_map(raw: Path) -> dict[str, int | None]:
    """Label otoritatif dari data.csv TBX11K: 1=active_tb, 0=no_tb, None=latent (dikeluarkan)."""
    m: dict[str, int | None] = {}
    for p in raw.rglob("data.csv"):
        if "tbx11k" not in str(p).lower():
            continue
        for r in csv.DictReader(p.open()):
            f = (r.get("fname") or "").strip()
            if not f:
                continue
            t = (r.get("tb_type") or "").lower()
            tgt = (r.get("target") or "").lower()
            if "latent" in t:
                m.setdefault(f, None)
            elif tgt == "tb":
                m[f] = 1
            else:
                m.setdefault(f, 0)
    return m

def classify(path: Path, raw: Path, tbx_map: dict[str, int | None]) -> tuple[int, str] | None:
    """Return (label, source) atau None kalau tidak bisa diklasifikasi."""
    rel = path.relative_to(raw)
    low = str(rel).lower()

    # --- NLM Shenzhen / Montgomery: konvensi suffix nama file ---
    m = re.search(r"(chncxr|mcucxr)_d+_(\d+)", path.stem.lower())
    if m:
        label = int(m.group(2))
        source = "shenzhen" if m.group(1) == "chncxr" else "montgomery"
```

```

        return label, source

toks = _tokens(rel)
if toks & EXCLUDE_TOKENS or any(t.startswith("latent") for t in toks):
    return None

if "tbx11k" in low:
    if tbx_map:
        lab = tbx_map.get(path.name, None)
        return None if lab is None else (lab, "tbx11k")
    # fallback bila data.csv tidak ada: prefix nama file
    stem = path.stem.lower()
    if stem.startswith("t"):
        return 1, "tbx11k"
    if stem.startswith(("h", "s")):
        return 0, "tbx11k"
    return None

if toks & POS_TOKENS:
    return 1, "rahman"
if toks & NEG_TOKENS:
    return 0, "rahman"
return None

class DSU:
    def __init__(self, n: int):
        self.parent = list(range(n))

    def find(self, x: int) -> int:
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def union(self, a: int, b: int) -> None:
        ra, rb = self.find(a), self.find(b)
        if ra != rb:
            self.parent[rb] = ra

_POP = np.array([bin(i).count("1") for i in range(256)], dtype=np.uint8)

def hamming_pairs(hashes: np.ndarray, threshold: int, chunk: int = 512) -> list[tuple[int, int]]:
    """Semua pasangan (i<j) dengan Hamming distance <= threshold (chunked, hemat memori)."""
    n = len(hashes)
    h8 = hashes.view(np.uint8).reshape(n, 8)
    pairs: list[tuple[int, int]] = []
    for start in range(0, n, chunk):
        stop = min(start + chunk, n)
        xor = np.bitwise_xor(h8[start:stop, None, :], h8[None, :, :])
        dist = _POP[xor].sum(axis=-1) # (chunk, n)
        rows, cols = np.where(dist <= threshold)
        for r, c in zip(rows, cols):
            i, j = start + int(r), int(c)
            if i < j:
                pairs.append((i, j))
    return pairs

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--raw", type=Path, default=Path("data/raw"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts"))
    ap.add_argument("--dup-threshold", type=int, default=3, help="Hamming pHash <= ini dianggap duplikat")
    ap.add_argument("--val-frac", type=float, default=0.15)
    ap.add_argument("--seed", type=int, default=42)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    raw = args.raw.resolve()
    root = raw.parent.parent # asumsi struktur <root>/data/raw

    # ----- 1) discovery -----
    labeled: list[dict] = []
    unlabeled = Counter()
    tbx_map = load_tbx11k_map(raw)
    print(f"[discovery] label TBX11K dari data.csv: {len(tbx_map)} entri")
    for p in sorted(raw.rglob("*")):
        if p.suffix.lower() not in IMG_EXTS:
            continue
        cls = classify(p, raw, tbx_map)
        if cls is None:
            unlabeled[str(p.relative_to(raw).parent)] += 1
            continue
        label, source = cls
        labeled.append({"path": _rel(p, root), "label": label, "source": source})

    print(f"[discovery] berlabel={len(labeled)} tak-terklasifikasi={sum(unlabeled.values())}")
    for k, v in unlabeled.most_common(10):
        print(f" skip {v:6d} {k}")

```

```

if not labeled:
    raise SystemExit("Tidak ada citra berlabel – cek struktur data/raw")

# ----- 2) pHash + kluster duplikat -----
print("[dedup] menghitung pHash...")
ok: list[dict] = []
for item in tqdm(labeled, unit="img"):
    try:
        with Image.open(item["path"]) as im:
            h = imagehash.phash(im.convert("L"), hash_size=8)
            item["phash"] = int(str(h), 16)
            ok.append(item)
    except Exception as e: # gambar korup
        print(f"warn: gagal baca {item['path']}: {e}")
labeled = ok

hashes = np.array([it["phash"] for it in labeled], dtype=np.uint64)
pairs = hamming_pairs(hashes, args.dup_threshold)
dsu = DSU(len(labeled))
for i, j in pairs:
    dsu.union(i, j)
for idx, it in enumerate(labeled):
    it["cluster"] = dsu.find(idx)

by_cluster: dict[int, list[dict]] = {}
for it in labeled:
    by_cluster.setdefault(it["cluster"], []).append(it)

# ----- 3) lock external, buang kebocoran dari train -----
ext = [it for it in labeled if it["source"] in ("shenzhen", "montgomery")]
ext_clusters = {it["cluster"] for it in ext}
train_pool, dropped_cross = [], []
for it in labeled:
    if it["source"] in ("shenzhen", "montgomery"):
        continue
    (dropped_cross if it["cluster"] in ext_clusters else train_pool).append(it)

# duplikat internal train: keep-first per kluster (deterministik via path)
keep, dropped_internal = [], []
seen_clusters: set[int] = set()
for it in sorted(train_pool, key=lambda x: x["path"]):
    if it["cluster"] in seen_clusters:
        dropped_internal.append(it)
    else:
        seen_clusters.add(it["cluster"])
        keep.append(it)
train_pool = keep

# ----- 4) split val per-kluster -----
rng = np.random.default_rng(args.seed)
sizes = Counter(it["cluster"] for it in train_pool)
pool_clusters = sorted(sizes)
rng.shuffle(pool_clusters)
target_val = int(len(train_pool) * args.val_frac)
val_clusters: set[int] = set()
acc = 0
for c in pool_clusters:
    if acc >= target_val:
        break
    val_clusters.add(c)
    acc += sizes[c]
train = [it for it in train_pool if it["cluster"] not in val_clusters]
val = [it for it in train_pool if it["cluster"] in val_clusters]

def dump(items: list[dict]) -> list[dict]:
    return [
        {k: it[k] for k in ("path", "label", "source", "cluster", "phash")}
        for it in sorted(items, key=lambda x: x["path"])
    ]

manifest = {
    "meta": {
        "root": str(root),
        "raw": _rel(raw, root),
        "dup_threshold": args.dup_threshold,
        "val_frac": args.val_frac,
        "seed": args.seed,
        "counts": {
            "train": len(train), "val": len(val), "external": len(ext),
            "dropped_cross_corpus": len(dropped_cross),
            "dropped_internal_dup": len(dropped_internal),
            "unlabeled": sum(unlabeled.values()),
        },
    },
    "train": dump(train),
    "val": dump(val),
    "external": dump(ext),
}
(args.out / "manifest.json").write_text(json.dumps(manifest, indent=1))
(args.out / "dedup_report.json").write_text(json.dumps({
    "pairs_within_threshold": len(pairs),
    "clusters": len(by_cluster),
}))

```

```

        "dropped_cross_corpus": dump(dropped_cross),
        "dropped_internal_dup": dump(dropped_internal),
        "unlabeled_by_dir": dict(unlabeled),
    }, indent=1))

for split in ("train", "val", "external"):
    with (args.out / f"{split}.csv").open("w") as f:
        f.write("path,label,source,cluster\n")
        for it in manifest[split]:
            f.write(f"{it['path']},{it['label']},{it['source']},{it['cluster']}\n")

def comp(items):
    c = Counter((it["source"], it["label"]) for it in items)
    return " ".join(f"{s}/{ 'TB' if l else 'NEG' }={n}" for (s, l), n in sorted(c.items()))

print("\n=== RINGKASAN ===")
print(f"train      n={len(train):6d}  {comp(train)}")
print(f"val        n={len(val):6d}    {comp(val)}")
print(f"external  n={len(ext):6d}    {comp(ext)}")
print(f"dibuang  cross-corpus={len(dropped_cross)}  internal-dup={len(dropped_internal)}")
print(f"manifest -> {args.out/'manifest.json'}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_train/loop-tb/ml/data/prepare_meta.py

```
#!/usr/bin/env python3
"""Ekstrak metadata klinis NLM (Montgomery + Shenzhen) → CSV untuk analisis subgroup.

Output: ml/data/artifacts/meta_nlm.csv
Kolom: filename,sex,age,reading,scar_flag
- Montgomery: "Patient's Sex: F / Patient's Age: 027Y / <reading>"
- Shenzhen: "male 45yrs / <reading>" (kadang terkontaminasi antar-file)
- scar_flag: menyebut scar/inactive/fibrosis → untuk pembahasan mode gagal
            "healed scar false positive" (selaras audit Bilal 2026).
"""
from __future__ import annotations

import argparse
import csv
import re
from pathlib import Path

SEX_M = re.compile(r"sex[:\s]+(m|male)\b", re.I)
SEX_F = re.compile(r"sex[:\s]+(f|female)\b", re.I)
SEX_FMT2 = re.compile(r"\b(male|female)\b", re.I)
AGE_MONT = re.compile(r"age[:\s]+(\d{1,3})\s*Y?\b", re.I)
AGE_FMT2 = re.compile(r"\b(\d{1,3})\s*yrs\b", re.I)
SCAR = re.compile(r"scar|inactive|fibro", re.I)

def parse_montgomery(text: str) -> dict | None:
    age = AGE_MONT.search(text)
    sex = "F" if SEX_F.search(text) else ("M" if SEX_M.search(text) else "")
    if not age:
        return None
    # reading = baris setelah Age
    lines = [ln.strip() for ln in text.splitlines() if ln.strip()]
    reading = lines[2] if len(lines) > 2 else ""
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": int(bool(SCAR.search(reading)))}

def parse_shenzhen(text: str) -> dict | None:
    age = AGE_FMT2.search(text)
    sex_m = SEX_FMT2.findall(text)
    sex = sex_m[0][0].upper() if sex_m else ""
    if not age:
        return None
    # reading = token huruf setelah "yrs"
    tail = text[age.end():]
    reading = ""
    m = re.search(r"[A-Za-z]{2,}", tail)
    if m:
        reading = m.group(0)
    return {"sex": sex, "age": int(age.group(1)), "reading": reading, "scar_flag": 0}

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--nlm", type=Path, default=Path("data/raw/nlm"))
    ap.add_argument("--out", type=Path, default=Path("ml/data/artifacts/meta_nlm.csv"))
    args = ap.parse_args()
    args.out.parent.mkdir(parents=True, exist_ok=True)

    rows, n_fail = [], 0
    for corpus, parser in (("montgomery", parse_montgomery), ("shenzhen", parse_shenzhen)):
        d = args.nlm / corpus / "ClinicalReadings"
        for p in sorted(d.glob("*.txt")):
            r = parser(p.read_text(errors="ignore"))
            if r is None:
                n_fail += 1
                continue
            r["filename"] = p.stem + ".png"
            r["corpus"] = corpus
            rows.append(r)

    with args.out.open("w", newline="") as f:
        w = csv.DictWriter(f, fieldnames=["filename", "corpus", "sex", "age", "reading", "scar_flag"])
        w.writeheader()
        w.writerows(rows)

    from collections import Counter
    print(f"meta -> {args.out} n={len(rows)} gagal-parse={n_fail}")
    print("per korpus:", Counter(r["corpus"] for r in rows))
    print("sex:", Counter(r["sex"] for r in rows))
    print("scar_flag:", Counter(r["scar_flag"] for r in rows))

if __name__ == "__main__":
    main()
```

ml/kaggle/out_train/loop-tb/ml/evaluate.py

```
#!/usr/bin/env python3
"""Evaluasi rigor: kalibrasi, CI bootstrap, transfer, band triase, subgroup.

Contoh (ensemble + temperature scaling):
python ml/evaluate.py --runs ml/runs/efficientnet_b0 ml/runs/convnext_tiny \
    --manifest ml/data/artifacts/manifest.json --split external --temp-scale --out ml/runs/eval_external
"""
from __future__ import annotations

import argparse
import csv
import json
from pathlib import Path

import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import brier_score_loss, roc_auc_score, roc_curve
from torch.utils.data import DataLoader
from tqdm import tqdm

import sys
sys.path.insert(0, str(Path(__file__).resolve().parent))
from train import CXRManifest # noqa: E402

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

def sigmoid(x: np.ndarray) -> np.ndarray:
    return 1 / (1 + np.exp(-x))

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 -> fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

def threshold_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    fpr, tpr, thr = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(thr[ok][np.argmax(tpr[ok])]) if ok.any() else 0.5

def ece(y: np.ndarray, p: np.ndarray, bins: int = 15) -> float:
    edges = np.linspace(0, 1, bins + 1)
    total = 0.0
    for i in range(bins):
        m = (p >= edges[i]) & ((p < edges[i + 1]) if i < bins - 1 else (p <= edges[i + 1]))
        if m.sum():
            total += m.mean() * abs(p[m].mean() - y[m].mean())
    return float(total)

def bootstrap_auroc(y: np.ndarray, p: np.ndarray, n: int = 1000, seed: int = 42) -> tuple[float, float]:
    rng = np.random.default_rng(seed)
    vals = []
    for _ in range(n):
        idx = rng.integers(0, len(y), len(y))
        if len(np.unique(y[idx])) > 1:
            vals.append(roc_auc_score(y[idx], p[idx]))
    return float(np.percentile(vals, 2.5)), float(np.percentile(vals, 97.5))

@torch.no_grad()
def infer(ckpt_path: Path, items: list[dict], img_size: int, device: torch.device, workers: int) -> np.ndarray:
    ckpt = torch.load(ckpt_path, map_location="cpu", weights_only=False)
    arch = ckpt["arch"]
    model = timm.create_model(arch, pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.to(device).eval()
    dl = DataLoader(CXRManifest(items, train=False, img_size=img_size), batch_size=64,
                    shuffle=False, num_workers=workers)

    out = []
    for x, _, _ in tqdm(dl, desc=f"infer {ckpt_path.parent.name}", leave=False):
        x = x.to(device)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            out.append(model(x).squeeze(1).float().cpu().numpy())
    return np.concatenate(out)
```

```

def fit_temperature(val_logits: list[np.ndarray], val_y: np.ndarray) -> float:
    """Cari T>0 yang meminimalkan NLL pada val (grid search halus)."""
    logits = torch.tensor(np.mean(val_logits, axis=0), dtype=torch.float32)
    y = torch.tensor(val_y, dtype=torch.float32)
    best_t, best_nll = 1.0, float("inf")
    for t in np.linspace(0.25, 10.0, 200):
        nll = nn.functional.binary_cross_entropy_with_logits(logits / t, y).item()
        if nll < best_nll:
            best_t, best_nll = float(t), nll
    return best_t

def reliability_plot(y: np.ndarray, p: np.ndarray, path: Path, title: str) -> None:
    edges = np.linspace(0, 1, 11)
    mids, accs = [], []
    for i in range(10):
        m = (p >= edges[i]) & (p < edges[i + 1] if i < 9 else p <= edges[i + 1])
        if m.sum():
            mids.append(p[m].mean())
            accs.append(y[m].mean())
    plt.figure(figsize=(4.2, 4.2))
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.plot(mids, accs, "o-", color="#d62728", lw=2)
    plt.xlabel("Confidence"), plt.ylabel("Empirical TB rate"), plt.title(title)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def roc_plot(curves: dict[str, tuple[np.ndarray, np.ndarray]], path: Path) -> None:
    plt.figure(figsize=(5, 5))
    for name, (y, p) in curves.items():
        fpr, tpr, _ = roc_curve(y, p)
        plt.plot(fpr, tpr, lw=2, label=f"{name} (AUROC {roc_auc_score(y, p):.3f})")
    plt.plot([0, 1], [0, 1], "--", color="#999", lw=1)
    plt.xlabel("FPR"), plt.ylabel("TPR (sensitivitas)"), plt.legend(fontsize=8)
    plt.tight_layout(), plt.savefig(path, dpi=160), plt.close()

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--runs", type=Path, nargs="+", required=True)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--split", default="external", choices=["val", "external"])
    ap.add_argument("--temp-scale", action="store_true")
    ap.add_argument("--meta", type=Path, default=None, help="CSV filename,sex,age untuk subgroup (opsional)")
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--workers", type=int, default=8)
    args = ap.parse_args()

    args.out.mkdir(parents=True, exist_ok=True)
    manifest = json.loads(args.manifest.read_text())
    items = manifest[args.split]
    y = np.array([it["label"] for it in items])

    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    per_run_target, per_run_val, tscales = {}, {}, {}
    for run in args.runs:
        ckpt = torch.load(run / "best.pt", map_location="cpu", weights_only=False)
        img_size = ckpt["config"]["img_size"]
        per_run_target[run.name] = infer(run / "best.pt", items, img_size, device, args.workers)
        val_items = manifest["val"]
        per_run_val[run.name] = infer(run / "best.pt", val_items, img_size, device, args.workers)
        if args.temp_scale:
            tscales[run.name] = fit_temperature([per_run_val[run.name]], np.array([it["label"] for it in val_items]))

    per_run_target = {k: sigmoid(v) for k, v in per_run_target.items()}
    per_run_val = {k: sigmoid(v) for k, v in per_run_val.items()}
    val_y = np.array([it["label"] for it in manifest["val"]])

    # kalibrasi per-run pada val, lalu terapkan T ke target (bukan val-fit pada target!)
    if args.temp_scale:
        for k in per_run_target:
            per_run_val[k] = sigmoid(np.log(np.clip(per_run_val[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_val[k], 1e-6, 1 - 1e-6)))
            / tscales[k])
            lg = np.log(np.clip(per_run_target[k], 1e-6, 1 - 1e-6) / (1 - np.clip(per_run_target[k], 1e-6, 1 - 1e-6)))
            per_run_target[k] = sigmoid(lg / tscales[k])

    # ensemble = rata-rata; ketidakpastian = std antar anggota
    stack_t = np.stack(list(per_run_target.values()))
    p_ens = stack_t.mean(axis=0)
    u_ens = stack_t.std(axis=0)
    stack_v = np.stack(list(per_run_val.values()))
    p_ens_v = stack_v.mean(axis=0)

    tau_high = threshold_at_spec(val_y, p_ens_v, 0.70) # spec>=0.70 (TPP WHO)
    tau_low = threshold_at_spec(val_y, p_ens_v, 0.95)
    if tau_low > tau_high:
        tau_low, tau_high = tau_high, tau_low

    band = np.where(p_ens >= tau_high, "rujuk_prioritas", np.where(p_ens < tau_low, "negatif_skrining", "ragu"))
    band_report = {}
    for b in ("rujuk_prioritas", "ragu", "negatif_skrining"):
        m = band == b

```

```

        band_report[b] = {"n": int(m.sum()), "frac": float(m.mean()),
                          "tb_rate": float(y[m].mean()) if m.sum() else None}

lo, hi = bootstrap_aucroc(y, p_ens)
metrics = {
    "split": args.split, "n": int(len(y)), "pos": int(y.sum()),
    "aucroc": float(roc_auc_score(y, p_ens)), "aucroc_ci95": [lo, hi],
    "sens_at_spec_90": sens_at_spec(y, p_ens, 0.90),
    "sens_at_spec_70": sens_at_spec(y, p_ens, 0.70),
    "ece": ece(y, p_ens), "brier": float(brier_score_loss(y, p_ens)),
    "temp_scales": tscales, "tau_low": tau_low, "tau_high": tau_high,
    "bands": band_report,
    "deferral": {"rate": float((band == "ragu").mean()),
                  "error_on_deferred": float(1 - (band[band == "ragu"] == y[band == "ragu"]).mean()) if (band == "ragu").any()
                  else None},
    "per_run_aucroc": {k: float(roc_auc_score(y, v)) for k, v in per_run_target.items()},
}

if args.meta and args.meta.exists():
    meta = {r["filename"]: r for r in csv.DictReader(args.meta.open())}
    for key in ("sex", "age"):
        groups: dict[str, list[int]] = {}
        for i, it in enumerate(items):
            row = meta.get(Path(it["path"]).name, {})
            g = row.get(key, "NA") or "NA"
            if key == "age":
                try:
                    a = float(g)
                    g = "0-30" if a < 30 else "30-50" if a < 50 else "50+"
                except ValueError:
                    g = "NA"
            groups.setdefault(g, []).append(i)
        metrics[f"auroc_by_{key}"] = {
            g: (float(roc_auc_score(y[idx], p_ens[idx])) if len(np.unique(y[idx])) > 1 else None)
              for g, idx in groups.items() if g != "NA" and len(idx) >= 20
        }

(args.out / "metrics.json").write_text(json.dumps(metrics, indent=1))
reliability_plot(y, p_ens, args.out / "reliability.png", f"Kalibrasi - {args.split}")
curves = {f"ens ({args.split})": (y, p_ens)}
if len(per_run_target) > 1:
    curves.update({k: (y, v) for k, v in per_run_target.items()})
roc_plot(curves, args.out / "roc.png")

print("\n=== METRICS ===")
for k, v in metrics.items():
    if isinstance(v, (int, float, str)):
        print(f"{k:18s} {v}")
print(f"bands          {band_report}")

if __name__ == "__main__":
    main()

```

ml/kaggle/out_train/loop-tb/ml/export_onnx.py

```
#!/usr/bin/env python3
"""Export checkpoint ke ONNX (opsi 17) + verifikasi onnxruntime.

Contoh:
python ml/export_onnx.py --ckpt ml/runs/efficientnet_b0/best.pt --out artifacts/tb_efficientnet_b0.onnx
"""
from __future__ import annotations

import argparse
from pathlib import Path

import numpy as np
import torch

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm")

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__)
    ap.add_argument("--ckpt", type=Path, required=True)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--opset", type=int, default=17)
    args = ap.parse_args()

    ckpt = torch.load(args.ckpt, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    model.eval()

    img_size = ckpt["config"].get("img_size", 224)
    dummy = torch.randn(1, 3, img_size, img_size)
    args.out.parent.mkdir(parents=True, exist_ok=True)
    torch.onnx.export(
        model, dummy, args.out,
        input_names=["input"], output_names=["logit"],
        dynamic_axes={"input": {0: "batch"}, "logit": {0: "batch"}},
        opset_version=args.opset, dynamo=False, # legacy: satu file mandiri, opset stabil
    )

    import onnxruntime as ort
    sess = ort.InferenceSession(str(args.out), providers=["CPUExecutionProvider"])
    ref = model(dummy).detach().numpy()
    got = sess.run(None, {"input": dummy.numpy()})[0]
    max_diff = float(np.abs(ref - got).max())
    ok = bool(np.allclose(ref, got, rtol=1e-3, atol=1e-3))
    size_mb = args.out.stat().st_size / 1e6
    print(f"ONNX ok: {args.out} ({size_mb:.1f} MB) max|diff|={max_diff:.2e} opset={args.opset}")
    assert ok, "Verifikasi ONNX gagal (allclose rtol/atol 1e-3)"

if __name__ == "__main__":
    main()
```

ml/kaggle/out_train/loop-tb/ml/kaggle/hello/hello.py

```
"""Kernel uji: verifikasi GPU, internet, versi lib, dan struktur dataset ter-mount."""
import os
import sys

lines: list[str] = []

def log(*a) -> None:
    print(*a, flush=True)
    lines.append(" ".join(str(x) for x in a))

log("python:", sys.version.split()[0])

try:
    import torch

    log("torch:", torch.__version__, "| cuda:", torch.cuda.is_available())
    if torch.cuda.is_available():
        log("gpu:", torch.cuda.get_device_name(0))
except Exception as e: # noqa: BLE001
    log("torch error:", e)

for mod in ("timm", "sklearn", "onnx", "onnxruntime"):
    try:
        m = __import__(mod)
        log(f"{mod}: {getattr(m, '__version__', '?')}")
    except Exception as e: # noqa: BLE001
        log(f"{mod}: MISSING ({e})")

log("--- input mounts ---")
for root, dirs, _ in os.walk("/kaggle/input"):
    log(root, "dirs:", dirs[:6])
    if root.count("/") > 3:
        dirs[:] = []

log("HELLO DONE")
with open("/kaggle/working/hello_result.txt", "w") as f:
    f.write("\n".join(lines) + "\n")
```

ml/kaggle/out_train/loop-tb/ml/kaggle/train/train_kernel.py

```
"""Kernel training loop-tb: clone repo → symlink dataset → train 2 arsitektur + baseline → export ONNX.

Output di /kaggle/working/loop-tb/{runs,artifacts}; diunduh via `kaggle kernels output`.
Manifest yang dipakai = split v1 frozen dari repo (deterministik; NLM external tidak dibutuhkan di sini).
"""
import json
import os
import subprocess
import sys

WORK = "/kaggle/working"
REPO = f"{WORK}/loop-tb"

def sh(cmd: list[str], **kw) -> None:
    print("+", " ".join(map(str, cmd)), flush=True)
    subprocess.run(cmd, check=True, **kw)

# 1) repo + deps
if not os.path.exists(REPO):
    sh(["git", "clone", "--depth", "1", "https://github.com/Alvnc/loop-tb.git", REPO])
os.chdir(REPO)
sh([sys.executable, "-m", "pip", "install", "-q", "timm", "onnxruntime"])

# 2) symlink dataset ke struktur yang diharapkan manifest (tahan variasi struktur mount)
os.makedirs("data/raw", exist_ok=True)

def pick(dst: str, cands: list[str], markers: tuple[str, ...]) -> None:
    """Symlink kandidat mount pertama yang punya 'marker' (images/Normal/Tuberculosis)."""
    if os.path.exists(dst):
        return
    for c in cands:
        if any(os.path.isdir(os.path.join(c, m)) for m in markers):
            os.symlink(c, dst)
            print("symlink", dst, "->", c, flush=True)
            return
    for c in cands:
        if os.path.exists(c):
            os.symlink(c, dst)
            print("symlink (fallback)", dst, "->", c, flush=True)
            return
    print("WARNING: tidak ada kandidat mount untuk", dst, flush=True)

TBX_CANDS = [
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified",
]
RAH_CANDS = [
    "/kaggle/input/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
    "/kaggle/input/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
]
pick("data/raw/tbx11k-simplified", TBX_CANDS, ("images",))
pick("data/raw/TB_Chest_Radiography_Database", RAH_CANDS, ("Normal", "Tuberculosis"))

for root, dirs, files in os.walk("/kaggle/input"):
    print("mount:", root, "dirs:", dirs[:6], "files:", len(files), flush=True)
    if root.count("/") > 4:
        dirs[:] = []

# 3) sanity: semua path train/val harus ada
manifest = json.load(open("ml/data/artifacts/manifest.json"))
missing = [it["path"] for s in ("train", "val") for it in manifest[s] if not os.path.exists(it["path"])]
print(f"train={len(manifest['train'])} val={len(manifest['val'])} missing={len(missing)}", flush=True)
print("missing contoh:", missing[:3], flush=True)
assert not missing, "data train/val tidak lengkap – cek mount dataset"

# 4) training (dua arsitektur + baseline fitur beku)
JOBS = [
    ("efficientnet_b0", "runs/eb0", ["--epochs", "14", "--batch", "64"]),
    ("convnext_tiny", "runs/convnext", ["--epochs", "12", "--batch", "48"]),
]
for arch, out, extra in JOBS:
    sh(
        [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
         "--arch", arch, "--balanced", "--out", out, *extra]
    )

sh(
    [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
     "--arch", "efficientnet_b0", "--freeze", "--lr", "1e-3", "--epochs", "8", "--batch", "64",
     "--out", "runs/linear_probe"]
)
```

```
# 5) export ONNX (satu file mandiri per model)
os.makedirs("artifacts", exist_ok=True)
for run, name in (("runs/eb0", "eb0"), ("runs/convnext", "convnext")):
    sh([sys.executable, "ml/export_onnx.py", "--ckpt", f"{run}/best.pt", "--out", f"artifacts/{name}.onnx"])

print("ALL DONE", flush=True)
```

ml/kaggle/out_train/loop-tb/ml/train.py

```
#!/usr/bin/env python3
"""Train TB CXR classifier dari manifest provenance-aware.

Contoh:
python ml/train.py --manifest ml/data/artifacts/manifest.json --arch efficientnet_b0 \
    --epochs 20 --out ml/runs/efficientnet_b0

Baseline linear probe (fitur beku):
python ml/train.py ... --freeze --lr 1e-3 --out ml/runs/linear_probe
"""
from __future__ import annotations

import argparse
import csv
import json
import time
from pathlib import Path

import numpy as np
import torch
import torch.nn as nn
from sklearn.metrics import roc_auc_score, roc_curve
from torch.utils.data import DataLoader, Dataset, WeightedRandomSampler
from torchvision import transforms as T
from tqdm import tqdm

try:
    import timm
except ImportError: # pragma: no cover
    raise SystemExit("Butuh timm: pip install timm")

MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

class CXRManifest(Dataset):
    def __init__(self, items: list[dict], train: bool, img_size: int):
        self.items = items
        if train:
            self.tx = T.Compose([
                T.RandomResizedCrop(img_size, scale=(0.75, 1.0), ratio=(0.9, 1.1)),
                T.RandomRotation(7, fill=0),
                T.ColorJitter(brightness=0.15, contrast=0.15),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])
        else:
            self.tx = T.Compose([
                T.Resize(int(img_size * 1.14)),
                T.CenterCrop(img_size),
                T.ToTensor(),
                T.Normalize(MEAN, STD),
            ])

    def __len__(self) -> int:
        return len(self.items)

    def __getitem__(self, i: int):
        from PIL import Image
        it = self.items[i]
        with Image.open(it["path"]) as im:
            x = self.tx(im.convert("RGB"))
        return x, torch.tensor(float(it["label"])), i

def sens_at_spec(y: np.ndarray, p: np.ndarray, spec: float) -> float:
    """Sensitivitas pada spesifisitas >= spec (mis. spec=0.90 → fpr <= 0.10)."""
    fpr, tpr, _ = roc_curve(y, p)
    ok = fpr <= 1 - spec + 1e-9
    return float(tpr[ok].max()) if ok.any() else float("nan")

@torch.no_grad()
def predict(model: nn.Module, loader: DataLoader, device: torch.device) -> tuple[np.ndarray, np.ndarray]:
    model.eval()
    logits_all, labels_all = [], []
    for x, y, _ in loader:
        x = x.to(device, non_blocking=True)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            logits = model(x).squeeze(1).float()
        logits_all.append(logits.cpu().numpy())
        labels_all.append(y.numpy())
    return np.concatenate(logits_all), np.concatenate(labels_all)

def evaluate(logits: np.ndarray, y: np.ndarray) -> dict:
    p = 1 / (1 + np.exp(-logits))
    out = {"n": int(len(y)), "pos": int(y.sum())}
    if len(np.unique(y)) > 1:
```

```

        out["auroc"] = float(roc_auc_score(y, p))
        out["sens_at_spec_90"] = sens_at_spec(y, p, 0.90)
        out["sens_at_spec_70"] = sens_at_spec(y, p, 0.70)
    return out

```

```

def main() -> None:
    ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("--manifest", type=Path, required=True)
    ap.add_argument("--arch", default="efficientnet_b0")
    ap.add_argument("--epochs", type=int, default=20)
    ap.add_argument("--batch", type=int, default=32)
    ap.add_argument("--lr", type=float, default=3e-4)
    ap.add_argument("--wd", type=float, default=1e-4)
    ap.add_argument("--img-size", type=int, default=224)
    ap.add_argument("--out", type=Path, required=True)
    ap.add_argument("--seed", type=int, default=42)
    ap.add_argument("--workers", type=int, default=8)
    ap.add_argument("--freeze", action="store_true", help="linear probe: backbone beku")
    ap.add_argument("--balanced", action="store_true", help="WeightedRandomSampler")
    ap.add_argument("--limit", type=int, default=0, help="smoke test: batasi n sampel (0=semua)")
    args = ap.parse_args()

    torch.manual_seed(args.seed)
    np.random.seed(args.seed)
    args.out.mkdir(parents=True, exist_ok=True)

    device = torch.device("cuda" if torch.cuda.is_available() else "mps" if torch.backends.mps.is_available() else "cpu")
    manifest = json.loads(args.manifest.read_text())

    def take(items: list[dict], n: int) -> list[dict]:
        """Subset berstratifikasi untuk smoke test."""
        if not n:
            return items
        pos = [it for it in items if it["label"] == 1]
        neg = [it for it in items if it["label"] == 0]
        n = min(n, len(items))
        kp = min(len(pos), max(1, n // 4))
        return pos[:kp] + neg[: max(n - kp, 0)]

    train_items = take(manifest["train"], args.limit)
    val_items = take(manifest["val"], args.limit)

    ds_tr = CXRMManifest(train_items, train=True, img_size=args.img_size)
    ds_va = CXRMManifest(val_items, train=False, img_size=args.img_size)
    sampler = None
    if args.balanced:
        labels = np.array([it["label"] for it in train_items])
        w = np.where(labels == 1, 1.0 / labels.sum(), 1.0 / (len(labels) - labels.sum()))
        sampler = WeightedRandomSampler(torch.from_numpy(w), num_samples=len(labels), replacement=True)
    dl_tr = DataLoader(ds_tr, batch_size=args.batch, shuffle=sampler is None, sampler=sampler,
                      num_workers=args.workers, pin_memory=(device.type == "cuda"), drop_last=True)
    dl_va = DataLoader(ds_va, batch_size=args.batch, shuffle=False, num_workers=args.workers, pin_memory=(device.type == "cuda"))

    model = timm.create_model(args.arch, pretrained=True, num_classes=1).to(device)
    if args.freeze:
        for n, p in model.named_parameters():
            p.requires_grad = "head" in n or "classifier" in n or "fc" in n
    params = [p for p in model.parameters() if p.requires_grad]
    opt = torch.optim.AdamW(params, lr=args.lr, weight_decay=args.wd)
    sched = torch.optim.lr_scheduler.CosineAnnealingLR(opt, T_max=max(args.epochs, 1))
    loss_fn = nn.BCEWithLogitsLoss()
    scaler = torch.amp.GradScaler(enabled=(device.type == "cuda"))

    print(f"device={device} arch={args.arch} freeze={args.freeze} "
          f"train={len(ds_tr)} val={len(ds_va)} params={sum(p.numel() for p in params)/1e6:.2f}M")

    hist = []
    best = {"auroc": -1.0, "epoch": -1}
    for epoch in range(1, args.epochs + 1):
        model.train()
        t0, losses = time.time(), []
        for x, y, _ in tqdm(dl_tr, desc=f"epoch {epoch:02d}", leave=False):
            x, y = x.to(device, non_blocking=True), y.to(device, non_blocking=True)
            opt.zero_grad(set_to_none=True)
            with torch.autocast(device.type, enabled=(device.type == "cuda")):
                logits = model(x).squeeze(1)
                loss = loss_fn(logits.float(), y)
            scaler.scale(loss).backward()
            scaler.step(opt)
            scaler.update()
            losses.append(loss.item())
        sched.step()

        logits, y_va = predict(model, dl_va, device)
        m = evaluate(logits, y_va)
        row = {"epoch": epoch, "loss": float(np.mean(losses)), "lr": opt.param_groups[0]["lr"],
              "sec": round(time.time() - t0, 1), **m}
        hist.append(row)
    print(f"epoch {epoch:02d} | loss {row['loss']:.4f} | AUROC {m.get('auroc', float('nan')):.4f} "
          f"| sens@spec90 {m.get('sens_at_spec_90', float('nan')):.3f} | {row['sec']}s")

```

```

torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
           "epoch": epoch, "val": m}, args.out / "last.pt")
if m.get("auroc", -1) > best["auroc"]:
    best = {"auroc": m["auroc"], "epoch": epoch}
    torch.save({"arch": args.arch, "state_dict": model.state_dict(), "config": vars(args),
               "epoch": epoch, "val": m}, args.out / "best.pt")
    with (args.out / "val_preds.csv").open("w", newline="") as f:
        w = csv.writer(f)
        w.writerow(["path", "label", "logit"])
        for it, lg in zip(val_items, logits):
            w.writerow([it["path"], it["label"], float(lg)])

(args.out / "history.csv").write_text(
    "epoch,loss,lr,sec,n,pos,auroc,sens_at_spec_90,sens_at_spec_70\n" + "\n".join(
        ", ".join(str(r.get(k, "")) for k in ("epoch", "loss", "lr", "sec", "n", "pos", "auroc", "sens_at_spec_90",
        "sens_at_spec_70"))
        for r in hist) + "\n")
(args.out / "metrics.json").write_text(json.dumps({"best": best, "history": hist, "config": vars(args)}, indent=1, default=str))
print(f"\nBEST epoch {best['epoch']} AUROC {best['auroc']:.4f} -> {args.out/'best.pt'}")

if __name__ == "__main__":
    main()

```

ml/kaggle/ssl/ssl_extract.py

```
"""Ekstraksi embedding SSL untuk studi representasi (ADR-007).

Encoder:
- dinov2 : timm vit_base_patch14_dinov2.lvd142m (SSL umum, 518 px)
- rad_dino: hf-hub:microsoft/rad-dino (DINOv2 pada X-ray dada, 518 px)

Output di /kaggle/working/loop-tb/ssl_embeddings/:
meta_{train,val,external}.json + emb_{encoder}_{split}.npy
Dipakai oleh ml/ssl_probe.py secara lokal (sklearn, tanpa GPU).
"""
from __future__ import annotations

import json
import os
import subprocess
import sys

import numpy as np

WORK = "/kaggle/working"
REPO = f"{WORK}/loop-tb"

def sh(cmd: list[str]) -> None:
    print("+", " ".join(map(str, cmd)), flush=True)
    subprocess.run(cmd, check=True)

if not os.path.exists(REPO):
    sh(["git", "clone", "--depth", "1", "https://github.com/Alvnc/loop-tb.git", REPO])
os.chdir(REPO)
sh([sys.executable, "-m", "pip", "install", "-q", "timm", "huggingface_hub", "transformers"])

# --- mount datasets (self-healing) ---
os.makedirs("data/raw", exist_ok=True)

def pick(dst: str, cands: list[str], markers: tuple[str, ...]) -> None:
    if os.path.exists(dst):
        return
    for c in cands:
        if any(os.path.isdir(os.path.join(c, m)) for m in markers):
            os.symlink(c, dst)
            print("symlink", dst, "->", c, flush=True)
            return
    for c in cands:
        if os.path.exists(c):
            os.symlink(c, dst)
            print("symlink (fallback)", dst, "->", c, flush=True)
            return
    print("WARNING: tidak ada mount untuk", dst, flush=True)

pick("data/raw/tbx11k-simplified",
     ["/kaggle/input/datasets/vbookshelf/tbx11k-simplified/tbx11k-simplified",
      "/kaggle/input/datasets/vbookshelf/tbx11k-simplified", "/kaggle/input/tbx11k-simplified"],
     ("images",))
pick("data/raw/TB_Chest_Radiography_Database",
     ["/kaggle/input/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
      "/kaggle/input/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database"],
     ("Normal", "Tuberculosis"))
pick("data/raw/nlm",
     ["/kaggle/input/datasets/alvinreba/nlm-tb-cxr-resized/nlm",
      "/kaggle/input/datasets/alvinreba/nlm-tb-cxr-resized",
      "/kaggle/input/nlm-tb-cxr-resized/nlm",
      "/kaggle/input/nlm-tb-cxr-resized"],
     ("shenzhen", "montgomery"))

# --- siapkan daftar item (external: .png -> .jpg hasil resize) ---
manifest = json.load(open("ml/data/artifacts/manifest.json"))
train_items = manifest["train"]
val_items = manifest["val"]
ext_items = []
for it in manifest["external"]:
    p = it["path"]
    jp = p[:-4] + ".jpg" if p.lower().endswith(".png") else p
    ext_items.append({**it, "path": jp})

for name, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
    missing = [it["path"] for it in items if not os.path.exists(it["path"])]
    print(f"{name}: n={len(items)} missing={len(missing)}", missing[:2], flush=True)
    assert not missing, f"data {name} tidak lengkap"

out_dir = "ssl_embeddings"
os.makedirs(out_dir, exist_ok=True)
for split, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
    with open(f"{out_dir}/meta_{split}.json", "w") as f:
        json.dump([{"path": it["path"], "label": it["label"], "source": it["source"]} for it in items], f)
```

```

# --- ekstraksi ---
import torch # noqa: E402
from PIL import Image # noqa: E402
from torch.utils.data import DataLoader, Dataset # noqa: E402
from torchvision import transforms as T # noqa: E402

MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

class Items(Dataset):
    def __init__(self, items: list[dict], size: int, mean: tuple = MEAN, std: tuple = STD):
        self.items = items
        self.tx = T.Compose([T.Resize(size), T.CenterCrop(size), T.ToTensor(), T.Normalize(mean, std)])

    def __len__(self) -> int:
        return len(self.items)

    def __getitem__(self, i: int):
        with Image.open(self.items[i]["path"]) as im:
            x = self.tx(im.convert("RGB"))
        return x, i

@torch.no_grad()
def extract(model, items: list[dict], size: int, device: torch.device, tag: str,
            mean: tuple = MEAN, std: tuple = STD) -> np.ndarray:
    dl = DataLoader(Items(items, size, mean, std), batch_size=32, shuffle=False, num_workers=4, pin_memory=True)
    out = np.zeros((len(items), 0), dtype=np.float32)
    embs = []
    for bi, (x, _) in enumerate(dl):
        x = x.to(device, non_blocking=True)
        with torch.autocast(device.type, enabled=(device.type == "cuda")):
            feats = model(x)
        feats = feats.float().cpu().numpy()
        embs.append(feats)
        if bi % 20 == 0:
            print(f" {tag} batch {bi}/{len(dl)}", flush=True)
    out = np.concatenate(embs, axis=0)
    return out

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("device:", device, flush=True)

ENCODERS = [
    ("dinov2", "vit_base_patch14_dinov2.lvd142m", 518),
    ("rad_dino", "hf-hub:microsoft/rad-dino", 518),
]

import timm # noqa: E402

for enc_name, model_id, size in ENCODERS:
    if all(os.path.exists(f"{out_dir}/emb_{enc_name}_{s}.npy") for s in ("train", "val", "external")):
        print(f"skip {enc_name} (sudah ada)", flush=True)
        continue
    try:
        print(f"=== encoder {enc_name} ({model_id}) ===", flush=True)
        mean, std = MEAN, STD
        if enc_name == "rad_dino":
            from transformers import AutoImageProcessor, AutoModel

            hf_id = model_id.replace("hf-hub:", "")

            class HFWrap(torch.nn.Module):
                def __init__(self, m):
                    super().__init__()
                    self.m = m

                def forward(self, x):
                    out = self.m(pixel_values=x)
                    return out.last_hidden_state[:, 0] if hasattr(out, "last_hidden_state") else out.pooler_output

            proc = AutoImageProcessor.from_pretrained(hf_id)
            print("rad_dino processor:", proc.image_mean, proc.image_std, proc.size, flush=True)
            if proc.image_mean:
                mean, std = tuple(proc.image_mean), tuple(proc.image_std)
            model = HFWrap(AutoModel.from_pretrained(hf_id))
        else:
            model = timm.create_model(model_id, pretrained=True, num_classes=0, dynamic_img_size=True)
        model.to(device).eval()
        for split, items in (("train", train_items), ("val", val_items), ("external", ext_items)):
            emb = extract(model, items, size, device, f"{enc_name}/{split}", mean, std)
            np.save(f"{out_dir}/emb_{enc_name}_{split}.npy", emb)
            print(f"{enc_name} {split}: {emb.shape}", flush=True)
        with open(f"{out_dir}/encmeta_{enc_name}.json", "w") as f:
            json.dump({
                "type": "hf" if enc_name == "rad_dino" else "timm",
                "id": model_id.replace("hf-hub:", ""),
                "size": size,
                "mean": list(mean),
                "std": list(std),
            }, f)
    except Exception as e:
        print(f"Error processing {enc_name} {model_id}: {e}")

```

```
    }, f)
    del model
    torch.cuda.empty_cache()
except Exception as e: # noqa: BLE001
    print(f"ERROR encoder {enc_name}: {e}", flush=True)

print("SSL EXTRACT DONE", flush=True)
```

ml/kaggle/train/train_kernel.py

```
"""Kernel training loop-tb: clone repo → symlink dataset → train 2 arsitektur + baseline → export ONNX.

Output di /kaggle/working/loop-tb/{runs,artifacts}; diunduh via `kaggle kernels output`.
Manifest yang dipakai = split v1 frozen dari repo (deterministik; NLM external tidak dibutuhkan di sini).
"""
import json
import os
import subprocess
import sys

WORK = "/kaggle/working"
REPO = f"{WORK}/loop-tb"

def sh(cmd: list[str], **kw) -> None:
    print("+", " ".join(map(str, cmd)), flush=True)
    subprocess.run(cmd, check=True, **kw)

# 1) repo + deps
if not os.path.exists(REPO):
    sh(["git", "clone", "--depth", "1", "https://github.com/Alvnc/loop-tb.git", REPO])
os.chdir(REPO)
sh([sys.executable, "-m", "pip", "install", "-q", "timm", "onnxruntime"])

# 2) symlink dataset ke struktur yang diharapkan manifest (tahan variasi struktur mount)
os.makedirs("data/raw", exist_ok=True)

def pick(dst: str, cands: list[str], markers: tuple[str, ...]) -> None:
    """Symlink kandidat mount pertama yang punya 'marker' (images/Normal/Tuberculosis)."""
    if os.path.exists(dst):
        return
    for c in cands:
        if any(os.path.isdir(os.path.join(c, m)) for m in markers):
            os.symlink(c, dst)
            print("symlink", dst, "->", c, flush=True)
            return
    for c in cands:
        if os.path.exists(c):
            os.symlink(c, dst)
            print("symlink (fallback)", dst, "->", c, flush=True)
            return
    print("WARNING: tidak ada kandidat mount untuk", dst, flush=True)

TBX_CANDS = [
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/datasets/vbookshelf/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified/tbx11k-simplified",
    "/kaggle/input/tbx11k-simplified",
]
RAH_CANDS = [
    "/kaggle/input/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
    "/kaggle/input/tuberculosis-tb-chest-xray-dataset/TB_Chest_Radiography_Database",
]
pick("data/raw/tbx11k-simplified", TBX_CANDS, ("images",))
pick("data/raw/TB_Chest_Radiography_Database", RAH_CANDS, ("Normal", "Tuberculosis"))

for root, dirs, files in os.walk("/kaggle/input"):
    print("mount:", root, "dirs:", dirs[:6], "files:", len(files), flush=True)
    if root.count("/") > 4:
        dirs[:] = []

# 3) sanity: semua path train/val harus ada
manifest = json.load(open("ml/data/artifacts/manifest.json"))
missing = [it["path"] for s in ("train", "val") for it in manifest[s] if not os.path.exists(it["path"])]
print(f"train={len(manifest['train'])} val={len(manifest['val'])} missing={len(missing)}", flush=True)
print("missing contoh:", missing[:3], flush=True)
assert not missing, "data train/val tidak lengkap – cek mount dataset"

# 4) training (dua arsitektur + baseline fitur beku)
JOBS = [
    ("efficientnet_b0", "runs/eb0", ["--epochs", "14", "--batch", "64"]),
    ("convnext_tiny", "runs/convnext", ["--epochs", "12", "--batch", "48"]),
]
for arch, out, extra in JOBS:
    sh(
        [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
         "--arch", arch, "--balanced", "--out", out, *extra]
    )

sh(
    [sys.executable, "ml/train.py", "--manifest", "ml/data/artifacts/manifest.json",
     "--arch", "efficientnet_b0", "--freeze", "--lr", "1e-3", "--epochs", "8", "--batch", "64",
     "--out", "runs/linear_probe"]
)
```

```
# 5) export ONNX (satu file mandiri per model)
os.makedirs("artifacts", exist_ok=True)
for run, name in (("runs/eb0", "eb0"), ("runs/convnext", "convnext")):
    sh([sys.executable, "ml/export_onnx.py", "--ckpt", f"{run}/best.pt", "--out", f"artifacts/{name}.onnx"])

print("ALL DONE", flush=True)
```

api/main.py

```
"""SIGAP API – inference ensemble (supervised ckpt &/atau probe SSL) + Grad-CAM + band triase.

Jalankan lokal:
    uvicorn api.main:app --host 0.0.0.0 --port 8000

Konfigurasi (default `api/model_config.json`, override env LOOPTB_CONFIG):
{
    "score_members": [ {"path": "ml/runs/ssl_eval/probe_dinov2.npz"} ],
    "uncertainty_members": [ {"path": "ml/runs/eb0/best.pt", "T": 1.2336},
                             {"path": "ml/runs/convnext/best.pt", "T": 1.3405} ],
    "bands": {"tau_low": ..., "tau_high": ..., "tau_uncertainty": 0.2}
}

Skor p = rata-rata prob anggota `score_members` (model terbaik untuk AUROC eksternal).
Ketidakpastian u = std antar SEMUA anggota unik (disagreement lintas-keluarga SSL-supervised).
Jika hanya ada satu anggota total, u = 0.
"""
from __future__ import annotations

import base64
import io
import json
import os
from pathlib import Path

import matplotlib

matplotlib.use("Agg")
import matplotlib.cm as cm
import numpy as np
import timm
import torch
import torch.nn as nn
from fastapi import FastAPI, File, HTTPException, UploadFile
from fastapi.middleware.cors import CORSMiddleware
from PIL import Image
from torchvision import transforms as T

CONFIG_PATH = Path(os.environ.get("LOOPTB_CONFIG", "api/model_config.json"))
MEAN, STD = (0.485, 0.456, 0.406), (0.229, 0.224, 0.225)

app = FastAPI(title="loop-tb (SIGAP)", version="0.4.0")
app.add_middleware(
    CORSMiddleware, allow_origins=["*"], allow_credentials=False,
    allow_methods=["*"], allow_headers=["*"],
)

_state: dict = {}

def sigmoid(z: float) -> float:
    return float(1.0 / (1.0 + np.exp(-np.clip(z, -60, 60))))

class HFViT(nn.Module):
    """Pembungkus model HF (mis. RAD-DINO) → CLS token sebagai fitur."""
    def __init__(self, model_id: str):
        super().__init__()
        from transformers import AutoModel

        self.m = AutoModel.from_pretrained(model_id)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        out = self.m(pixel_values=x)
        return out.last_hidden_state[:, 0] if hasattr(out, "last_hidden_state") else out.pooler_output

def _load_member(spec: dict) -> dict:
    path = spec["path"]
    if path.endswith(".npz"):
        d = np.load(path, allow_pickle=False)
        enc_type, enc_id = str(d["encoder_type"][0]), str(d["encoder_id"][0])
        model: nn.Module = HFViT(enc_id) if enc_type == "hf" else timm.create_model(
            enc_id, pretrained=True, num_classes=0, dynamic_img_size=True
        )
    return {
        "kind": "probe",
        "model": model.eval(),
        "T": float(d["T"][0]),
        "size": int(d["size"][0]),
        "pixel_mean": tuple(float(v) for v in d["pixel_mean"]),
        "pixel_std": tuple(float(v) for v in d["pixel_std"]),
        "scaler_mean": d["scaler_mean"].astype(np.float32),
        "scaler_scale": d["scaler_scale"].astype(np.float32),
        "coef": d["coef"].astype(np.float32),
        "intercept": float(d["intercept"][0]),
        "path": path,
    }
```

```

    }
    ckpt = torch.load(path, map_location="cpu", weights_only=False)
    model = timm.create_model(ckpt["arch"], pretrained=False, num_classes=1)
    model.load_state_dict(ckpt["state_dict"])
    return {
        "kind": "torch",
        "model": model.eval(),
        "T": float(spec.get("T", 1.0)),
        "size": int(ckpt["config"].get("img_size", 224)),
        "path": path,
    }

def _load() -> None:
    cfg = json.loads(CONFIG_PATH.read_text())
    score_specs = cfg.get("score_members") or cfg.get("members") or []
    unc_specs = cfg.get("uncertainty_members") or score_specs
    cache: dict[str, dict] = {}

    def get(spec: dict) -> dict:
        p = spec["path"]
        if p not in cache:
            cache[p] = _load_member(spec)
        return cache[p]

    _state.update(
        score_members=[get(s) for s in score_specs],
        unc_members=[get(s) for s in unc_specs],
        bands={"tau_high": 0.5, "tau_low": 0.5, **cfg.get("bands", {})},
    )
    print(f"[SIGAP] score={len(_state['score_members'])} unc={len(_state['unc_members'])} bands={_state['bands']}")

def _names(members: list[dict]) -> list[str]:
    return [f"{Path(m['path']).parent.name}/{Path(m['path']).name}" for m in members]

def preprocess(img: Image.Image, size: int, mean=MEAN, std=STD) -> torch.Tensor:
    tx = T.Compose([T.Resize(int(size * 1.14)), T.CenterCrop(size), T.ToTensor(), T.Normalize(mean, std)])
    return tx(img.convert("RGB")).unsqueeze(0)

def band_of(p: float, u: float, bands: dict) -> str:
    if u > float(bands.get("tau_uncertainty", 1.0)):
        return "ragu"
    if p >= bands["tau_high"]:
        return "rujuk_prioritas"
    if p < bands["tau_low"]:
        return "negatif_skrining"
    return "ragu"

def infer_prob(member: dict, img: Image.Image) -> float:
    if member["kind"] == "probe":
        x = preprocess(img, member["size"], member.get("pixel_mean", MEAN), member.get("pixel_std", STD))
        with torch.no_grad():
            feats = member["model"](x).float().numpy()[0]
            z = float(((feats - member["scaler_mean"]) / member["scaler_scale"]) @ member["coef"] + member["intercept"])
    else:
        x = preprocess(img, member["size"])
        with torch.no_grad():
            z = float(member["model"](x).squeeze(1).item())
    return sigmoid(z / member["T"])

def _cam_from_tokens(grads: torch.Tensor, acts: torch.Tensor, out_hw: tuple[int, int]) -> torch.Tensor | None:
    g, a = grads[1:], acts[1:]
    n = g.shape[0]
    side = int(round(n ** 0.5))
    if side * side != n:
        return None
    g = g.reshape(side, side, -1)
    a = a.reshape(side, side, -1)
    cam = (g * a).sum(-1).clamp(min=0)[None, None]
    return nn.functional.interpolate(cam, size=out_hw, mode="bilinear", align_corners=False)[0, 0]

def gradcam(member: dict, img: Image.Image) -> np.ndarray | None:
    model = member["model"]
    if member["kind"] == "torch":
        x = preprocess(img, member["size"])
        model.zero_grad(set_to_none=True)
        feats = model.forward_features(x)
        feats.retain_grad()
        model.forward_head(feats).squeeze(1).backward()
        grads, acts = feats.grad, feats
        if grads.dim() == 4:
            w = grads.mean(dim=(2, 3), keepdim=True)
            cam = (w * acts).sum(1, keepdim=True).clamp(min=0)
            cam = nn.functional.interpolate(cam, size=x.shape[-2:], mode="bilinear", align_corners=False)[0, 0]
        else:
            cam = _cam_from_tokens(grads[0], acts[0], tuple(x.shape[-2:]))

```

```

else:
    mean, std = member.get("pixel_mean", MEAN), member.get("pixel_std", STD)
    x = preprocess(img, member["size"], mean, std)
    model.zero_grad(set_to_none=True)
    if hasattr(model, "forward_features"):
        feats = model.forward_features(x)
        pooled = model.forward_head(feats)
    else:
        feats = model.m(pixel_values=x).last_hidden_state
        pooled = feats[:, 0]
    feats = feats if feats.dim() == 3 else feats.flatten(2).transpose(1, 2)
    feats.retain_grad()
    mt = torch.tensor(member["scaler_mean"])
    st = torch.tensor(member["scaler_scale"])
    cf = torch.tensor(member["coef"])
    z = ((pooled[0].float() - mt) / st) @ cf + member["intercept"]
    z.backward()
    cam = _cam_from_tokens(feats.grad[0], feats[0], tuple(x.shape[-2:]))
if cam is None:
    return None
cam = cam.detach()
return (cam / (cam.amax() + 1e-8)).numpy()

def overlay_png(img: Image.Image, cam: np.ndarray) -> str:
    img_np = np.asarray(img.convert("RGB").resize((cam.shape[1], cam.shape[0])), dtype=np.float32) / 255.0
    heat = cm.jet(cam)[..., :3].astype(np.float32)
    alpha = cam[..., None] * 0.45
    blend = img_np * (1 - alpha) + heat * alpha
    out = Image.fromarray((np.clip(blend, 0, 1) * 255).astype(np.uint8))
    buf = io.BytesIO()
    out.save(buf, format="PNG")
    return base64.b64encode(buf.getvalue()).decode()

@app.on_event("startup")
def startup() -> None:
    _load()

@app.get("/health")
def health() -> dict:
    return {
        "status": "ok" if _state.get("score_members") else "no-model",
        "score_members": _names(_state.get("score_members", [])),
        "uncertainty_members": _names(_state.get("unc_members", [])),
        "bands": _state.get("bands"),
    }

@app.post("/predict")
async def predict(file: UploadFile = File(...)) -> dict:
    if not _state.get("score_members"):
        raise HTTPException(503, "model belum siap")
    raw = await file.read()
    try:
        img = Image.open(io.BytesIO(raw))
        img.load()
    except Exception as e:
        raise HTTPException(400, f"bukan citra valid: {e}")

    score_probs = [infer_prob(m, img) for m in _state["score_members"]]
    extra = [m for m in _state["unc_members"] if m not in _state["score_members"]]
    all_probs = score_probs + [infer_prob(m, img) for m in extra]

    p = float(np.mean(score_probs))
    u = float(np.std(all_probs)) if len(all_probs) > 1 else 0.0
    band = band_of(p, u, _state["bands"])

    cam = gradcam(_state["score_members"][0], img)
    heatmap_b64 = overlay_png(img, cam) if cam is not None else None

    return {
        "p_tb": round(p, 4),
        "uncertainty": round(u, 4),
        "band": band,
        "bands": _state["bands"],
        "members": _names(_state["score_members"]),
        "uncertainty_members": _names(_state["unc_members"]),
        "disclaimer": "Alat triase skrining – bukan diagnosis. Wajib konfirmasi (GeneXpert) oleh tenaga kesehatan.",
        "heatmap_png_b64": heatmap_b64,
    }

```

api/model_config.json

```
{
  "score_members": [{ "path": "ml/runs/ssl_eval/probe_rad_dino.npz" }],
  "uncertainty_members": [
    { "path": "ml/runs/ssl_eval/probe_rad_dino.npz" },
    { "path": "ml/runs/ssl_eval/probe_dinov2.npz" },
    { "path": "ml/runs/eb0/best.pt", "T": 1.2336 },
    { "path": "ml/runs/convnext/best.pt", "T": 1.3405 }
  ],
  "bands": { "tau_low": 0.0233, "tau_high": 0.0977, "tau_uncertainty": 0.25 }
}
```

api/model_config_probe.json

```
{  
  "members": [{ "path": "ml/runs/ssl_eval/probe_dinov2.npz" }],  
  "bands": { "tau_low": 0.0346, "tau_high": 0.0863, "tau_uncertainty": 1.0 }  
}
```

api/model_config_space.json

```
{
  "score_members": [{ "path": "models/probe_rad_dino.npz" }],
  "uncertainty_members": [
    { "path": "models/probe_rad_dino.npz" },
    { "path": "models/probe_dinov2.npz" },
    { "path": "models/eb0.pt", "T": 1.2336 },
    { "path": "models/convnext.pt", "T": 1.3405 }
  ],
  "bands": { "tau_low": 0.0233, "tau_high": 0.0977, "tau_uncertainty": 0.25 }
}
```

scripts/fetch_models.sh

```
#!/usr/bin/env bash
# Unduh artefak model dari GitHub Release `models-v1` ke lokasi yang diharapkan
# api/model_config.json dan ml/evaluate.py (ml/runs/...).
set -euo pipefail
REPO="{REPO:-Alvnc/loop-tb}"
TAG="{TAG:-models-v1}"
BASE="https://github.com/$REPO/releases/download/$TAG"

mkdir -p ml/runs/eb0 ml/runs/convnext ml/runs/linear_probe ml/runs/ssl_eval

fetch() { # $1 = nama file, $2 = tujuan
  if [ -f "$2" ]; then echo "skip (ada): $2"; return; fi
  echo "unduh $1 -> $2"
  curl -fL --retry 3 -o "$2" "$BASE/$1"
}

fetch eb0.pt ml/runs/eb0/best.pt
fetch convnext.pt ml/runs/convnext/best.pt
fetch linear_probe.pt ml/runs/linear_probe/best.pt
fetch probe_rad_dino.npz ml/runs/ssl_eval/probe_rad_dino.npz
fetch probe_dinov2.npz ml/runs/ssl_eval/probe_dinov2.npz

echo "Model siap. Jalankan API:"
echo " .venv/bin/python -m uvicorn api.main:app --port 8000"
echo " (atau LOOPTB_CONFIG=api/model_config_probe.json untuk probe DINOv2 saja)"
```

tools/e2e_screen.py

```
#!/usr/bin/env python3
"""Uji integrasi UI+API: unggah citra X-ray lewat halaman /skrining, ambil screenshot hasil.

Jalankan dengan helper server (web 3210 + api 8000):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
--server "cd web && npm start -- --port 3210" --port 3210 \
--server ".venv/bin/python -m uvicorn api.main:app --port 8000" --port 8000 \
-- .venv/bin/python tools/e2e_screen.py
"""
from __future__ import annotations

import sys
from pathlib import Path

from playwright.sync_api import sync_playwright

OUT = Path("submission/screenshots")
OUT.mkdir(parents=True, exist_ok=True)

CASES = [
    ("data/raw/nlm/shenzhen/CXR_png/CHNCXR_0327_1.png", "tb", "S-014"),
    ("data/raw/nlm/shenzhen/CXR_png/CHNCXR_0001_0.png", "normal", "S-015"),
]

def run_case(page, img: str, tag: str, pid: str) -> None:
    page.goto("http://localhost:3210/skrining", wait_until="networkidle")
    page.set_input_files('input[type="file"]', img)
    page.fill('input[placeholder="mis. S-014"]', pid)
    page.fill('input[placeholder="34"]', "42")
    page.click("text=Analisis citra")
    page.wait_for_selector("text=Hasil analisis", timeout=120_000)
    page.wait_for_timeout(600)
    page.screenshot(path=str(OUT / f"skrining_hasil_{tag}_mobile.png"), full_page=True)
    verdict = page.locator("section >> text=Hasil analisis").first
    print(f"[ok] {tag}: {img}")

def main() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        ctx = browser.new_context(viewport={"width": 390, "height": 844}, device_scale_factor=2)
        page = ctx.new_page()
        console: list[str] = []
        page.on("console", lambda m: console.append(f"{m.type}: {m.text}") if m.type == "error" else None)
        for img, tag, pid in CASES:
            if not Path(img).exists():
                print("[skip]", img)
                continue
            run_case(page, img, tag, pid)
        if console:
            print("[console errors]", console[:6])
        ctx.close()
        browser.close()
    print("E2E SCREENSHOTS DONE")

if __name__ == "__main__":
    main()
```

tools/make_pdfs.py

```
#!/usr/bin/env python3
"""Hasilkan PDF submission: one-pager (dari HTML) + code PDF (dari repo).

Jalankan:
.venv/bin/python tools/make_pdfs.py
"""
from __future__ import annotations

import html
import sys
from pathlib import Path

from playwright.sync_api import sync_playwright

ROOT = Path(__file__).resolve().parent.parent
OUT = ROOT / "submission"

CODE_GLOBS = [
    "ml/*.py",
    "ml/data/*.py",
    "ml/kaggle/**/*.py",
    "api/*.py",
    "api/*.json",
    "scripts/*.sh",
    "tools/*.py",
    "web/src/**/*.ts",
    "web/src/**/*.tsx",
    "web/src/**/*.css",
    "web/*.json",
    "web/*.ts",
    "docs/*.md",
]

EXCLUDE_PARTS = {"node_modules", ".next", "__pycache__", ".venv"}

def collect_files() -> list[Path]:
    files: list[Path] = []
    seen: set[Path] = set()
    for g in CODE_GLOBS:
        for p in sorted(ROOT.glob(g)):
            if not p.is_file() or any(part in EXCLUDE_PARTS for part in p.parts):
                continue
            if p in seen or p.suffix == ".png":
                continue
            seen.add(p)
            files.append(p.relative_to(ROOT))
    return files

def build_code_html(files: list[Path]) -> str:
    parts = [
        """<!DOCTYPE html><html><head><meta charset="utf-8"><title>SIGAP – Code</title>

<style>
@page { size: A4; margin: 14mm 12mm; }
body { font-family: "DejaVu Sans Mono", monospace; font-size: 8.4px; color: #14181d; }
h1 { font-family: sans-serif; font-size: 20px; margin-bottom: 6px; }
h2 { font-family: sans-serif; font-size: 12px; margin: 14px 0 6px; border-bottom: 1px solid #ccc; padding-bottom: 3px; }
.filetree { font-size: 9px; line-height: 1.5; margin-bottom: 10px; }
pre { white-space: pre-wrap; word-break: break-word; border: 1px solid #ddd; border-radius: 4px; padding: 6px 8px; margin-bottom: 10px; }
.filehead { font-family: sans-serif; font-weight: bold; font-size: 10px; margin: 14px 0 4px; page-break-before: always; }
</style></head><body>""",
        "<h1>SIGAP – complete code listing</h1>",
        "<p class='filetree'><b>Repository:</b> github.com/Alvncnc/loop-tb – prototipe riset UnivaBio 2026.<br>"
        "<Berisi: pipeline data (audit dedup), training, studi representasi SSL, evaluasi rigor, API, dan UI.</p>",
        "<h2>Daftar file</h2><div class='filetree'>" + "<br>".join(html.escape(str(f)) for f in files) + "</div>",
    ]

    for f in files:
        text = (ROOT / f).read_text(errors="ignore")
        parts.append(f"<div class='filehead'>{html.escape(str(f))}</div>")
        parts.append(f"<pre>{html.escape(text)}</pre>")
    parts.append("</body></html>")
    return "".join(parts)

def main() -> None:
    OUT.mkdir(exist_ok=True)
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        page = browser.new_page()

        # 1) One-pager
        page.goto((OUT / "onepager.html").as_uri(), wait_until="networkidle")
        page.pdf(path=str(OUT / "onepager.pdf"), format="A4", print_background=True,
            margin={"top": "0", "right": "0", "bottom": "0", "left": "0"})
        print("onepager.pdf OK")

        # 2) Code PDF
```

```
files = collect_files()
code_html = build_code_html(files)
tmp = Path("/tmp/opencode/code_listing.html")
tmp.write_text(code_html)
page.goto(tmp.as_uri(), wait_until="load")
page.pdf(path=str(OUT / "code_listing.pdf"), format="A4", print_background=True)
print(f"code_listing.pdf OK - {len(files)} file")

browser.close()
```

```
if __name__ == "__main__":
    main()
```

tools/make_video.py

```
#!/usr/bin/env python3
"""Produksi video demo SIGAP secara otomatis.

Tahapan: render slide → rekam screencast (Playwright) → sintesis narasi (Piper TTS)
→ rakit dengan ffmpeg → submission/sigap_demo.mp4.

Jalankan lewat helper (web 3210 + api 8000):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
  --server "cd web && npm start -- --port 3210" --port 3210 \
  --server ".venv/bin/python -m uvicorn api.main:app --port 8000" --port 8000 \
  -- .venv/bin/python tools/make_video.py
"""
from __future__ import annotations

import subprocess
import wave
from pathlib import Path

from playwright.sync_api import sync_playwright

ROOT = Path(__file__).resolve().parent.parent
ASSETS = ROOT / "submission" / "video_assets"
ASSETS.mkdir(parents=True, exist_ok=True)
FINAL = ROOT / "submission" / "sigap_demo.mp4"
VOICE_MODEL = "/tmp/opcode/piper/en/en_US/lessac/medium/en_US-lessac-medium.onnx"
BASE = "http://localhost:3210"
W, H = 1920, 1080

SLIDE_CSS = """
<style>
@import url('https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght@400;600;800&display=swap');
* { margin:0; padding:0; box-sizing:border-box; }
body { width:1920px; height:1080px; background:#0b0f13; color:#e9e7e0;
      font-family:'Plus Jakarta Sans',sans-serif; display:flex; align-items:center; }
.pad { padding: 0 140px; width:100%; }
h1 { font-size:96px; font-weight:800; letter-spacing:-0.03em; line-height:1.04; }
p { font-size:38px; line-height:1.45; color:rgba(233,231,224,.85); margin-top:36px; max-width:1500px; }
.small { font-size:26px; color:rgba(233,231,224,.55); margin-top:44px; }
.accent { color:#ee8b7e; } .defer { color:#e5b95c; } .clear { color:#83c7a4; }
table { border-collapse:collapse; margin-top:40px; font-size:34px; }
td, th { text-align:left; padding:14px 28px 14px 0; border-bottom:1px solid rgba(233,231,224,.18); }
th { font-size:26px; color:rgba(233,231,224,.55); font-weight:600; }
.num { font-variant-numeric: tabular-nums; font-weight:800; }
</style>"""

SLIDES = {
    "slide_problem": f"""{SLIDE_CSS}<div class="pad">
    <h1>≈200,000 people a year<br>transmit TB <span class="accent">before ever being detected</span>.</h1>
    <p>Indonesia carries 10% of the world's TB burden. The care loop – detect, refer, treat, monitor –
    moves slower than the disease transmits.</p>
    <div class="small">Source: WHO Global TB Report 2025 · Kemenkes 2024</div></div>""",
    "slide_solution": f"""{SLIDE_CSS}<div class="pad">
    <h1>One calibrated risk state.<br>Three interfaces.</h1>
    <p><b>Screen</b> – CXR triage with a <span class="defer">deferral band</span>. &nbsp;<b>Refer</b> – action band + printable referral letter. &nbsp;<b>Support</b> – adherence loop with escalation.</p>
    <p>When model families disagree, SIGAP says <i>"uncertain – human re-read"</i> instead of forcing a verdict.</p></div>""",
    "slide_evidence": f"""{SLIDE_CSS}<div class="pad">
    <h1>We report the number most projects hide.</h1>
    <table>
    <tr><th>Evaluation</th><th>AUROC</th><th>sens@spec90</th></tr>
    <tr><td>Internal random split <i>(the illusion)</i></td><td class="num">0.98 – 0.9999</td><td class="num">–</td></tr>
    <tr><td>External hospitals · supervised fine-tune</td><td class="num">0.675</td><td class="num">0.27</td></tr>
    <tr><td><b>External · RAD-DINO self-supervised probe (final)</b></td><td class="num clear">0.887</td><td class="num
    clear">0.71</td></tr>
    </table>
    <div class="small">Data audit removed 338 duplicate images and 19 training images that leaked into the external test set.</div>
</div>""",
    "slide_close": f"""{SLIDE_CSS}<div class="pad">
    <h1>SIGAP – screen faster<br>than the disease spreads.</h1>
    <p>Research triage prototype – not a diagnostic device. WHO sensitivity target not yet met; reported as-is.
    Next: prospective validation, per-site calibration, cough audio.</p>
    <div class="small">Code, frozen splits &amp; evaluation: github.com/Alvncnc/loop-tb</div></div>""",
}

# (jenis, id, teks narasi)
SEGMENTS = [
    ("slide", "slide_problem",
     "Indonesia carries ten percent of the world's tuberculosis burden. Around two hundred thousand people "
     "each year transmit TB before ever being detected. The root cause is latency. The care loop moves "
     "slower than the disease."),
    ("slide", "slide_solution",
     "SIGAP compresses that latency with one calibrated risk state. It drives screening, referral, and "
     "treatment support. And when model families disagree, it says so, instead of forcing a decision."),
    ("clip", "clip_tb",
     "A health worker uploads a chest X-ray. For a real TB case, the model families agree: refer today, "
     "with a referral letter for GeneXpert confirmation. The attention map shows where the model looked."),
    ("clip", "clip_normal",
```

```

    "For a normal X-ray from an unseen hospital, the families disagree – so SIGAP answers: uncertain, "
    "human re-read. That is the deferral band doing its job on a case the model should not decide."),
    ("clip", "clip_pasien",
     "The same risk state follows the patient home. Missed doses and symptoms raise the score and trigger "
     "escalation, so the loop keeps moving until the patient is cured."),
    ("slide", "slide_evidence",
     "Here is the evidence, honestly. On an internal split the model looks perfect. On external hospitals, "
     "supervised fine-tuning falls to zero point six eight AUROC. Our final model, a self-supervised chest "
     "X-ray representation called RAD-DINO, holds zero point eight nine. We also removed nineteen training "
     "images that had leaked into the test set."),
    ("slide", "slide_close",
     "This is a research triage prototype, not a diagnostic device. The WHO sensitivity target is not met "
     "yet, and we report that. Next steps: prospective validation, per-site threshold calibration, and "
     "cough audio. SIGAP – screen faster than the disease spreads."),
]

def sh(cmd: list[str], **kw) -> None:
    subprocess.run(cmd, check=True, **kw)

def wav_duration(p: Path) -> float:
    with wave.open(str(p)) as w:
        return w.getnframes() / w.getframerate()

def render_slides() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        page = browser.new_page(viewport={"width": W, "height": H, "device_scale_factor": 1})
        for name, html in SLIDES.items():
            tmp = ASSETS / f"{name}.html"
            tmp.write_text(f"<!DOCTYPE html><html><head><meta charset='utf-8'></head><body>{html}</body></html>")
            page.goto(tmp.resolve().as_uri(), wait_until="networkidle")
            page.wait_for_timeout(400)
            page.screenshot(path=str(ASSETS / f"{name}.png"))
            print("slide:", name)
        browser.close()

def record_clips() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)

        def new_ctx(tag: str):
            return browser.new_context(
                viewport={"width": W, "height": H},
                record_video_dir=str(ASSETS / "raw"),
                record_video_size={"width": W, "height": H},
            )

        # --- clip TB ---
        ctx = new_ctx("tb")
        page = ctx.new_page()
        page.goto(f"{BASE}/skrining", wait_until="networkidle")
        page.wait_for_timeout(1200)
        page.set_input_files('input[type="file"]', str(ROOT / "data/raw/nlm/shenzhen/CXR_png/CHNCXR_0327_1.png"))
        page.fill('input[placeholder="mis. S-014"]', "S-014")
        page.wait_for_timeout(800)
        page.click("text=Analisis citra")
        page.wait_for_selector("text=Hasil analisis", timeout=120_000)
        page.wait_for_timeout(2500)
        page.click("text=Citra asli")
        page.wait_for_timeout(1400)
        page.click("text=Peta perhatian")
        page.wait_for_timeout(1200)
        page.mouse.wheel(0, 420)
        page.wait_for_timeout(1400)
        video = page.video
        ctx.close()
        video.save_as(ASSETS / "clip_tb.webm")
        print("clip_tb ok")

        # --- clip normal ---
        ctx = new_ctx("normal")
        page = ctx.new_page()
        page.goto(f"{BASE}/skrining", wait_until="networkidle")
        page.wait_for_timeout(900)
        page.set_input_files('input[type="file"]', str(ROOT / "data/raw/nlm/shenzhen/CXR_png/CHNCXR_0001_0.png"))
        page.fill('input[placeholder="mis. S-014"]', "S-015")
        page.click("text=Analisis citra")
        page.wait_for_selector("text=Hasil analisis", timeout=120_000)
        page.wait_for_timeout(3200)
        page.mouse.wheel(0, 380)
        page.wait_for_timeout(1600)
        video = page.video
        ctx.close()
        video.save_as(ASSETS / "clip_normal.webm")
        print("clip_normal ok")

        # --- clip pasien ---
        ctx = new_ctx("pasien")

```

```

page = ctx.new_page()
page.goto(f"{BASE}/pasien", wait_until="networkidle")
page.wait_for_timeout(1200)
page.locator('input[type="checkbox"]').first.click() # batalkan "minum obat"
page.select_option("select", "2")
page.wait_for_timeout(700)
page.click("text=Simpan check-in")
page.wait_for_timeout(2600)
page.mouse.wheel(0, 300)
page.wait_for_timeout(1400)
video = page.video
ctx.close()
video.save_as(ASSETS / "clip_pasien.webm")
print("clip_pasien ok")

browser.close()

```

```

def narrate() -> list[Path]:
    from piper import PiperVoice

    voice = PiperVoice.load(VOICE_MODEL)
    wavs = []
    for i, (_, _, text) in enumerate(SEGMENTS):
        w = ASSETS / f"seg{i}.wav"
        with wave.open(str(w), "wb") as f:
            voice.synthesize_wav(text, f)
        wavs.append(w)
        print(f"narasi seg{i}: {wav_duration(w):.1f}s")
    return wavs

def clip_duration(p: Path) -> float:
    out = subprocess.run(
        ["ffmpeg", "-v", "error", "-show_entries", "format=duration", "-of", "csv=p=0", str(p)],
        capture_output=True, text=True, check=True,
    )
    return float(out.stdout.strip())

def assemble(wavs: list[Path]) -> None:
    parts = []
    for i, ((kind, ident, _), wav) in enumerate(zip(SEGMENTS, wavs)):
        dur = wav_duration(wav) + 0.6
        out = ASSETS / f"part{i}.mp4"
        if kind == "slide":
            sh(["ffmpeg", "-y", "-loglevel", "error",
                "-loop", "1", "-i", str(ASSETS / f"{ident}.png"),
                "-i", str(wav),
                "-af", "apad=pad_dur=0.6",
                "-t", f"{dur:.2f}",
                "-r", "30", "-c:v", "libx264", "-pix_fmt", "yuv420p", "-preset", "medium", "-crf", "20",
                "-c:a", "aac", "-ar", "44100", "-ac", "2", str(out)])
        else:
            clip = ASSETS / f"{ident}.webm"
            cd = clip_duration(clip)
            pad = max(0.0, dur - cd)
            vf = f"tpad=stop_mode=clone:stop_duration={pad:.2f},scale={W}:{H},fps=30,format=yuv420p"
            sh(["ffmpeg", "-y", "-loglevel", "error",
                "-i", str(clip), "-i", str(wav),
                "-filter_complex", f"[0:v]{vf}[v]",
                "-map", "[v]", "-map", "1:a",
                "-af", "apad=pad_dur=0.6",
                "-t", f"{dur:.2f}",
                "-c:v", "libx264", "-preset", "medium", "-crf", "20",
                "-c:a", "aac", "-ar", "44100", "-ac", "2", str(out)])
            parts.append(out)
        print(f"part{i} ok ({kind}:{ident}, {dur:.1f}s)")

    lst = ASSETS / "concat.txt"
    lst.write_text("\n".join(f"file '{p.name}'\n" for p in parts))
    sh(["ffmpeg", "-y", "-loglevel", "error", "-f", "concat", "-safe", "0",
        "-i", str(lst), "-c", "copy", str(FINAL)])
    print("FINAL:", FINAL, f"({clip_duration(FINAL):.1f}s)")

def main() -> None:
    render_slides()
    record_clips()
    wavs = narrate()
    assemble(wavs)

if __name__ == "__main__":
    main()

```

tools/print_check.py

```
#!/usr/bin/env python3
"""Uji tampilan cetak surat rujukan (print media emulation) – screenshot bukti.

Jalankan dengan helper server (web 3210 + api 8000):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
--server "cd web && npm start -- --port 3210" --port 3210 \
--server ".venv/bin/python -m uvicorn api.main:app --port 8000" --port 8000 \
-- .venv/bin/python tools/print_check.py
"""
from __future__ import annotations

from pathlib import Path

from playwright.sync_api import sync_playwright

OUT = Path("submission/screenshots/referral_letter_print.png")
IMG = "data/raw/nlm/shenzhen/CXR_png/CHNCXR_0327_1.png"

def main() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        page = browser.new_page(viewport={"width": 794, "height": 1123}, device_scale_factor=2)
        page.goto("http://localhost:3210/skrining", wait_until="networkidle")
        page.set_input_files('input[type="file"]', IMG)
        page.fill('input[placeholder="mis. S-014"]', "S-014")
        page.fill('input[placeholder="34"]', "42")
        page.click("text=Analisis citra")
        page.wait_for_selector("text=Hasil analisis", timeout=120_000)
        page.emulate_media(media="print")
        page.wait_for_timeout(500)
        page.screenshot(path=str(OUT), full_page=True)
        print("print check ok ->", OUT)
        browser.close()

if __name__ == "__main__":
    main()
```

tools/screenshot_ui.py

```
#!/usr/bin/env python3
"""QA visual + screenshot semua halaman SIGAP (mobile & desktop).

Jalankan lewat helper server (lihat ml/kaggle/README atau riwayat commit):
python /home/alvn/.agents/skills/webapp-testing/scripts/with_server.py \
--server "cd web && npm run dev -- --port 3210" --port 3210 \
-- .env/bin/python tools/screenshot_ui.py http://localhost:3210
"""
from __future__ import annotations

import sys
from pathlib import Path

from playwright.sync_api import sync_playwright

BASE = sys.argv[1] if len(sys.argv) > 1 else "http://localhost:3210"
OUT = Path("submission/screenshots")
OUT.mkdir(parents=True, exist_ok=True)

ROUTES = [
    ("", "landing"),
    ("skrining", "skrining"),
    ("pasien", "pasien"),
    ("klinisi", "klinisi"),
    ("tentang", "tentang"),
]
VIEWPORTS = [("mobile", 390, 844), ("desktop", 1280, 900)]

def main() -> None:
    with sync_playwright() as p:
        browser = p.chromium.launch(headless=True)
        for vp_name, w, h in VIEWPORTS:
            ctx = browser.new_context(viewport={"width": w, "height": h}, device_scale_factor=2)
            page = ctx.new_page()
            console_errors: list[str] = []
            page.on("console", lambda m: console_errors.append(m.text) if m.type == "error" else None)
            for route, name in ROUTES:
                page.goto(f"{BASE}/{route}", wait_until="networkidle")
                page.wait_for_timeout(400)
                page.screenshot(path=str(OUT / f"{name}_{vp_name}.png"), full_page=True)
                print(f"[ok] {vp_name:7s} {name}")
            if console_errors:
                print(f"[warn] console errors ({vp_name}): {console_errors[:5]}")
            ctx.close()
        browser.close()
    print(f"SCREENSHOTS -> {OUT}")

if __name__ == "__main__":
    main()
```

web/src/lib/api.ts

```
export type Band = "rujuk_prioritas" | "ragu" | "negatif_skrining";

export interface PredictResponse {
  p_tb: number;
  band: Band;
  bands: { tau_high: number; tau_low: number };
  arch?: string;
  disclaimer?: string;
  heatmap_png_b64?: string;
}

const API_URL = (process.env.NEXT_PUBLIC_API_URL ?? "http://localhost:8000").replace(/\/$/, "");

export async function predict(file: File): Promise<PredictResponse> {
  const fd = new FormData();
  fd.append("file", file);
  const res = await fetch(`${API_URL}/predict`, { method: "POST", body: fd });
  if (!res.ok) throw new Error(`API ${res.status}`);
  return (await res.json()) as PredictResponse;
}

export const BAND_UI: Record<
  Band,
  { label: string; title: string; action: string; text: string; dot: string }
> = {
  rujuk_prioritas: {
    label: "Prioritas rujukan",
    title: "Rujuk – prioritas hari ini",
    action:
      "Kirim pasien untuk pemeriksaan GeneXpert di fasilitas terdekat hari ini. Catat sebagai suspek TB pada buku register.",
    text: "text-[#ee8b7e]",
    dot: "bg-refer",
  },
  ragu: {
    label: "Perlu pembacaan ulang",
    title: "Ragu – perlu pembacaan ulang",
    action:
      "Jangan diputuskan sendiri. Minta pembacaan oleh petugas/radiolog, atau ulangi skrining dengan citra yang lebih baik.",
    text: "text-[#e5b95c]",
    dot: "bg-defer",
  },
  negatif_skrining: {
    label: "Negatif skrining",
    title: "Negatif skrining",
    action:
      "Edukasi gejala TB. Minta pasien kembali jika gejala berlanjut lebih dari 2 minggu atau memburuk.",
    text: "text-[#83c7a4]",
    dot: "bg-clear",
  },
};

export function fmtPct(x: number, digits = 0): string {
  return new Intl.NumberFormat("id-ID", {
    style: "percent",
    maximumFractionDigits: digits,
  }).format(x);
}

export function fmtDate(iso: string): string {
  return new Intl.DateTimeFormat("id-ID", { dateStyle: "long" }).format(new Date(iso));
}
```

```

"use client";

import { useEffect, useMemo, useState } from "react";
import { BAND_UI, fmtPct, type Band } from "@lib/api";

type CheckIn = { date: string; med: boolean; symptoms: number; danger: string[] };
type Case = {
  id: string;
  date: string;
  name: string;
  age?: string;
  sex?: string;
  p: number;
  band: Band;
  history: CheckIn[];
};

export default function KlinisiPage() {
  const [cases, setCases] = useState<Case[]>([]);
  const [selectedId, setSelectedId] = useState("");

  useEffect(() => {
    try {
      const stored = JSON.parse(localStorage.getItem("sigap.cases") ?? "[]") as Case[];
      setCases(stored);
      if (stored[0]) setSelectedId(stored[0].id);
    } catch {
      /* abaikan */
    }
  }, []);

  const selected = useMemo(() => cases.find((c) => c.id === selectedId), [cases, selectedId]);
  const adherence = useMemo(() => {
    if (!selected || selected.history.length === 0) return null;
    const taken = selected.history.filter((h) => h.med).length;
    return taken / selected.history.length;
  }, [selected]);

  return (
    <main className="mx-auto max-w-3xl px-5">
      <section className="pt-10 print:hidden">
        <h1 className="text-[26px] font-extrabold tracking-tight">Ringkasan klinisi</h1>
        <p className="mt-2 max-w-xl text-[15px] leading-relaxed text-ink/70">
          Satu halaman untuk pembacaan ulang dan tindak lanjut: hasil skrining, kepatuhan, dan
          tanda eskalasi.
        </p>
      </section>

      {cases.length === 0 ? (
        <p className="mt-8 text-[15px] text-ink/60">
          Belum ada kasus tersimpan. Simpan hasil dari halaman <a className="underline" href="/skrining">Skrining</a>{" "}
          terlebih dahulu.
        </p>
      ) : (
        <>
          <div className="mt-8 flex flex-wrap gap-2 print:hidden">
            {cases.map((c) => (
              <button
                key={c.id}
                type="button"
                onClick={() => setSelectedId(c.id)}
                aria-pressed={selectedId === c.id}
                className={`rounded-[8px] border px-3.5 py-2 text-[13px] font-semibold transition-colors ${
                  selectedId === c.id ? "border-ink bg-ink text-paper" : "border-ink/20 hover:border-ink/45"
                }`}
              >
                {c.name}
              </button>
            ))}
          </div>

          {selected && (
            <section className="reveal mt-6 rounded-[12px] border border-ink/15 bg-white/70 p-5 sm:p-6 print:border-0">
              <header className="flex flex-wrap items-start justify-between gap-4">
                <div>
                  <h2 className="text-[20px] font-extrabold">{selected.name}</h2>
                  <p className="mt-0.5 text-[13px] text-ink/60">
                    {selected.age || "-"} th · {selected.sex || "-"} · skrining{" "}
                    {new Intl.DateTimeFormat("id-ID", { dateStyle: "long" }).format(new Date(selected.date))}
                  </p>
                </div>
                <div className="text-right">
                  <p className="flex items-center justify-end gap-2 text-[13px] font-semibold">
                    <span className={`inline-block h-2 w-2 rounded-full ${BAND_UI[selected.band].dot}` } />
                    {BAND_UI[selected.band].title}
                  </p>
                  <p className="text-[13px] text-ink/60 tabular-nums">p(TB) awal {fmtPct(selected.p)}</p>
                </div>
              </header>
            </section>
          )}
        </>
      )}
    </main>
  );
}

```

```

</header>

<dl className="mt-6 grid grid-cols-2 gap-4 border-t border-ink/10 pt-4 sm:grid-cols-4">
  <div>
    <dt className="text-[12px] text-ink/55">Check-in tercatat</dt>
    <dd className="text-[18px] font-bold tabular-nums">{selected.history.length}</dd>
  </div>
  <div>
    <dt className="text-[12px] text-ink/55">Kepatuhan obat</dt>
    <dd className="text-[18px] font-bold tabular-nums">
      {adherence === null ? "-" : fmtPct(adherence)}
    </dd>
  </div>
  <div>
    <dt className="text-[12px] text-ink/55">Gejala terakhir</dt>
    <dd className="text-[18px] font-bold tabular-nums">
      {selected.history[0] ? `${selected.history[0].symptoms}/3` : "-"}
    </dd>
  </div>
  <div>
    <dt className="text-[12px] text-ink/55">Tanda bahaya</dt>
    <dd className="text-[18px] font-bold">
      {selected.history.some((h) => h.danger.length > 0) ? (
        <span className="text-refer">Ada</span>
      ) : (
        <span className="text-clear">Tidak</span>
      )}
    </dd>
  </div>
</dl>

{selected.history.some((h) => h.danger.length > 0) && (
  <p className="mt-5 rounded-[8px] border border-refer/35 bg-refer/5 px-4 py-3 text-[14px] font-semibold text-refer">
    Perhatian: pasien melaporkan tanda bahaya ({selected.history.flatMap((h) => h.danger).slice(0, 3).join(", ")}).
    Pertimbangkan evaluasi klinis segera.
  </p>
)}

<div className="mt-6 flex flex-wrap gap-3 print:hidden">
  <button
    type="button"
    onClick={() => window.print()}
    className="rounded-[8px] bg-ink px-5 py-3 text-[14px] font-semibold text-paper hover:bg-ink/90"
  >
    Cetak ringkasan
  </button>
</div>

<p className="mt-6 text-[12px] leading-relaxed text-ink/50">
  Ringkasan prototipe riset – bukan rekam medis. Sumber angka: state risiko skrining +
  log peragaan check-in.
</p>
</section>
)}
</>
)}
</main>
);
}

```

web/src/app/layout.tsx

```
import type { Metadata } from "next";
import { Plus_Jakarta_Sans } from "next/font/google";
import "../globals.css";
import SiteHeader from "@/components/SiteHeader";

const jakarta = Plus_Jakarta_Sans({
  subsets: ["latin"],
  variable: "--font-jakarta",
  display: "swap",
});

export const metadata: Metadata = {
  title: "SIGAP – skrining TB yang tahu kapan harus ragu",
  description:
    "Prototipe riset triase tuberkulosis: X-ray terkalibrasi → rujukan → pendampingan minum obat.",
};

export default function RootLayout({ children }: LayoutProps) {
  return (
    <html lang="id" className={` ${jakarta.variable} h-full antialiased`} >
      <body className="flex min-h-full flex-col">
        <SiteHeader />
        <div className="flex-1">{children}</div>
        <footer className="no-print mt-16 border-t border-ink/10">
          <div className="mx-auto max-w-3xl px-5 py-6 text-[13px] leading-relaxed text-ink/55">
            SIGAP adalah prototipe riset triase skrining – <strong>bukan alat diagnosis</strong>. Setiap
            hasil wajib dikonfirmasi tenaga kesehatan (GeneXpert). Angka beban penyakit: WHO Global TB
            Report 2025 dan Kemenkes 2024.{ " "}
            <a
              className="underline decoration-ink/30 underline-offset-2 hover:text-ink"
              href="https://github.com/Alvnvnc/loop-tb"
              target="_blank"
              rel="noreferrer"
            >
              Kode, split data, dan evaluasi terbuka
            </a>
          </div>
        </footer>
      </body>
    </html>
  );
}
```

```

export default function Home() {
  const steps = [
    {
      title: "Kenali lebih cepat",
      body: "Kader mengunggah X-ray pasien. Model tersaring dari kebocoran data memisahkan TB dari paru tidak normal lain – dengan pita ragu ketika bukti belum cukup.",
    },
    {
      title: "Rujuk tanpa menunggu",
      body: "Skor risiko diterjemahkan menjadi band tindakan dan surat rujukan siap cetak: prioritas GeneXpert hari ini, pembacaan ulang, atau edukasi.",
    },
    {
      title: "Dampingi sampai sembuh",
      body: "Kartu pasien mencatat check-in minum obat. Risiko diperbarui dari waktu ke waktu; bila memburuk, muncul tanda eskalasi untuk petugas.",
    },
  ];

  return (
    <main className="mx-auto max-w-3xl px-5">
      <section className="pt-14 pb-12">
        <p className="text-[13px] font-medium text-ink/50">
          Tuberkulosis · Indonesia · data 2024
        </p>
        <h1 className="mt-3 text-[33px] leading-[1.14] font-extrabold tracking-tight sm:text-[42px]">
          Sekitar 200 ribu orang menularkan TB tanpa pernah terdeteksi. Loop perawatannya lebih lambat daripada penularannya.
        </h1>
        <p className="mt-5 max-w-2xl text-[16px] leading-relaxed text-ink/75">
          SIGAP memampatkan jeda itu dengan satu state risiko yang menggerakkan{" "}
          <em className="not-italic font-semibold">skrining → rujukan → pendampingan</em>.
          Bukan skor lain – triase terkalibrasi yang tahu kapan dirinya harus ragu.
        </p>
        <div className="mt-7 flex flex-wrap gap-3">
          <a
            href="/skrining"
            className="rounded-[8px] bg-ink px-5 py-3 text-[15px] font-semibold text-paper transition-colors hover:bg-ink/90"
          >
            Mulai skrining
          </a>
          <a
            href="/pasien"
            className="rounded-[8px] border border-ink/20 px-5 py-3 text-[15px] font-semibold transition-colors hover:border-ink/45"
          >
            Lihat alur pendampingan
          </a>
        </div>
      </section>

      <section className="border-t border-ink/10 py-10">
        <h2 className="text-[15px] font-bold">Satu loop, tiga titik kompresi</h2>
        <ol className="mt-7 space-y-7">
          {steps.map((s, i) => (
            <li key={s.title} className="grid grid-cols-[44px_1fr] gap-4">
              <span className="pt-0.5 text-[15px] font-bold tabular-nums text-ink/30">
                {String(i + 1).padStart(2, "0")}
              </span>
              <div>
                <h3 className="text-[17px] font-bold">{s.title}</h3>
                <p className="mt-1 max-w-xl text-[15px] leading-relaxed text-ink/70">{s.body}</p>
              </div>
            </li>
          ))}
        </ol>
      </section>

      <section className="border-t border-ink/10 py-10">
        <h2 className="text-[15px] font-bold">Yang terbuka, yang belum</h2>
        <div className="mt-5 grid gap-7 sm:grid-cols-2">
          <div>
            <h3 className="text-[14px] font-semibold text-clear">Terukur dan terbuka</h3>
            <ul className="mt-2 space-y-2 text-[14px] leading-relaxed text-ink/70">
              <li>
                Audit kebocoran dataset: 19 citra train yang duplikat dengan test set eksternal dibuang sebelum training – angka yang jarang dilaporkan.
              </li>
              <li>
                Evaluasi lintas-rumah-sakit dilaporkan apa adanya: AUROC, sensitivitas pada spesifisitas tetap, dan kalibrasi (ECE).
              </li>
              <li>Split data, model, dan kode dapat direproduksi dari repositori.</li>
            </ul>
          </div>
          <div>
            <h3 className="text-[14px] font-semibold text-defer">Belum beres – jujur</h3>
            <ul className="mt-2 space-y-2 text-[14px] leading-relaxed text-ink/70">
              <li>Belum divalidasi prospektif di Indonesia; ini prototipe riset, bukan alat medis.</li>
            </ul>
          </div>
        </div>
      </section>
    </main>
  );
}

```

```
        <li>Ambang triase dikalibrasi pada data publik, belum pada populasi puskesmas.</li>
        <li>Analisis suara batuk dan integrasi SITB belum dikerjakan.</li>
    </ul>
</div>
</div>
</section>
</main>
);
}
```

```

"use client";

import { useEffect, useMemo, useState } from "react";
import { BAND_UI, fmtPct, type Band } from "@lib/api";

type CheckIn = { date: string; med: boolean; symptoms: number; danger: string[]; note?: string };
type Case = {
  id: string;
  date: string;
  name: string;
  age?: string;
  sex?: string;
  p: number;
  band: Band;
  history: CheckIn[];
};

const DANGER = ["Batuk darah", "Sesak napas", "Demam tinggi"];

const DEMO_CASE: Case = {
  id: "demo",
  date: new Date(Date.now() - 9 * 864e5).toISOString(),
  name: "Contoh peragaan – Ibu S",
  age: "45",
  sex: "P",
  p: 0.42,
  band: "ragu",
  history: [
    { date: new Date(Date.now() - 1 * 864e5).toISOString(), med: true, symptoms: 1, danger: [] },
    { date: new Date(Date.now() - 3 * 864e5).toISOString(), med: true, symptoms: 2, danger: [] },
    { date: new Date(Date.now() - 5 * 864e5).toISOString(), med: false, symptoms: 2, danger: [] },
  ],
};

function step(prev: number, h: CheckIn): number {
  const delta = (h.med ? -0.08 : 0.18) + 0.09 * h.symptoms + (h.danger.length ? 0.3 : 0);
  return Math.min(0.99, Math.max(0.01, prev + delta));
}

function seriesOf(c: Case): number[] {
  const out = [c.p];
  for (const h of [...c.history].reverse()) out.push(step(out[out.length - 1], h));
  return out;
}

function Sparkline({ values }: { values: number[] }) {
  const w = 160;
  const h = 40;
  const pts = values
    .map((v, i) => {
      const x = values.length === 1 ? w : (i / (values.length - 1)) * w;
      const y = h - v * h;
      return `${x.toFixed(1)},${y.toFixed(1)}`;
    })
    .join(" ");
  return (
    <svg viewBox={`0 0 ${w} ${h}`} className="h-10 w-40" role="img" aria-label="Tren risiko">
      <line x1="0" y1={h - 0.61 * h} x2={w} y2={h - 0.61 * h} stroke="#b7791f" strokeDasharray="3 3" strokeWidth="1" />
      <polyline points={pts} fill="none" stroke="#14181d" strokeWidth="2" />
    </svg>
  );
}

export default function PasienPage() {
  const [cases, setCases] = useState<Case[]>([]);
  const [selectedId, setSelectedId] = useState<string>("demo");
  const [loaded, setLoaded] = useState(false);
  const [med, setMed] = useState(true);
  const [symptoms, setSymptoms] = useState(1);
  const [danger, setDanger] = useState<string[]>([]);
  const [note, setNote] = useState("");

  useEffect(() => {
    try {
      const stored = JSON.parse(localStorage.getItem("sigap.cases") ?? "[]") as Case[];
      setCases(stored);
      if (stored[0]) setSelectedId(stored[0].id);
    } catch {
      /* penyimpanan diblokir – tetap tampilkan contoh */
    }
    setLoaded(true);
  }, []);

  const all = useMemo<Case[]>(() => [DEMO_CASE, ...cases], [cases]);
  const selected = all.find((c) => c.id === selectedId) ?? DEMO_CASE;
  const risk = seriesOf(selected).at(-1) ?? selected.p;
  const escalate = risk >= 0.61;
  const tone = BAND_UI[selected.band];

```

```

function persist(next: Case) {
  if (next.id === "demo") return;
  const updated = cases.map((c) => (c.id === next.id ? next : c));
  setCases(updated);
  localStorage.setItem("sigap.cases", JSON.stringify(updated));
}

function addCheckIn() {
  const h: CheckIn = { date: new Date().toISOString(), med, symptoms, danger, note: note.trim() || undefined };
  const base = selected.id === "demo" ? { ...DEMO_CASE, id: crypto.randomUUID(), name: DEMO_CASE.name + " (salinan)" } : selected;
  const next: Case = { ...base, history: [h, ...base.history] };
  if (selected.id === "demo") {
    const updated = [next, ...cases];
    setCases(updated);
    try {
      localStorage.setItem("sigap.cases", JSON.stringify(updated));
    } catch {
      /* abaikan */
    }
    setSelectedId(next.id);
  } else {
    persist(next);
  }
  setNote("");
  setDanger([]);
}

return (
  <main className="mx-auto max-w-3xl px-5">
    <section className="pt-10">
      <h1 className="text-[26px] font-extrabold tracking-tight">Pendampingan pasien</h1>
      <p className="mt-2 max-w-xl text-[15px] leading-relaxed text-ink/70">
        Kartu pasien menyambung skrining ke pengobatan: check-in harian memperbarui risiko dan
        memicu eskalasi bila memburuk.
      </p>
    </section>

    <div className="mt-8 flex flex-wrap gap-2">
      {all.map((c) => (
        <button
          key={c.id}
          type="button"
          onClick={() => setSelectedId(c.id)}
          aria-pressed={selected.id === c.id}
          className={`rounded-[8px] border px-3.5 py-2 text-[13px] font-semibold transition-colors ${
            selected.id === c.id ? "border-ink bg-ink text-paper" : "border-ink/20 hover:border-ink/45"
          }`}
        >
          {c.name}
        </button>
      ))}
      {loaded && cases.length === 0 && (
        <span className="py-2 text-[13px] text-ink/50">
          Belum ada kartu tersimpan – menyimpan dari halaman Skrining akan muncul di sini.
        </span>
      )}
    </div>

    <section className="reveal mt-6 rounded-[12px] border border-ink/15 bg-white/70 p-5 sm:p-6">
      <div className="flex flex-wrap items-start justify-between gap-4">
        <div>
          <h2 className="text-[18px] font-extrabold">{selected.name}</h2>
          <p className="mt-0.5 text-[13px] text-ink/60">
            {selected.age ? `${selected.age} th` : "usia -"} · {selected.sex || "-"} · skrining{" "}
            {new Intl.DateTimeFormat("id-ID", { dateStyle: "medium" }).format(new Date(selected.date))}
          </p>
        </div>
        <div className="text-right">
          <p className="text-[12px] text-ink/55">Risiko saat ini</p>
          <p className="text-[24px] font-extrabold tabular-nums">{fmtPct(risk)}</p>
        </div>
      </div>

      <div className="mt-4 flex flex-wrap items-center gap-5 border-t border-ink/10 pt-4">
        <Sparkline values={seriesOf(selected)} />
        <div className="text-[13px] leading-relaxed text-ink/70">
          <p className="flex items-center gap-2 font-semibold">
            <span className={`inline-block h-2 w-2 rounded-full ${tone.dot}`} />
            {tone.title}
          </p>
          <p className="mt-1">
            Garis putus-putus = ambang rujuk (61%).{" "}
            {selected.history.length < 2 ? "Data check-in masih sedikit." : ""}
          </p>
        </div>
      </div>

      {escalate && (
        <p className="mt-4 rounded-[8px] border border-refer/35 bg-refer/5 px-4 py-3 text-[14px] font-semibold text-refer">
          Eskalasi: risiko menembus ambang. Kunjungi atau telepon pasien hari ini; pertimbangkan
          rujukan ulang.
        </p>
      )}
    </section>
  </main>
)

```

```

    </p>
  })

  { /* Check-in */
  <div className="mt-6 border-t border-ink/10 pt-5">
    <h3 className="text-[14px] font-bold">Check-in hari ini</h3>
    <div className="mt-3 flex flex-wrap items-center gap-5">
      <label className="flex items-center gap-2 text-[14px]">
        <input type="checkbox" checked={med} onChange={(e) => setMed(e.target.checked)} className="h-4 w-4 accent-ink" />
        Minum obat sesuai jadwal
      </label>
      <label className="flex items-center gap-2 text-[14px]">
        Gejala (0-3)
        <select
          value={symptoms}
          onChange={(e) => setSymptoms(Number(e.target.value))}
          className="rounded-[6px] border border-ink/20 bg-white px-2 py-1.5 text-[14px]"
        >
          {[0, 1, 2, 3].map((n) => (
            <option key={n} value={n}>
              {n}
            </option>
          ))}
        </select>
      </label>
    </div>
    <div className="mt-3 flex flex-wrap gap-4">
      {DANGER.map((d) => (
        <label key={d} className="flex items-center gap-2 text-[14px]">
          <input
            type="checkbox"
            checked={danger.includes(d)}
            onChange={(e) =>
              setDanger(e.target.checked ? [...danger, d] : danger.filter((x) => x !== d))
            }
            className="h-4 w-4 accent-ink"
          />
          {d}
        </label>
      ))}
    </div>
    <div className="mt-3 flex flex-wrap gap-3">
      <input
        value={note}
        onChange={(e) => setNote(e.target.value)}
        placeholder="Catatan (opsional)"
        className="min-w-0 flex-1 rounded-[8px] border border-ink/20 bg-white px-3 py-2.5 text-[14px] outline-none
focus:border-ink/50"
      />
      <button
        type="button"
        onClick={addCheckIn}
        className="rounded-[8px] bg-ink px-5 py-2.5 text-[14px] font-semibold text-paper hover:bg-ink/90"
      >
        Simpan check-in
      </button>
    </div>
  </div>

  { /* Riwayat */
  {selected.history.length > 0 && (
    <div className="mt-6 border-t border-ink/10 pt-5">
      <h3 className="text-[14px] font-bold">Riwayat check-in</h3>
      <ul className="mt-3 space-y-2">
        {selected.history.map((h, i) => (
          <li key={i} className="flex flex-wrap items-baseline gap-x-3 gap-y-1 text-[14px] text-ink/75">
            <span className="tabular-nums text-ink/50">
              {new Intl.DateTimeFormat("id-ID", { dateStyle: "medium" }).format(new Date(h.date))}
            </span>
            <span className={h.med ? "text-clear" : "font-semibold text-refer"}>
              {h.med ? "obat ✓" : "obat terlewat"}
            </span>
            <span>gejala {h.symptoms}/3</span>
            {h.danger.map((d) => (
              <span key={d} className="font-semibold text-refer">
                {d}
              </span>
            ))}
            {h.note && <span className="text-ink/55">{h.note}</span>}
          </li>
        ))}
      </ul>
    </div>
  )}

  <p className="mt-6 text-[12px] leading-relaxed text-ink/50">
    Catatan metodologis: pembaruan risiko di halaman ini memakai aturan peragaan sederhana
    (kepatuhan -8%, gejala +9%/poin, tanda bahaya +30%) untuk mendemonstrasikan loop – bukan
    model terlatih. Versi berikutnya memakai state risiko yang sama dengan mesin skring.
  </p>
</section>
</main>

```


web/src/app/skrining/page.tsx

```
import type { Metadata } from "next";
import ScreeningFlow from "@/components/ScreeningFlow";

export const metadata: Metadata = {
  title: "Skrining – SIGAP",
};

export default function SkriningPage() {
  return (
    <main className="mx-auto max-w-3xl px-5">
      <section className="pt-10 print:hidden">
        <h1 className="text-[26px] font-extrabold tracking-tight">Skrining X-ray</h1>
        <p className="mt-2 max-w-xl text-[15px] leading-relaxed text-ink/70">
          Unggah citra X-ray dada, jalankan analisis, lalu ikuti band tindakan. Kurang dari tiga
          menit per pasien.
        </p>
      </section>
      <ScreeningFlow />
    </main>
  );
}
```

```

import type { Metadata } from "next";

export const metadata: Metadata = {
  title: "Tentang – SIGAP",
};

export default function TentangPage() {
  return (
    <main className="mx-auto max-w-3xl px-5">
      <section className="pt-10">
        <h1 className="text-[26px] font-extrabold tracking-tight">Tentang SIGAP</h1>
        <p className="mt-3 max-w-2xl text-[16px] leading-relaxed text-ink/75">
          Indonesia adalah negara dengan beban tuberkulosis terbesar kedua di dunia: sekitar 10%
          dari kasus global 2024 (WHO Global TB Report 2025), dengan ±200 ribu kasus per tahun yang
          diperkirakan tidak pernah terdeteksi. SIGAP adalah prototipe riset yang menyerang akar
          masalah itu: <strong>loop perawatan yang lebih lambat daripada penularannya</strong>.
        </p>
      </section>

      <section className="mt-10 border-t border-ink/10 pt-8">
        <h2 className="text-[16px] font-bold">Cara kerja</h2>
        <ol className="mt-4 space-y-3 text-[15px] leading-relaxed text-ink/75">
          <li>
            <strong>Skrining.</strong> Citra X-ray diproses model klasifikasi TB yang dilatih pada
            dataset publik (TBX11K dan basis Rahman) dan diuji secara eksternal pada korpus NLM
            Shenzhen + Montgomery yang tidak pernah tersentuh training.
          </li>
          <li>
            <strong>Kalibrasi & pita ragu.</strong> Skor dikalibrasi (temperature scaling) dan
            diterjemahkan menjadi tiga band: prioritas rujukan, ragu (perlu pembacaan manusia), dan
            negatif skrining. Ambang diambil dari titik kerja sensitivitas-target pada data validasi,
            mengacu target TPP WHO.
          </li>
          <li>
            <strong>Loop.</strong> Hasil skrining menjadi state risiko awal pada kartu pasien;
            check-in harian memperbarui risiko dan memunculkan eskalasi bila memburuk.
          </li>
        </ol>
      </section>

      <section className="mt-10 border-t border-ink/10 pt-8">
        <h2 className="text-[16px] font-bold">Rigor yang bisa diperiksa</h2>
        <ul className="mt-4 space-y-3 text-[15px] leading-relaxed text-ink/75">
          <li>
            Audit kebocoran: deduplikasi perseptual (pHash) menemukan 338 duplikat internal dan 19
            citra train yang duplikat dengan test set eksternal – semuanya dibuang sebelum training
            (split manifest tersedia di repositori).
          </li>
          <li>
            Evaluasi eksternal lintas-rumah-sakit dengan AUROC + interval bootstrap, sensitivitas
            pada spesifisitas 90%/70%, kalibrasi ECE, dan analisis subgrup usia/jenis kelamin dari
            metadata Montgomery.
          </li>
          <li>
            Keterbatasan dilaporkan, bukan disembunyikan: data publik bersifat radiografis (bukan
            konfirmasi bakteriologis), reprocessing agresif dapat lolos dari deduplikasi, dan
            belum ada data Indonesia.
          </li>
        </ul>
      </section>

      <section className="mt-10 border-t border-ink/10 pt-8 pb-4">
        <h2 className="text-[16px] font-bold">Kontak & sumber</h2>
        <p className="mt-3 text-[15px] leading-relaxed text-ink/75">
          Dibuat untuk UnivaBio 2026. Kode, split data, dan hasil evaluasi:{" "}
          <a
            className="underline decoration-ink/30 underline-offset-2 hover:text-ink"
            href="https://github.com/Alvnc/loop-tb"
            target="_blank"
            rel="noreferrer"
          >
            github.com/Alvnc/loop-tb
          </a>
          . Referensi utama: WHO consolidated guidelines on TB (Module 2: Screening, 2021) dan WHO
          policy statement on CAD for TB screening (2025); Kemenkes (TOSS-TB).
        </p>
        <p className="mt-6 rounded-[10px] border border-ink/15 bg-white/70 p-4 text-[14px] leading-relaxed text-ink/70">
          Pernyataan jujur: ini prototipe riset untuk kompetisi, <strong>bukan alat medis</strong>.
          Jangan dipakai untuk keputusan klinis tanpa validasi dan pengawasan tenaga kesehatan.
        </p>
      </section>
    </main>
  );
}

```

web/src/components/ScreeningFlow.tsx

```
"use client";

import { useRef, useState, type ReactNode } from "react";
import { BAND_UI, fmtPct, predict, type PredictResponse } from "@lib/api";

type Patient = { name: string; age: string; sex: string; complaint: string };

const EMPTY_PATIENT: Patient = { name: "", age: "", sex: "", complaint: "" };

/** Contoh keluaran statis untuk peragaan tanpa server ML – selalu diberi label jelas. */
const DEMO_RESULT: PredictResponse = {
  p_tb: 0.42,
  band: "ragu",
  bands: { tau_low: 0.21, tau_high: 0.61 },
  arch: "arsip peragaan (bukan dari citra Anda)",
};

function StepTitle({ n, children }: { n: number; children: ReactNode }) {
  return (
    <h2 className="flex items-center gap-3 text-[16px] font-bold">
      <span className="grid h-7 w-7 place-items-center rounded-[6px] border border-ink/20 text-[13px] tabular-nums">
        {n}
      </span>
      {children}
    </h2>
  );
}

export default function ScreeningFlow() {
  const [file, setFile] = useState<File | null>(null);
  const [preview, setPreview] = useState("");
  const [patient, setPatient] = useState<Patient>(EMPTY_PATIENT);
  const [demo, setDemo] = useState(false);
  const [loading, setLoading] = useState(false);
  const [result, setResult] = useState<PredictResponse | null>(null);
  const [error, setError] = useState("");
  const [overlay, setOverlay] = useState<"heatmap" | "original">("heatmap");
  const [saved, setSaved] = useState(false);
  const inputRef = useRef<HTMLInputElement>(null);

  function onPick(f: File | null | undefined) {
    if (!f) return;
    if (!f.type.startsWith("image/")) {
      setError("File harus berupa citra (PNG/JPG).");
      return;
    }
    setError("");
    setResult(null);
    setSaved(false);
    setFile(f);
    setPreview(URL.createObjectURL(f));
  }

  async function analyze() {
    if (!file || loading) return;
    setLoading(true);
    setError("");
    setResult(null);
    try {
      const r = demo ? DEMO_RESULT : await predict(file);
      setResult(r);
      setOverlay(r.heatmap_png_b64 ? "heatmap" : "original");
    } catch {
      setError(
        "Server model tidak terjangkau. Nyalakan mode peragaan untuk mendemokan alurnya, atau periksa koneksi API."
      );
    } finally {
      setLoading(false);
    }
  }

  function saveToCard() {
    if (!result) return;
    try {
      const cases = JSON.parse(localStorage.getItem("sigap.cases") ?? "[]");
      cases.unshift({
        id: crypto.randomUUID(),
        date: new Date().toISOString(),
        name: patient.name.trim() || "Tanpa nama",
        age: patient.age,
        sex: patient.sex,
        p: result.p_tb,
        band: result.band,
        history: [],
      });
      localStorage.setItem("sigap.cases", JSON.stringify(cases));
      setSaved(true);
    } catch {}
  }
}
```

```

        setError("Tidak bisa menyimpan di perangkat ini (penyimpanan lokal diblokir).");
    }
}

const heat = result?.heatmap_png_b64 ? `data:image/png;base64,${result.heatmap_png_b64}` : "";
const tone = result ? BAND_UI[result.band] : null;
const today = new Intl.DateTimeFormat("id-ID", { dateStyle: "long" }).format(new Date());

return (
    <>
    <div className="print:hidden">
        {/* 1 - Citra */}
        <section className="mt-9">
            <StepTitle n={1}>Citra X-ray dada</StepTitle>
            <input
                ref={inputRef}
                type="file"
                accept="image/*"
                className="sr-only"
                aria-label="Pilih citra X-ray"
                onChange={(e) => onPick(e.target.files?.[0])}
            />
            {!file ? (
                <button
                    type="button"
                    onClick={() => inputRef.current?.click()}
                    onDragOver={(e) => e.preventDefault()}
                    onDrop={(e) => {
                        e.preventDefault();
                        onPick(e.dataTransfer.files?.[0])};
                    }}
                    className="mt-4 w-full rounded-[12px] border border-dashed border-ink/25 bg-white/60 px-6 py-12 text-center transition-
colors hover:border-ink/45"
                >
                    <span className="block text-[15px] font-semibold">Pilih foto X-ray</span>
                    <span className="mt-1 block text-[13px] text-ink/55">
                        PNG atau JPG, tampak depan (PA/AP). Tarik file ke sini atau ketuk untuk memilih.
                    </span>
                </button>
            ) : (
                <div className="mt-4 rounded-[12px] border border-ink/15 bg-white/70 p-3">
                    <div className="flex items-start gap-4">
                        {/* eslint-disable-next-line @next/next/no-img-element */}
                        <img
                            src={preview}
                            alt="Pratinjau X-ray"
                            className="h-28 w-28 rounded-[8px] bg-film object-contain"
                        />
                        <div className="min-w-0 flex-1">
                            <p className="truncate text-[14px] font-semibold">{file.name}</p>
                            <p className="mt-0.5 text-[13px] text-ink/55">
                                {(file.size / 1e6).toFixed(1)} MB
                            </p>
                            <button
                                type="button"
                                onClick={() => inputRef.current?.click()}
                                className="mt-2 text-[13px] font-semibold underline decoration-ink/30 underline-offset-2 hover:text-ink"
                            >
                                Ganti citra
                            </button>
                        </div>
                    </div>
                </div>
            )}
        </section>

        {/* 2 - Data pasien */}
        <section className="mt-9">
            <StepTitle n={2}>Data pasien (opsional)</StepTitle>
            <div className="mt-4 grid gap-3 sm:grid-cols-2">
                <label className="block text-[13px] font-medium text-ink/70">
                    Nama atau kode pasien
                    <input
                        value={patient.name}
                        onChange={(e) => setPatient({ ...patient, name: e.target.value })}
                        className="mt-1 w-full rounded-[8px] border border-ink/20 bg-white px-3 py-2.5 text-[15px] outline-none focus:border-
ink/50"
                        placeholder="mis. S-014"
                    />
                </label>
                <div className="grid grid-cols-2 gap-3">
                    <label className="block text-[13px] font-medium text-ink/70">
                        Usia
                        <input
                            value={patient.age}
                            onChange={(e) => setPatient({ ...patient, age: e.target.value })}
                            inputMode="numeric"
                            className="mt-1 w-full rounded-[8px] border border-ink/20 bg-white px-3 py-2.5 text-[15px] outline-none
focus:border-ink/50"
                            placeholder="34"
                        />
                    </label>

```

```

<label className="block text-[13px] font-medium text-ink/70">
  Jenis kelamin
  <select
    value={patient.sex}
    onChange={(e) => setPatient({ ...patient, sex: e.target.value })}
    className="mt-1 w-full rounded-[8px] border border-ink/20 bg-white px-3 py-2.5 text-[15px] outline-none
focus:border-ink/50"
    >
      <option value=""></option>
      <option value="L">Laki-laki</option>
      <option value="P">Perempuan</option>
    </select>
  </label>
</div>
<label className="block text-[13px] font-medium text-ink/70 sm:col-span-2">
  Keluhan utama
  <input
    value={patient.complaint}
    onChange={(e) => setPatient({ ...patient, complaint: e.target.value })}
    className="mt-1 w-full rounded-[8px] border border-ink/20 bg-white px-3 py-2.5 text-[15px] outline-none focus:border-
ink/50"
    placeholder="mis. batuk 3 minggu, keringat malam"
  />
</label>
</div>
</section>

{/* Aksi */}
<div className="mt-9 flex flex-wrap items-center gap-4">
  <button
    type="button"
    onClick={analyze}
    disabled={!file || loading}
    className="rounded-[8px] bg-ink px-6 py-3 text-[15px] font-semibold text-paper transition-colors hover:bg-ink/90
disabled:cursor-not-allowed disabled:opacity-40"
    >
    {loading ? "Menganalisis..." : "Analisis citra"}
  </button>
  <label className="flex items-center gap-2 text-[13px] text-ink/65">
    <input
      type="checkbox"
      checked={demo}
      onChange={(e) => setDemo(e.target.checked)}
      className="h-4 w-4 accent-ink"
    />
    Mode peragaan (hasil contoh, bukan dari citra Anda)
  </label>
</div>

{loading && (
  <div className="mt-5 h-1 w-full overflow-hidden rounded-full bg-ink/10" role="status" aria-label="Menganalisis">
    <div className="h-full w-1/3 animate-pulse rounded-full bg-ink/60" />
  </div>
)}

{error && (
  <p role="alert" className="mt-5 rounded-[8px] border border-refer/30 bg-refer/5 px-4 py-3 text-[14px] text-refer">
    {error}
  </p>
)}

{/* Hasil – lightbox */}
{result && tone && (
  <section className="reveal mt-10" aria-live="polite">
    <h2 className="text-[15px] font-bold">Hasil analisis</h2>
    <div className="mt-4 overflow-hidden rounded-[12px] bg-film text-bone">
      <div className="relative aspect-[4/3]">
        {/* eslint-disable-next-line @next/next/no-img-element */}
        <img
          src={overlay === "heatmap" && heat ? heat : preview}
          alt="X-ray dengan peta perhatian model"
          className="absolute inset-0 h-full w-full object-contain"
        />
        {heat && (
          <div className="absolute top-3 right-3 flex gap-1 rounded-[8px] bg-film/80 p-1">
            {(
              [
                ["heatmap", "Peta perhatian"],
                ["original", "Citra asli"],
              ] as const
            ).map(([k, label]) => (
              <button
                key={k}
                type="button"
                onClick={() => setOverlay(k)}
                aria-pressed={overlay === k}
                className={`rounded-[6px] px-2.5 py-1.5 text-[12px] font-semibold transition-colors ${
                  overlay === k ? "bg-bone text-film" : "text-bone/70 hover:text-bone"
                }`}
              >
                {label}
              </button>
            ))}
          </div>
        )}
      </div>
    </div>
  </section>
)}

```

```

    ))}
  </div>
)}
</div>
<div className="border-t border-bone/10 p-5 sm:p-6">
  <p className={`flex items-center gap-2 text-[13px] font-semibold ${tone.text}`}>
    <span className={`inline-block h-2 w-2 rounded-full ${tone.dot}`} />
    {tone.label}
  </p>
  <h3 className="mt-1.5 text-[26px] leading-tight font-extrabold sm:text-[32px]">
    {tone.title}
  </h3>
  <p className="mt-3 max-w-2xl text-[15px] leading-relaxed text-bone/85">{tone.action}</p>
  <dl className="mt-5 grid grid-cols-3 gap-3 border-t border-bone/10 pt-4">
    <div>
      <dt className="text-[12px] text-bone/50">Estimasi p(TB)</dt>
      <dd className="mt-0.5 text-[20px] font-bold tabular-nums">{fmtPct(result.p_tb)}</dd>
    </div>
    <div>
      <dt className="text-[12px] text-bone/50">Ambang ragu</dt>
      <dd className="mt-0.5 text-[20px] font-bold tabular-nums">
        {fmtPct(result.bands.tau_low)}
      </dd>
    </div>
    <div>
      <dt className="text-[12px] text-bone/50">Ambang rujuk</dt>
      <dd className="mt-0.5 text-[20px] font-bold tabular-nums">
        {fmtPct(result.bands.tau_high)}
      </dd>
    </div>
  </dl>
  <p className="mt-4 text-[12px] leading-relaxed text-bone/50">
    {result.arch ? `Model: ${result.arch}. ` : ""}
    Ambang diambil dari titik sensitivitas-target pada data validasi (TPP WHO: sensitivitas
    &gt;90%, spesifisitas &gt;70%).
  </p>
</div>
</div>

/* Penjelasan */
<div className="mt-6 grid gap-5 sm:grid-cols-2">
  <div className="rounded-[10px] border border-ink/12 bg-white/70 p-4">
    <h3 className="text-[14px] font-bold">Yang dilihat model</h3>
    <p className="mt-1.5 text-[14px] leading-relaxed text-ink/70">
      Area yang paling memengaruhi skor disorot pada peta perhatian – untuk diverifikasi
      manusia, bukan dikutip sebagai bukti lesi.
    </p>
  </div>
  <div className="rounded-[10px] border border-defer/35 bg-defer/5 p-4">
    <h3 className="text-[14px] font-bold text-defer">Yang tidak ditunjukkan</h3>
    <p className="mt-1.5 text-[14px] leading-relaxed text-ink/75">
      Peta perhatian bukan bukti lesi. Model belum bisa membedakan bekas TB (scar) dari TB
      aktif – karena itu pita ragu ada dan pembacaan manusia tetap wajib.
    </p>
  </div>
</div>

<div className="mt-6 flex flex-wrap gap-3">
  <button
    type="button"
    onClick={() => window.print()}
    className="rounded-[8px] bg-ink px-5 py-3 text-[14px] font-semibold text-paper hover:bg-ink/90"
  >
    Cetak surat rujukan
  </button>
  <button
    type="button"
    onClick={saveToCard}
    disabled={saved}
    className="rounded-[8px] border border-ink/20 px-5 py-3 text-[14px] font-semibold hover:border-ink/45
disabled:opacity-50"
  >
    {saved ? "Tersimpan di kartu pasien" : "Simpan ke kartu pasien"}
  </button>
  <button
    type="button"
    onClick={() => {
      setResult(null);
      setSaved(false);
      setFile(null);
      setPreview("");
    }}
    className="rounded-[8px] px-5 py-3 text-[14px] font-semibold text-ink/60 hover:bg-ink/5 hover:text-ink"
  >
    Skrining baru
  </button>
</div>

<p className="mt-5 text-[13px] leading-relaxed text-ink/55">
  Alat triase riset – <strong>bukan diagnosis</strong>. Hasil wajib dikonfirmasi tenaga
  kesehatan melalui pemeriksaan GeneXpert.
</p>

```

```

    </section>
  })
</div>

{/* Surat rujukan (hanya saat cetak) */}
<div className="hidden print:block">
  <h1 className="text-[22px] font-extrabold">Surat Rujukan Skrining Tuberkulosis</h1>
  <p className="mt-1 text-[12px] text-ink/60">
    Prototipe riset SIGAP – bukan dokumen medis resmi.
  </p>
  <table className="mt-6 w-full border-collapse text-[14px]">
    <tbody>
      <tr>
        <td>
          [
            ["Tanggal", today],
            ["Fasilitas pengirim", "Puskesmas / posyandu: _____"],
            ["Pasien", patient.name || "_____"],
            ["Usia / jenis kelamin", `${patient.age || "-"} / ${patient.sex || "-"}`],
            ["Keluhan", patient.complaint || "-"],
            ["Estimasi risiko TB", result ? fmtPct(result.p_tb) : "-"],
            ["Band triase", result ? BAND_UI[result.band].title : "-"],
            [
              "Ambang (ragu / rujuk)",
              result
                ? `${fmtPct(result.bands.tau_low)} / ${fmtPct(result.bands.tau_high)}`
                : "-",
            ],
          ]
        <td>
          .map(([k, v]) => (
            <tr key={k as string} className="border-b border-ink/15">
              <td className="w-56 py-2 pr-4 align-top font-semibold">{k}</td>
              <td className="py-2">{v}</td>
            </tr>
          ))
        </td>
      </tr>
    </tbody>
  </table>
  <p className="mt-6 text-[14px] leading-relaxed">
    Mohon dilakukan pemeriksaan GeneXpert dan evaluasi klinis sesuai algoritma program TB nasional (TOSS-TB). Hasil skrining ini adalah alat bantu triase, bukan diagnosis.
  </p>
  <div className="mt-10 flex justify-between text-[14px]">
    <div>
      Petugas pengirim
      <div className="mt-14 w-56 border-t border-ink/40" />
    </div>
    <div>
      Penerima rujukan
      <div className="mt-14 w-56 border-t border-ink/40" />
    </div>
  </div>
</div>
</>
);
}

```

web/src/components/SiteHeader.tsx

```
"use client";

import Link from "next/link";
import { usePathname } from "next/navigation";

const NAV = [
  { href: "/skrining", label: "Skrining" },
  { href: "/pasien", label: "Pasien" },
  { href: "/klinisi", label: "Klinisi" },
  { href: "/tentang", label: "Tentang" },
];

export default function SiteHeader() {
  const path = usePathname();
  return (
    <header className="no-print sticky top-0 z-40 border-b border-ink/10 bg-paper">
      <div className="mx-auto flex max-w-3xl items-center gap-2 px-4 py-2.5 sm:gap-3 sm:px-5 sm:py-3">
        <Link href="/" className="flex shrink-0 items-baseline gap-2">
          <span className="text-[18px] font-extrabold tracking-tight sm:text-[19px]">SIGAP</span>
          <span className="hidden text-[12px] text-ink/50 sm:block">skrining TB</span>
        </Link>
        <nav aria-label="Navigasi utama" className="ml-auto flex gap-0.5 overflow-x-auto sm:gap-1">
          {NAV.map((n) => {
            const active = path?.startsWith(n.href);
            return (
              <Link
                key={n.href}
                href={n.href}
                aria-current={active ? "page" : undefined}
                className={`whitespace-nowrap rounded-[8px] px-2.5 py-2 text-[13px] font-medium transition-colors sm:px-3 sm:text-[14px] ${
                  active
                    ? "bg-ink text-paper"
                    : "text-ink/65 hover:bg-ink/5 hover:text-ink"
                }`}
              >
                {n.label}
              </Link>
            );
          })}
        </nav>
      </div>
    </header>
  );
}
```

web/src/app/globals.css

```
@import "tailwindcss";

@theme {
  --color-paper: #f2f4f5;
  --color-ink: #14181d;
  --color-film: #0b0f13;
  --color-bone: #e9e7e0;
  --color-refer: #c0392b;
  --color-defer: #b7791f;
  --color-clear: #2e7d5b;
  --font-sans: var(--font-jakarta), system-ui, -apple-system, "Segoe UI", sans-serif;
}

html {
  -webkit-text-size-adjust: 100%;
}

body {
  background: var(--color-paper);
  color: var(--color-ink);
  font-family: var(--font-sans);
}

::selection {
  background: var(--color-ink);
  color: var(--color-paper);
}

:focus-visible {
  outline: 2px solid var(--color-defer);
  outline-offset: 2px;
  border-radius: 2px;
}

@keyframes rise {
  from {
    opacity: 0;
    transform: translateY(10px);
  }
  to {
    opacity: 1;
    transform: none;
  }
}

.reveal {
  animation: rise 0.4s cubic-bezier(0.2, 0.7, 0.2, 1) both;
}

@media (prefers-reduced-motion: reduce) {
  *,
  *::before,
  *::after {
    animation-duration: 0.01ms !important;
    transition-duration: 0.01ms !important;
  }
}

@media print {
  .no-print {
    display: none !important;
  }
  body {
    background: #fff;
  }
}
```

```
{
  "name": "web",
  "version": "0.1.0",
  "lockfileVersion": 3,
  "requires": true,
  "packages": {
    "": {
      "name": "web",
      "version": "0.1.0",
      "dependencies": {
        "next": "16.3.8",
        "react": "19.2.8",
        "react-dom": "19.2.8"
      },
      "devDependencies": {
        "@tailwindcss/postcss": "^4",
        "@types/node": "^20",
        "@types/react": "^19",
        "@types/react-dom": "^19",
        "tailwindcss": "^4",
        "typescript": "^5"
      }
    },
    "node_modules/@alloc/quick-lru": {
      "version": "5.3.0",
      "resolved": "https://registry.npmjs.org/@alloc/quick-lru/-/quick-lru-5.3.0.tgz",
      "integrity": "sha512-U4+70PcS5ZS9osnCBCE5Jha/ciHM+Yp+CNMNC/7HvYbNRk1Ldd+f7q065W5qfhu/TCv+/ozljXXe9Nj8419DMA==",
      "dev": true,
      "license": "MIT",
      "engines": {
        "node": ">=10"
      },
      "funding": {
        "url": "https://github.com/sponsors/sindresorhus"
      }
    },
    "node_modules/@emnapi/runtime": {
      "version": "1.11.3",
      "resolved": "https://registry.npmjs.org/@emnapi/runtime/-/runtime-1.11.3.tgz",
      "integrity": "sha512-QzXGvUxOwEg8tV3pFdYX4ZCQUDNYOu68so5zoDwOkiZl2nzJGJl6uzpE5+oaj8Neu8mke1hLezAhiOIvdA==",
      "license": "MIT",
      "optional": true,
      "dependencies": {
        "tslib": "^2.4.0"
      }
    },
    "node_modules/@img/colour": {
      "version": "1.1.0",
      "resolved": "https://registry.npmjs.org/@img/colour/-/colour-1.1.0.tgz",
      "integrity": "sha512-Td76q7j57o/tLVdgS746cYARfSyxk8iEfRxewL9h40MZYhbW4TAcpp10mT4eyqXddh6L/jwoM75mo7ixa/pCeQ==",
      "license": "MIT",
      "optional": true,
      "engines": {
        "node": ">=18"
      }
    },
    "node_modules/@img/sharp-darwin-arm64": {
      "version": "0.35.5",
      "resolved": "https://registry.npmjs.org/@img/sharp-darwin-arm64/-/sharp-darwin-arm64-0.35.5.tgz",
      "integrity": "sha512-QRUIFQ0WxdWyqqG/WtI3iupfD5rBzmCHXSdPsY9IsAtVtTo7Q4cb6z0ccZ3gqEqkr0f1As1ehLqmEpDsRf+lq==",
      "cpu": [
        "arm64"
      ],
      "license": "Apache-2.0",
      "optional": true,
      "os": [
        "darwin"
      ],
      "engines": {
        "node": ">=20.9.0"
      },
      "funding": {
        "url": "https://opencollective.com/libvips"
      },
      "optionalDependencies": {
        "@img/sharp-libvips-darwin-arm64": "1.3.4"
      }
    },
    "node_modules/@img/sharp-darwin-x64": {
      "version": "0.35.5",
      "resolved": "https://registry.npmjs.org/@img/sharp-darwin-x64/-/sharp-darwin-x64-0.35.5.tgz",
      "integrity": "sha512-BR255RhDlpYgUpOc/JdtInT6DQ3XG/ERoSwbcd0f5Q320dKtPCCKPLR1LJs9VGXRaMa81luUa0tkCN0XiAxZUw==",
      "cpu": [
        "x64"
      ],
      "license": "Apache-2.0",
      "optional": true,
      "os": [

```

```
{
  "name": "web",
  "version": "0.1.0",
  "lockfileVersion": 3,
  "requires": true,
  "packages": {
    "": {
      "name": "web",
      "version": "0.1.0",
      "dependencies": {
        "next": "16.3.8",
        "react": "19.2.8",
        "react-dom": "19.2.8"
      },
      "devDependencies": {
        "@tailwindcss/postcss": "^4",
        "@types/node": "^20",
        "@types/react": "^19",
        "@types/react-dom": "^19",
        "tailwindcss": "^4",
        "typescript": "^5"
      }
    },
    "node_modules/@alloc/quick-lru": {
      "version": "5.3.0",
      "resolved": "https://registry.npmjs.org/@alloc/quick-lru/-/quick-lru-5.3.0.tgz",
      "integrity": "sha512-U441QsW7YUcUk+YMUT4OFkw3qjxiP+bD8I1K9FI02HV2JY+/PtS6Vz61y8Hx9+ys9sGPxYpL5MsPtN0tRkA==",
      "dev": true,
      "license": "MIT",
      "engines": {
        "node": ">=10"
      }
    },
    "node_modules/@emnapi/runtime": {
      "version": "1.11.3",
      "resolved": "https://registry.npmjs.org/@emnapi/runtime/-/runtime-1.11.3.tgz",
      "integrity": "sha512-Q09X0MkPpKz53Xa2R0aYz76YXJ99Dj91H8d968477dCE8ZB4j98We1cH810Yt69DNR68Vtzc9D8z1uQ==",
      "optional": true,
      "dependencies": {
        "tslib": "^2.4.0"
      }
    },
    "node_modules/@img/colour": {
      "version": "1.1.0",
      "resolved": "https://registry.npmjs.org/@img/colour/-/colour-1.1.0.tgz",
      "integrity": "sha512-Td76q7j57o/tLVdgS746cYARfSyxk8iEfRxewL9h40MzYhbW4TAcpl10mT4eyqXddh6L/jwoM75mo7ixa/pCeQ==",
      "optional": true,
      "engines": {
        "node": ">=18"
      }
    },
    "node_modules/@img/sharp-darwin-arm64": {
      "version": "0.35.5",
      "resolved": "https://registry.npmjs.org/@img/sharp-darwin-arm64/-/sharp-darwin-arm64-0.35.5.tgz",
      "integrity": "sha512-QRULFQ0WxvdWyqqG/WtI3iupfD5rBzmCHXSDPsY91sAtVtTo7Q4cb6z0ccZ3ggEqkr0f1As1ehLqmEpDsRf+lg==",
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      ],
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      "optional": true,
      "os": [
        "darwin"
      ],
      "engines": {
        "node": ">=20.9.0"
      }
    },
    "node_modules/@img/sharp-darwin-x64": {
      "version": "0.35.5",
      "resolved": "https://registry.npmjs.org/@img/sharp-darwin-x64/-/sharp-darwin-x64-0.35.5.tgz",
      "integrity": "sha512-BR255RhDlpygUp0c/Jdt1nT6DQ3XG/ER05wbcd0f5Q320dKtPCKPLR1LJs9VGXRaMa811uUa0tkCNOXiAxZUw==",
      "cpu": [
        "x64"
      ],
      "license": "Apache-2.0",
      "optional": true,
      "os": [

```

```

    "darwin"
  ],
  "engines": {
    "node": ">=20.9.0"
  },
  "funding": {
    "url": "https://opencollective.com/libvips"
  },
  "optionalDependencies": {
    "@img/sharp-libvips-darwin-x64": "1.3.4"
  }
},
"node_modules/@img/sharp-freebsd-wasm32": {
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  "resolved": "https://registry.npmjs.org/@img/sharp-freebsd-wasm32/-/sharp-freebsd-wasm32-0.35.5.tgz",
  "integrity": "sha512-Y/z91nEZ4uIBX5X3nfTovjU9lHNKFYbL2lpHCLVnmXQK03VIZvXBBt0KxbPGp2SdGSF+2mQU4e+hQaW0t86iAw=",
  "license": "Apache-2.0",
  "optional": true,
  "os": [
    "freebsd"
  ],
  "dependencies": {
    "@img/sharp-wasm32": "0.35.5"
  },
  "engines": {
    "node": ">=20.9.0"
  },
  "funding": {
    "url": "https://opencollective.com/libvips"
  }
},
"node_modules/@img/sharp-libvips-darwin-arm64": {
  "version": "1.3.4",
  "resolved": "https://registry.npmjs.org/@img/sharp-libvips-darwin-arm64/-/sharp-libvips-darwin-arm64-1.3.4.tgz",
  "integrity": "sha512-5R89nBYiRdUlsWJxPh0+GVtaXzXSxKnRu/xqMn3KTA3L9EB90y/P+Nn2f2v1hPuUdy/Zusb2DarbyTpGCfEduw=",
  "cpu": [
    "arm64"
  ],
  "license": "LGPL-3.0-or-later",
  "optional": true,
  "os": [
    "darwin"
  ],
  "funding": {
    "url": "https://opencollective.com/libvips"
  }
},
"node_modules/@img/sharp-libvips-darwin-x64": {
  "version": "1.3.4",
  "resolved": "https://registry.npmjs.org/@img/sharp-libvips-darwin-x64/-/sharp-libvips-darwin-x64-1.3.4.tgz",
  "integrity": "sha512-iR20KH80yi0U+dUplyh3/xdpFvps6YkCwsXenIJxqR1v9o+xtKTGbS9H7cps+2Vxjc8B1j96p75NmTGjIhtpQ=",
  "cpu": [
    "x64"
  ],
  "license": "LGPL-3.0-or-later",
  "optional": true,
  "os": [
    "darwin"
  ],
  "funding": {
    "url": "https://opencollective.com/libvips"
  }
},
"node_modules/@img/sharp-libvips-linux-arm": {
  "version": "1.3.4",
  "resolved": "https://registry.npmjs.org/@img/sharp-libvips-linux-arm/-/sharp-libvips-linux-arm-1.3.4.tgz",
  "integrity": "sha512-LmRtT0HuvM2+w102Db37d5MiZhB0FvSunciw48Yjd0kZz9KAiRbm8ujeMOA1INqmei5NapFxEK1D1ZSidmw=",
  "cpu": [
    "arm"
  ],
  "license": "LGPL-3.0-or-later",
  "optional": true,
  "os": [
    "linux"
  ],
  "funding": {
    "url": "https://opencollective.com/libvips"
  }
},
"node_modules/@img/sharp-libvips-linux-arm64": {
  "version": "1.3.4",
  "resolved": "https://registry.npmjs.org/@img/sharp-libvips-linux-arm64/-/sharp-libvips-linux-arm64-1.3.4.tgz",
  "integrity": "sha512-Y3dgX/61E2QhQb+Gxy0WZxfG9MEml/JBjamZpS2Ik1P7xIQoKN4hzAm7KcMVGtaVdt3neE90KBC7vAfonA/Lr1A=",
  "cpu": [
    "arm64"
  ],
  "license": "LGPL-3.0-or-later",
  "optional": true,
  "os": [
    "linux"
  ],
  "funding": {

```

```

    "url": "https://opencollective.com/libvips"
  },
  "node_modules/@img/sharp-libvips-linux-ppc64": {
    "version": "1.3.4",
    "resolved": "https://registry.npmjs.org/@img/sharp-libvips-linux-ppc64/-/sharp-libvips-linux-ppc64-1.3.4.tgz",
    "integrity": "sha512-Le6boB8Tai0Nis+gIxIpKx68UDVVIQdR8Tin5Yf1z2LJJQLDJvCDRqRu+jC2qCoD+eIomonm0wB4smBRxfVpYQ==",
    "cpu": [
      "ppc64"
    ],
    "license": "LGPL-3.0-or-later",
    "optional": true,
    "os": [
      "linux"
    ],
    "funding": {
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    "url": "https://opencollective.com/libvips"
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  "optionalDependencies": {

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"license": "Apache-2.0 AND LGPL-3.0-or-later AND MIT",
"optional": true,
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"engines": {
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"funding": {
  "url": "https://opencollective.com/libvips"
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  }
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  "optional": true,
  "os": [
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  "optional": true,
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"dev": true,
"license": "MIT",
"dependencies": {
  "@jridgewell/sourcemap-codec": "^1.5.0",
  "@jridgewell/trace-mapping": "^0.3.24"
}
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"license": "MIT",
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  "@jridgewell/trace-mapping": "^0.3.24"
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  "@jridgewell/sourcemap-codec": "^1.4.14"
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"integrity": "sha512-51SsM0TXURPglDfVuAQUqGek9Hg2Eks0Yay2m0+XR++b2NWYL/4sWyuVBIIs8oKnJaxkdi9whaL/sqN13afw==",
"license": "Apache-2.0",
"dependencies": {
  "tslib": "^2.8.0"
}
},
"node_modules/@tailwindcss/node": {
"version": "4.3.3",
"resolved": "https://registry.npmjs.org/@tailwindcss/node/-/node-4.3.3.tgz",
"integrity": "sha512-/T8IKesf9VTU6tLjgC7+sv2m0PtQxzE2jMw7u4Tt40Tx+QSZxpzh95/H6cMKoja9XuW7iMdLJYBB0o9G1CaAgg==",
"dev": true,
"license": "MIT",
"dependencies": {
  "@jridgewell/remapping": "^2.3.5",
  "enhanced-resolve": "^5.24.1",
  "jiti": "^2.7.0",
  "lightningcss": "1.32.0",
  "magic-string": "^0.30.21",
  "source-map-js": "^1.2.1",
  "tailwindcss": "4.3.3"
}
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"node_modules/@tailwindcss/oxide": {
"version": "4.3.3",
"resolved": "https://registry.npmjs.org/@tailwindcss/oxide/-/oxide-4.3.3.tgz",
"integrity": "sha512-kzXjAikiaFSPaK/FkAQT5UTx3VormQaiZ5hBF1JZ9UFGb/rwg1MZIhHAG9smMQRTdyJxP6Qt5MwMtdyU5FWrA==",
"dev": true,
"license": "MIT",
"engines": {
  "node": ">= 20"
}
},
"optionalDependencies": {
  "@tailwindcss/oxide-android-arm64": "4.3.3",
  "@tailwindcss/oxide-darwin-arm64": "4.3.3",
  "@tailwindcss/oxide-darwin-x64": "4.3.3",
  "@tailwindcss/oxide-freebsd-x64": "4.3.3",
  "@tailwindcss/oxide-linux-arm-gnueabi": "4.3.3",
  "@tailwindcss/oxide-linux-arm64-gnu": "4.3.3",
  "@tailwindcss/oxide-linux-arm64-musl": "4.3.3",
  "@tailwindcss/oxide-linux-x64-gnu": "4.3.3",
  "@tailwindcss/oxide-linux-x64-musl": "4.3.3",
  "@tailwindcss/oxide-wasm32-wasi": "4.3.3",
  "@tailwindcss/oxide-win32-arm64-msvc": "4.3.3",
  "@tailwindcss/oxide-win32-x64-msvc": "4.3.3"
}
},
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"cpu": [
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],
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"optional": true,
"os": [
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],
"engines": {
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}
},
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"cpu": [
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],
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"optional": true,
"os": [
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"engines": {
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}
},
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"cpu": [
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],
"dev": true,
"license": "MIT",
"optional": true,

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    "os": [
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    "engines": {
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    }
  },
  "node_modules/@tailwindcss/oxide-freebsd-x64": {
    "version": "4.3.3",
    "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-freebsd-x64/-/oxide-freebsd-x64-4.3.3.tgz",
    "integrity": "sha512-iyf5bV6+wnA1f1VeEy7R25dupxTNECZN5QMI0qNT6eT+EgaGdZcKhGkr5SdoaWiLJ3spLqIY9VCeSGrwtg4kw=",
    "cpu": [
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    ],
    "dev": true,
    "license": "MIT",
    "optional": true,
    "os": [
      "freebsd"
    ],
    "engines": {
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    }
  },
  "node_modules/@tailwindcss/oxide-linux-arm-gnueabihf": {
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    "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-linux-arm-gnueabihf/-/oxide-linux-arm-gnueabihf-4.3.3.tgz",
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    "cpu": [
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    ],
    "dev": true,
    "license": "MIT",
    "optional": true,
    "os": [
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    ],
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    }
  },
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    "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-linux-arm64-gnu/-/oxide-linux-arm64-gnu-4.3.3.tgz",
    "integrity": "sha512-nDxldcEEN0xZRzC2uu9jrtZDAAQtb+8WWDCSnWL1zvBk1+FN+x6MtDViPB5AJMfttVCUhehGWus3XBPgatM/w=",
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    "optional": true,
    "os": [
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    "integrity": "sha512-Md44bD6veX/PC5iyF8cDVnw4HBIAZepRZZ7a8DQ0vkfo5WUBwcp6iAuCUz23u+4SukhJlD3eL7hNdW8ezd/kA=",
    "cpu": [
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    "dev": true,
    "license": "MIT",
    "optional": true,
    "os": [
      "linux"
    ],
    "engines": {
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    }
  },
  "node_modules/@tailwindcss/oxide-linux-x64-gnu": {
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    "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-linux-x64-gnu/-/oxide-linux-x64-gnu-4.3.3.tgz",
    "integrity": "sha512-tx7us1muwOKAKWao2v/GaafFeQboE6aj88vC6ziN2NCGcRm8gWUhwjzg+YdVB1e4boAtdtma4L43onunI6NS4w=",
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    "dev": true,
    "license": "MIT",
    "optional": true,
    "os": [
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    }
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  "cpu": [
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  "dev": true,
  "license": "MIT",
  "optional": true,
  "os": [
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},
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  "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-wasm32-wasi/-/oxide-wasm32-wasi-4.3.3.tgz",
  "integrity": "sha512-jx1+rPhY/5YmpkktD656HBWEbLxP7dH06losBLjff5vgCODXvi9KhtftWcMIwTFIDqBr7cRnQkdLnAG+I0lGvQ=",
  "bundleDependencies": [
    "@napi-rs/wasm-runtime",
    "@emnapi/core",
    "@emnapi/runtime",
    "@tybys/wasm-util",
    "@emnapi/wasi-threads",
    "tslib"
  ],
  "cpu": [
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  ],
  "dev": true,
  "license": "MIT",
  "optional": true,
  "dependencies": {
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    "@emnapi/runtime": "^1.11.1",
    "@emnapi/wasi-threads": "^1.2.2",
    "@napi-rs/wasm-runtime": "^1.1.4",
    "@tybys/wasm-util": "^0.10.2",
    "tslib": "^2.8.1"
  },
  "engines": {
    "node": ">=14.0.0"
  }
},
"node_modules/@tailwindcss/oxide-win32-arm64-msvc": {
  "version": "4.3.3",
  "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-win32-arm64-msvc/-/oxide-win32-arm64-msvc-4.3.3.tgz",
  "integrity": "sha512-3rc292Ca2ceK6Ulc/bAVnTs/3nDtpPhyEKlGpV+yQJQi/JS/AMJlqzxv1DacL1nekbrcf6bTqp/jv4qgnPxNQ=",
  "cpu": [
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  "license": "MIT",
  "optional": true,
  "os": [
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  "engines": {
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  "resolved": "https://registry.npmjs.org/@tailwindcss/oxide-win32-x64-msvc/-/oxide-win32-x64-msvc-4.3.3.tgz",
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  "cpu": [
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  "dev": true,
  "license": "MIT",
  "optional": true,
  "os": [
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  ],
  "engines": {
    "node": ">= 20"
  }
},
"node_modules/@tailwindcss/postcss": {
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  "resolved": "https://registry.npmjs.org/@tailwindcss/postcss/-/postcss-4.3.3.tgz",
  "integrity": "sha512-JTSZZGQi1AyKirbLN3azmjVzef92tcX7hiSqPdaeStyFpGpDlKvvpXe0E8njhbUanbRwr3z8DyZhICWnMtQeg=",
  "dev": true,
  "license": "MIT",
  "dependencies": {
    "@alloc/quick-lru": "^5.2.0",
    "@tailwindcss/node": "4.3.3",
    "@tailwindcss/oxide": "4.3.3",
    "postcss": "^8.5.16",
  }
}

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    "tailwindcss": "4.3.3"
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    "resolved": "https://registry.npmjs.org/@types/node/-/node-20.19.43.tgz",
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    "dev": true,
    "license": "MIT",
    "dependencies": {
      "undici-types": "~6.21.0"
    }
  },
  "node_modules/@types/react": {
    "version": "19.3.0",
    "resolved": "https://registry.npmjs.org/@types/react/-/react-19.3.0.tgz",
    "integrity": "sha512-N0rFcuH9YoxG9/m6119MfpJKfmLOVU0em7ipIz6TRgSSkvReLB9vL85GB+yr8Bs51eqpv96JSwF4Z51s4viQg==",
    "dev": true,
    "license": "MIT",
    "dependencies": {
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    }
  },
  "node_modules/@types/react-dom": {
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    "resolved": "https://registry.npmjs.org/@types/react-dom/-/react-dom-19.3.0.tgz",
    "integrity": "sha512-ZI7bU42mZXXKhN/qNLEw2IrbIINU7X5+vf9gdxBHkCNPYWXjKgFQ/P+uyGb5Cj0LB9UcnTeg3rylQtV2hym44Q==",
    "dev": true,
    "license": "MIT",
    "peerDependencies": {
      "@types/react": "^19.3.0"
    }
  },
  "node_modules/baseline-browser-mapping": {
    "version": "2.11.27",
    "resolved": "https://registry.npmjs.org/baseline-browser-mapping/-/baseline-browser-mapping-2.11.27.tgz",
    "integrity": "sha512-ElY12DaR0Guan+lMmZ8Cvo/ZUbXPe7Enc/9VU/b1T3Kp4dwytrCnR8DoSJN5SNJT/CuvCA0DHDVmM0CePdRQ==",
    "license": "Apache-2.0",
    "bin": {
      "baseline-browser-mapping": "dist/cli.cjs"
    },
    "engines": {
      "node": ">=6.0.0"
    }
  },
  "node_modules/caniuse-lite": {
    "version": "1.0.30001814",
    "resolved": "https://registry.npmjs.org/caniuse-lite/-/caniuse-lite-1.0.30001814.tgz",
    "integrity": "sha512-Uaf11Azr59XcMpw0o96WoEfr+VXK20X4U9AgFoi5ShsVJ4HppnIFUjtYzsyDH2+tgANaQb2/oxYGwCPapN1FpA==",
    "funding": [
      {
        "type": "opencollective",
        "url": "https://opencollective.com/browserslist"
      },
      {
        "type": "tidelift",
        "url": "https://tidelift.com/funding/github/npm/caniuse-lite"
      },
      {
        "type": "github",
        "url": "https://github.com/sponsors/ai"
      }
    ],
    "license": "CC-BY-4.0"
  },
  "node_modules/client-only": {
    "version": "0.0.1",
    "resolved": "https://registry.npmjs.org/client-only/-/client-only-0.0.1.tgz",
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    "license": "MIT"
  },
  "node_modules/csstype": {
    "version": "3.2.3",
    "resolved": "https://registry.npmjs.org/csstype/-/csstype-3.2.3.tgz",
    "integrity": "sha512-z1HGKcYy2xA8AGQfwrn0PAY+PB7X/GSj3UVJW9qKyn43Wag15nXmU4qqLMRzWVLCFC8KusUX8T/0kCi0YpAIQ==",
    "dev": true,
    "license": "MIT"
  },
  "node_modules/detect-libc": {
    "version": "2.1.2",
    "resolved": "https://registry.npmjs.org/detect-libc/-/detect-libc-2.1.2.tgz",
    "integrity": "sha512-6xAg6UkMvDwKsP0Shrh/59489UcH3Uo6L203xkD6F5LZ42y7VX6Ud6M4KwT437594tOeWdi0znDfE6Q==",
    "devOptional": true,
    "license": "Apache-2.0",
    "engines": {
      "node": ">=8"
    }
  },
  "node_modules/enhanced-resolve": {
    "version": "5.26.0",
    "resolved": "https://registry.npmjs.org/enhanced-resolve/-/enhanced-resolve-5.26.0.tgz",
    "integrity": "sha512-9vhYk1cKBS08lJR3t71Uw+0S6l7wI0xYl1F788vF8lWpLW0V6VwDp0T+yS0dYvYQbLQ0aV1KdOm45Wg==",

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    "dev": true,
    "license": "MIT",
    "dependencies": {
      "graceful-fs": "^4.2.4",
      "tapable": "^2.3.3"
    },
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    }
  },
  "node_modules/graceful-fs": {
    "version": "4.2.11",
    "resolved": "https://registry.npmjs.org/graceful-fs/-/graceful-fs-4.2.11.tgz",
    "integrity": "sha512-RbJ5/jmFcNNCcdV5o9eTnBLJ/HszwV0P73bc+F4nS/rJj+YaS6IGyiOL0VoBYX+1lWrl3k63h/KrH+nhJ0XvQ==",
    "dev": true,
    "license": "ISC"
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    "version": "2.7.0",
    "resolved": "https://registry.npmjs.org/jiti/-/jiti-2.7.0.tgz",
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    "dev": true,
    "license": "MIT",
    "bin": {
      "jiti": "lib/jiti-cli.mjs"
    }
  },
  "node_modules/lightningcss": {
    "version": "1.32.0",
    "resolved": "https://registry.npmjs.org/lightningcss/-/lightningcss-1.32.0.tgz",
    "integrity": "sha512-NXYBzinNrb1fraPGyrbPoD19C1h91fI/1mzgWYvXUte414Gz/X1FD2XBZSZM7rRTMA8JL30tAaGifrIKhQ5yQ==",
    "dev": true,
    "license": "MPL-2.0",
    "dependencies": {
      "detect-libc": "^2.0.3"
    },
    "engines": {
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    },
    "funding": {
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      "url": "https://opencollective.com/parcel"
    },
    "optionalDependencies": {
      "lightningcss-android-arm64": "1.32.0",
      "lightningcss-darwin-arm64": "1.32.0",
      "lightningcss-darwin-x64": "1.32.0",
      "lightningcss-freebsd-x64": "1.32.0",
      "lightningcss-linux-arm-gnueabihf": "1.32.0",
      "lightningcss-linux-arm64-gnu": "1.32.0",
      "lightningcss-linux-arm64-musl": "1.32.0",
      "lightningcss-linux-x64-gnu": "1.32.0",
      "lightningcss-linux-x64-musl": "1.32.0",
      "lightningcss-win32-arm64-msvc": "1.32.0",
      "lightningcss-win32-x64-msvc": "1.32.0"
    }
  },
  "node_modules/lightningcss-android-arm64": {
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    "resolved": "https://registry.npmjs.org/lightningcss-android-arm64/-/lightningcss-android-arm64-1.32.0.tgz",
    "integrity": "sha512-YK7/ClTt4kAK0vo6w3X+Pnm0D2cf2vPHbh0XdoNti1Ga0al1P4TBzhwjATvjNwLEBCnKvjJc2jQgHXH0NEw1Ag==",
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    "dev": true,
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    "optional": true,
    "os": [
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    ]
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    "resolved": "https://registry.npmjs.org/lightningcss-darwin-arm64/-/lightningcss-darwin-arm64-1.32.0.tgz",
    "integrity": "sha512-RzeG9Ju5bag2Bv1/lw1VJvBE3q6TtXskdZLLCyfg5pt+HLz9Bq1IC07LZM7VHNTn/5PRhHFBsjk51c4cmscPQ==",
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    ],
    "dev": true,
    "license": "MPL-2.0",
    "optional": true,
    "os": [
      "darwin"
    ]
  },

```

```

    "node": ">= 12.0.0"
  },
  "funding": {
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    "url": "https://opencollective.com/parcel"
  }
},
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  "resolved": "https://registry.npmjs.org/lightningcss-darwin-x64/-/lightningcss-darwin-x64-1.32.0.tgz",
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  "dev": true,
  "license": "MPL-2.0",
  "optional": true,
  "os": [
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  ],
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    "url": "https://opencollective.com/parcel"
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},
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  "optional": true,
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  ],
  "engines": {
    "node": ">= 12.0.0"
  },
  "funding": {
    "type": "opencollective",
    "url": "https://opencollective.com/parcel"
  }
},
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  "version": "1.32.0",
  "resolved": "https://registry.npmjs.org/lightningcss-linux-arm64-gnu/-/lightningcss-linux-arm64-gnu-1.32.0.tgz",
  "integrity": "sha512-0nnMyoyOLRXfbM0ilaSRcLH3Jw5z9HDNGfT/gwCPgaDjnx0i8w7vBzFLFR1f6CMLKF8gVbebmKUN3fa/kJpQ==",
  "cpu": [
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  ],
  "dev": true,
  "license": "MPL-2.0",
  "optional": true,
  "os": [
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  ],
  "engines": {
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  },
  "funding": {
    "type": "opencollective",
    "url": "https://opencollective.com/parcel"
  }
},
"node_modules/lightningcss-linux-arm64-musl": {

```

```

"version": "1.32.0",
"resolved": "https://registry.npmjs.org/lightningcss-linux-arm64-musl/-/lightningcss-linux-arm64-musl-1.32.0.tgz",
"integrity": "sha512-UpQkoenr4UJEzgVIYpI80lDFvRmPVg6oqboNHfoH4CQIfNA+H0rZ7Mo7KZP02dC6LjghPQJeBsvXhJod/wnIBg==",
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],
"dev": true,
"license": "MPL-2.0",
"optional": true,
"os": [
  "linux"
],
"engines": {
  "node": ">= 12.0.0"
},
"funding": {
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  "url": "https://opencollective.com/parcel"
}
},
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"integrity": "sha512-V7Qr52IhZmdKPVr+Vtw8o+WLSQJYCTd8l0IfpDaMRWGUZfBOYEJeyJIkqGIDMZPwPx24pUMfwSxxI8phr/Mb0A==",
"cpu": [
  "x64"
],
"dev": true,
"license": "MPL-2.0",
"optional": true,
"os": [
  "linux"
],
"engines": {
  "node": ">= 12.0.0"
},
"funding": {
  "type": "opencollective",
  "url": "https://opencollective.com/parcel"
}
},
"node_modules/lightningcss-linux-x64-musl": {
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"cpu": [
  "x64"
],
"dev": true,
"license": "MPL-2.0",
"optional": true,
"os": [
  "linux"
],
"engines": {
  "node": ">= 12.0.0"
},
"funding": {
  "type": "opencollective",
  "url": "https://opencollective.com/parcel"
}
},
"node_modules/lightningcss-win32-arm64-msvc": {
"version": "1.32.0",
"resolved": "https://registry.npmjs.org/lightningcss-win32-arm64-msvc/-/lightningcss-win32-arm64-msvc-1.32.0.tgz",
"integrity": "sha512-8SbC8BR40pS6baCM8sbtYDSwEVQd4JlFT0laD3gWGHfThTcABnNDBda6eTZeQbofaliJhFx0qKzghJmcPtNdw==",
"cpu": [
  "arm64"
],
"dev": true,
"license": "MPL-2.0",
"optional": true,
"os": [
  "win32"
],
"engines": {
  "node": ">= 12.0.0"
},
"funding": {
  "type": "opencollective",
  "url": "https://opencollective.com/parcel"
}
},
"node_modules/lightningcss-win32-x64-msvc": {
"version": "1.32.0",
"resolved": "https://registry.npmjs.org/lightningcss-win32-x64-msvc/-/lightningcss-win32-x64-msvc-1.32.0.tgz",
"integrity": "sha512-Amq9B/SoZyDi1kFrojnoqPLxYhQ4Wo5XiL8EVJrVsB8ARoC1PWW6VGtT0WKCemjy8aC+louJnjS7U18x3b06Q==",
"cpu": [
  "x64"
],
"dev": true,
"license": "MPL-2.0",

```

```

    "optional": true,
    "os": [
      "win32"
    ],
    "engines": {
      "node": ">= 12.0.0"
    },
    "funding": {
      "type": "opencollective",
      "url": "https://opencollective.com/parcel"
    }
  },
  "node_modules/magic-string": {
    "version": "0.30.21",
    "resolved": "https://registry.npmjs.org/magic-string/-/magic-string-0.30.21.tgz",
    "integrity": "sha512-udRcpGjhVtUoW4zgB+NK09OAdPvE1H5jJl8Zi6eRI+Q3xrdLkMg1aVDEAY8ySLLWCI2hWxdXCL1mhdTBFGw==",
    "dev": true,
    "license": "MIT",
    "dependencies": {
      "@jridgewell/source-map-codec": "^1.5.5"
    }
  },
  "node_modules/nanoid": {
    "version": "3.3.19",
    "resolved": "https://registry.npmjs.org/nanoid/-/nanoid-3.3.19.tgz",
    "integrity": "sha512-Y2tUNY4ouW6tq5oDSKeQYG0yhkUBhN0cGV/02KC+6kd9eDGqdZd++mjMiIDilrBYvjEnCYvVtsuHCuP+okSfug=",
    "funding": [
      {
        "type": "github",
        "url": "https://github.com/sponsors/ai"
      }
    ],
    "license": "MIT",
    "bin": {
      "nanoid": "bin/nanoid.cjs"
    },
    "engines": {
      "node": "^10 || ^12 || ^13.7 || ^14 || >=15.0.1"
    }
  },
  "node_modules/next": {
    "version": "16.3.8",
    "resolved": "https://registry.npmjs.org/next/-/next-16.3.8.tgz",
    "integrity": "sha512-U7QEZAini6WKrb8A8hqLLqYQyCetegKjCpJOyxk642vWoMoU1x5PyZCJFvgYgiptA8xc5j/9xY1ZF07w9Sjmw=",
    "license": "MIT",
    "dependencies": {
      "@next/env": "16.3.8",
      "@swc/helpers": "0.5.23",
      "baseline-browser-mapping": "^2.9.19",
      "caniuse-lite": "^1.0.30001579",
      "postcss": "8.5.23",
      "styled-jsx": "5.1.6"
    },
    "bin": {
      "next": "dist/bin/next"
    },
    "engines": {
      "node": ">=20.9.0"
    },
    "optionalDependencies": {
      "@next/swc-darwin-arm64": "16.3.8",
      "@next/swc-darwin-x64": "16.3.8",
      "@next/swc-linux-arm64-gnu": "16.3.8",
      "@next/swc-linux-arm64-musl": "16.3.8",
      "@next/swc-linux-x64-gnu": "16.3.8",
      "@next/swc-linux-x64-musl": "16.3.8",
      "@next/swc-win32-arm64-msvc": "16.3.8",
      "@next/swc-win32-x64-msvc": "16.3.8",
      "sharp": "^0.35.4"
    },
    "peerDependencies": {
      "@opentelemetry/api": "^1.1.0",
      "@playwright/test": ">=1.51.1",
      "babel-plugin-react-compiler": "*",
      "react": ">=18.2.0 || 19.0.0-rc-de68d2f4-20241204 || ^19.0.0",
      "react-dom": ">=18.2.0 || 19.0.0-rc-de68d2f4-20241204 || ^19.0.0",
      "sass": ">=1.3.0"
    },
    "peerDependenciesMeta": {
      "@opentelemetry/api": {
        "optional": true
      },
      "@playwright/test": {
        "optional": true
      },
      "babel-plugin-react-compiler": {
        "optional": true
      },
      "sass": {
        "optional": true
      }
    }
  }
}

```

```

},
"node_modules/next/node_modules/postcss": {
  "version": "8.5.23",
  "resolved": "https://registry.npmjs.org/postcss/-/postcss-8.5.23.tgz",
  "integrity": "sha512-g5058Zr4bZmwFiTlflMu8E0bDTb5I5gertgwAKmsdULTQIhZtunzUld1WSzwcVWPoAVpsrA6v1fCD7oXvRwgg==",
  "funding": [
    {
      "type": "opencollective",
      "url": "https://opencollective.com/postcss/"
    },
    {
      "type": "tidelift",
      "url": "https://tidelift.com/funding/github/npm/postcss"
    },
    {
      "type": "github",
      "url": "https://github.com/sponsors/ai"
    }
  ],
  "license": "MIT",
  "dependencies": {
    "nanoid": "^3.3.16",
    "picocolors": "^1.1.1",
    "source-map-js": "^1.2.1"
  },
  "engines": {
    "node": ">=10 || ^12 || >=14"
  }
},
"node_modules/picocolors": {
  "version": "1.1.1",
  "resolved": "https://registry.npmjs.org/picocolors/-/picocolors-1.1.1.tgz",
  "integrity": "sha512-xcH2sln+Dy60wU8801H3338qUy+YKf1Bj1WwL293hQz9N8H21H4Wrf0vYtFVqEfwOHdYd3T6Z+8Tg==",
  "license": "ISC"
},
"node_modules/postcss": {
  "version": "8.5.28",
  "resolved": "https://registry.npmjs.org/postcss/-/postcss-8.5.28.tgz",
  "integrity": "sha512-UzI1VtZ4Zq2uV9HkynwLDEnFqZd3P4TpL5pQdVM72e1ZwIOYnWDRHw2d7nqZMNGWqIZy370x1Y4wvxMg==",
  "dev": true,
  "funding": [
    {
      "type": "opencollective",
      "url": "https://opencollective.com/postcss/"
    },
    {
      "type": "tidelift",
      "url": "https://tidelift.com/funding/github/npm/postcss"
    },
    {
      "type": "github",
      "url": "https://github.com/sponsors/ai"
    }
  ],
  "license": "MIT",
  "dependencies": {
    "nanoid": "^3.3.18",
    "picocolors": "^1.1.1",
    "source-map-js": "^1.2.1"
  },
  "engines": {
    "node": ">=10 || ^12 || >=14"
  }
},
"node_modules/react": {
  "version": "19.2.8",
  "resolved": "https://registry.npmjs.org/react/-/react-19.2.8.tgz",
  "integrity": "sha512-PWaYA1L/q9u2u7xYQi+Y3L3Yfnie7XyLeaJICV1MGD6LprsBxcAqGjYyr0eY3p+QdsA+x/Irkt4Qif8D63+Sbw==",
  "license": "MIT",
  "engines": {
    "node": ">=0.10.0"
  }
},
"node_modules/react-dom": {
  "version": "19.2.8",
  "resolved": "https://registry.npmjs.org/react-dom/-/react-dom-19.2.8.tgz",
  "integrity": "sha512-1WU2D7Wb3x73DwZ53UyZ6B0U0bU1915Uv3C1b50Z75Zd5W5S4E5ZQ9dOAH9VLIzZJnsC+52j8Q09A==",
  "license": "MIT",
  "dependencies": {
    "scheduler": "^0.27.0"
  },
  "peerDependencies": {
    "react": "^19.2.8"
  }
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  "version": "0.27.0",
  "resolved": "https://registry.npmjs.org/scheduler/-/scheduler-0.27.0.tgz",
  "integrity": "sha512-4P5XWpZ15X7Y3ziR0sWb6HDNOMJ1mYa4lWiPwfm9HFKYlUkIb0AQ6rvHsCAkshTZJlRUVIbbaLW29ruFRg==",
  "license": "MIT"
},
"node_modules/semver": {

```

```

"version": "7.8.5",
"resolved": "https://registry.npmjs.org/semver/-/semver-7.8.5.tgz",
"integrity": "sha512-Y7/KDsb8LjooZpwaqGyu106DQlksGcncchHGk+sZIY4SBvUocMBEFH5Ur1fI4dV+Jv10w6cjvucaIi40puRioA==",
"license": "ISC",
"optional": true,
"bin": {
  "semver": "bin/semver.js"
},
"engines": {
  "node": ">=10"
}
},
"node_modules/sharp": {
"version": "0.35.5",
"resolved": "https://registry.npmjs.org/sharp/-/sharp-0.35.5.tgz",
"integrity": "sha512-Ywn40nzGukp7CDMrp08RQ50YKmuwG47brZgIVPTvBaaAfQlRlygrRqSrxdcil9M+LlZLBIJ68IR1QqvzHyjC7g==",
"license": "Apache-2.0",
"optional": true,
"dependencies": {
  "@img/colour": "^1.1.0",
  "detect-libc": "^2.1.2",
  "semver": "^7.8.5"
},
"engines": {
  "node": ">=20.9.0"
},
"funding": {
  "url": "https://opencollective.com/libvips"
},
"optionalDependencies": {
  "@img/sharp-darwin-arm64": "0.35.5",
  "@img/sharp-darwin-x64": "0.35.5",
  "@img/sharp-freebsd-wasm32": "0.35.5",
  "@img/sharp-libvips-darwin-arm64": "1.3.4",
  "@img/sharp-libvips-darwin-x64": "1.3.4",
  "@img/sharp-libvips-linux-arm": "1.3.4",
  "@img/sharp-libvips-linux-arm64": "1.3.4",
  "@img/sharp-libvips-linux-ppc64": "1.3.4",
  "@img/sharp-libvips-linux-riscv64": "1.3.4",
  "@img/sharp-libvips-linux-s390x": "1.3.4",
  "@img/sharp-libvips-linux-x64": "1.3.4",
  "@img/sharp-libvips-linuxmusl-arm64": "1.3.4",
  "@img/sharp-libvips-linuxmusl-x64": "1.3.4",
  "@img/sharp-linux-arm": "0.35.5",
  "@img/sharp-linux-arm64": "0.35.5",
  "@img/sharp-linux-ppc64": "0.35.5",
  "@img/sharp-linux-riscv64": "0.35.5",
  "@img/sharp-linux-s390x": "0.35.5",
  "@img/sharp-linux-x64": "0.35.5",
  "@img/sharp-linuxmusl-arm64": "0.35.5",
  "@img/sharp-linuxmusl-x64": "0.35.5",
  "@img/sharp-webcontainers-wasm32": "0.35.5",
  "@img/sharp-win32-arm64": "0.35.5",
  "@img/sharp-win32-ia32": "0.35.5",
  "@img/sharp-win32-x64": "0.35.5"
},
"peerDependenciesMeta": {
  "@types/node": {
    "optional": true
  }
}
},
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"resolved": "https://registry.npmjs.org/source-map-js/-/source-map-js-1.2.2.tgz",
"integrity": "sha512-8Y433333aZVDtt+J4mK1hoLGHULMYfSkODJNQjNDC3oW1PqPoxMwo0pLUswM/UEGzON/NxeHywEfNXPN3Vw==",
"license": "BSD-3-Clause",
"engines": {
  "node": ">=0.10.0"
}
},
"node_modules/styled-jsx": {
"version": "5.1.6",
"resolved": "https://registry.npmjs.org/styled-jsx/-/styled-jsx-5.1.6.tgz",
"integrity": "sha512-qSVyDTeMotdvQYoHwLNGwRFJHC+i+ZvdBRYosOFgC+Wg1vx4frN2/RG/NA7SYqqvKNLf39P2LSRA2pu6n0XYZA==",
"license": "MIT",
"dependencies": {
  "client-only": "0.0.1"
},
"engines": {
  "node": ">= 12.0.0"
},
"peerDependencies": {
  "react": ">= 16.8.0 || 17.x.x || ^18.0.0-0 || ^19.0.0-0"
},
"peerDependenciesMeta": {
  "@babel/core": {
    "optional": true
  },
  "babel-plugin-macros": {
    "optional": true
  }
}
}

```

```

    },
    "node_modules/tailwindcss": {
      "version": "4.3.3",
      "resolved": "https://registry.npmjs.org/tailwindcss/-/tailwindcss-4.3.3.tgz",
      "integrity": "sha512-g0hV3P7ufE62QDGg1zVaTgCR+EtPv92k2nIhVcVKcLmxT1sUBsQGhnZj175j+MqRt4zLF7ic+sCYjfhxMxj7YQ==",
      "dev": true,
      "license": "MIT"
    },
    "node_modules/tapable": {
      "version": "2.3.3",
      "resolved": "https://registry.npmjs.org/tapable/-/tapable-2.3.3.tgz",
      "integrity": "sha512-tapable",
      "dev": true,
      "license": "MIT",
      "engines": {
        "node": ">=6"
      },
      "funding": {
        "type": "opencollective",
        "url": "https://opencollective.com/webpack"
      }
    },
    "node_modules/tslib": {
      "version": "2.8.1",
      "resolved": "https://registry.npmjs.org/tslib/-/tslib-2.8.1.tgz",
      "integrity": "sha512-0JFu94HQB+KVduSUQL7wnpmqnfmls0A/nAh6b6EH0wCEoK0/mPeXU6c3wKDV83Mk0uHPRHtSXXKU99IBazS/2w==",
      "license": "0BSD"
    },
    "node_modules/typescript": {
      "version": "5.9.3",
      "resolved": "https://registry.npmjs.org/typescript/-/typescript-5.9.3.tgz",
      "integrity": "sha512-jl1vZzPDinLr9eUt3J/t7V6FgNEw9QjvBPdysz9KfQDD41fQrC2Y4vKQdiaUpFT4bX1b1RHHLpp8wtm6M5TgSw==",
      "dev": true,
      "license": "Apache-2.0",
      "bin": {
        "tsc": "bin/tsc",
        "tsserver": "bin/tsserver"
      },
      "engines": {
        "node": ">=14.17"
      }
    },
    "node_modules/undici-types": {
      "version": "6.21.0",
      "resolved": "https://registry.npmjs.org/undici-types/-/undici-types-6.21.0.tgz",
      "integrity": "sha512-iwDZqg0QAgr9Rav5H4n0M64c3mkR59cJ6wQp+7C4nI0gsmExaedaYLN044eT4AtBBWjbTiGPM1t2Md0T9H9JQ==",
      "dev": true,
      "license": "MIT"
    }
  }
}

```

web/package.json

```
{
  "name": "web",
  "version": "0.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start"
  },
  "dependencies": {
    "next": "16.3.8",
    "react": "19.2.8",
    "react-dom": "19.2.8"
  },
  "devDependencies": {
    "@tailwindcss/postcss": "^4",
    "@types/node": "^20",
    "@types/react": "^19",
    "@types/react-dom": "^19",
    "tailwindcss": "^4",
    "typescript": "^5"
  }
}
```

web/tsconfig.json

```
{
  "compilerOptions": {
    "target": "ES2017",
    "lib": ["dom", "dom.iterable", "esnext"],
    "allowJs": true,
    "skipLibCheck": true,
    "strict": true,
    "noEmit": true,
    "esModuleInterop": true,
    "module": "esnext",
    "moduleResolution": "bundler",
    "resolveJsonModule": true,
    "isolatedModules": true,
    "jsx": "react-jsx",
    "incremental": true,
    "plugins": [
      {
        "name": "next"
      }
    ],
    "paths": {
      "@/*": [".src/*"]
    }
  },
  "include": [
    "next-env.d.ts",
    "**/*.ts",
    "**/*.tsx",
    ".next/types/**/*.ts",
    ".next/dev/types/**/*.ts",
    "**/*.mts"
  ],
  "exclude": ["node_modules"]
}
```

web/next-env.d.ts

```
/// <reference types="next" />
/// <reference types="next/image-types/global" />
import "../next/types/routes.d.ts";
import "../next/types/root-params.d.ts";

// NOTE: This file should not be edited
// see https://nextjs.org/docs/app/api-reference/config/typescript for more information.
```

web/next.config.ts

```
import type { NextConfig } from "next";

const nextConfig: NextConfig = {
  /* config options here */
};

export default nextConfig;
```

```
# Charter – Proyek `loop-tb` (UnivaBio 2026)

> **Status: v1 FROZEN – disetujui & dikunci 2 Okt 2026.**
> Perubahan hanya via ADR baru (lihat §5).

## 1. Problem statement

Indonesia adalah negara dengan beban TB terbesar **ke-2 dunia** (10% dari 10,7 juta kasus global 2024 – WHO Global TB Report 2025). Diperkirakan **~1,07–1,09 juta orang jatuh sakit dan ~125 ribu meninggal per tahun (~14 kematian/jam)**, sementara pada 2024 hanya **~885 ribu kasus ditemukan** (Kemenkes) → **~200 ribu orang menularkan tanpa terdeteksi**.
```

Program nasional TOSS-TB memperluas skrining CXR + distribusi X-ray portabel, tetapi tiga hambatan bertemu:

1. Volume bacaan CXR melampaui kapasitas pembaca ahli (puskesmas tanpa dokter/radiolog).
2. Alat CAD yang ada bersifat **komersial** (qXR, Cad4TB) atau masih **tahap awal tanpa kalibrasi & penanganan ketidakpastian** (TBScreen.AI UGM, pilot terbatas 2025).
3. Skrining berhenti di **skor** – tidak menyambung ke rujukan, kepatuhan, dan tindak lanjut.

Frasa inti: loop perawatan TB bergerak lebih lambat daripada penularan TB.

```
## 2. Target user

| Prioritas | Siapa | Kebutuhan |
|---|---|---|
| Utama (MVP) | Kader/petugas TB puskesmas di daerah terpencil & fasilitas tanpa dokter | Skrining CXR → keputusan triase + surat rujukan dalam < 3 menit |
| Sekunder | Pasien TB & keluarga | Kepatuhan obat & pemantauan harian, bahasa manusia |
| Tersier | Klinisi/pengelola program | Ringkasan trajektori risiko untuk tindak lanjut |

## 3. Win conditions (dipetakan dari rubrik 25 poin)

| Kriteria | Ambisi kita |
|---|---|
| Idea & Innovation | Bukan skor CAD ke-1000: **loop triase terkalibrasi + honest evaluation + last-mile design** |
| Implementation | Model nyata terlatih (bukan wrapper) + evaluasi eksternal + aplikasi live ter-deploy |
| Health Impact & Rigor | Angka WHO/Kemenkes + guideline resmi + protokol evaluasi ala TPP WHO + limitations jujur |
| Design & Usability | UI ID-first, mobile-first untuk kader; alur skrining < 3 menit |
| Presentation | Demo nyata + video 3 menit + slide jujur "selesai vs roadmap" |

## 4. Constraints & non-goals

- **Kapasitas:** solo, 6–8 jam/hari efektif. Tanpa GPU lokal → training di Colab/Kaggle (ADR-006).
- **Deadline policy:** platform = 13 Okt 23:45 EDT; halaman Rules (basi) menulis 7 Okt. **Target internal: submit ≤ 7 Okt WIB** (buffer). Verifikasi via email ke biocataanalysis@gmail.com (draft di `03-eksekusi.md`).
- **Data:** publik saja (daftar di `02-arsitektur.md`). Tanpa validator klinis → mitigasi: guideline resmi + framing "alat triase, bukan diagnosis" + limitations jujur.
- **NON-GOALS (dilarang masuk MVP):** alat diagnosis; menggantikan GeneXpert; CAD untuk <15 tahun (di luar rekomendasi WHO); analisis suara batuk; multi-penyakit; auth multi-user/RBAC; integrasi SITB otomatis; fitur apapun tanpa baris di coverage matrix (`02-arsitektur.md`).

## 5. Log keputusan (ADR)

| ID | Keputusan | Status |
|---|---|---|
| ADR-001 | Domain = **TB paru** (beban #2 dunia; loop lengkap; dataset publik; wedge latensi jelas) | accepted – 2 Okt 2026 |
| ADR-002 | Invarian inti = **state risiko terkalibrasi R(x,t)** sebagai satu-satunya mesin; semua fitur = proyeksinya | accepted – 2 Okt 2026 |
| ADR-003 | **Scope freeze MVP:** Recognition=DEEP, Access=MEDIUM, Interpretation=MEDIUM, Monitoring=LIGHT, Continuity=LIGHT (detail di `02`) | accepted – 2 Okt 2026 |
| ADR-004 | Nama produk | ditunda sampai UI freeze |
| ADR-005 | Deadline policy: target internal 7 Okt, submit awal, buffer 8–13 Okt hanya untuk polish/stretch tercatat | accepted – 2 Okt 2026 |
| ADR-006 | Jalur training: **Colab/Kaggle GPU (T4)** – tidak ada GPU lokal; model kecil + checkpoint tiap epoch wajib | accepted – 2 Okt 2026 |
| ADR-007 | **Studi representasi (dalam scope Recognition-DEEP):** bandingkan supervised fine-tune (EB0/ConvNeXt) vs probe SSL (DINOv2 umum + RAD-DINO khusus X-ray dada) di AUROC eksternal + kalibrasi; pilih pemenang untuk API. Bukan scope baru – ini pemilihan inti mesin. | accepted – 2 Okt 2026 |

**Kebijakan perubahan (anti-berubah-ubah):**

1. Setelah freeze, setiap penambahan scope **wajib menghapus item setara** (zero-sum) dan dicatat sebagai ADR superseding.
2. Tidak ada fitur yang tidak punya baris di coverage matrix.
3. Semua perubahan besar harus selesai dalam < 30 menit diskusi; kalau tidak, ditolak default.
```

```
# Pohon Masalah TB → Pemilihan Wedge

## 1. Akar (root cause)

**Loop perawatan TB (deteksi → diagnosis → pengobatan → pemantauan) bereaksi lebih lambat daripada laju penularan dan progresi penyakit.**

Bukan "kurang dokter", bukan "kurang alat" – itu gejala. Mekanismenya: latensi setiap tahap loop > waktu yang dibutuhkan penyakit untuk menular ke orang lain / memburuk pada pasien.

**Bukti delay (terverifikasi):**
- Median diagnostic delay di primary care Indonesia 4 hari (IQR 1-11), **tetapi 19,3% episode >14 hari dan 7,6% >30 hari** (Annals of Global Health, 2025, aogh.5369).
- Patient pathway analysis: median 20 hari (IQR 7-72), **57,3% mengalami long diagnostic delay** (Clinical Microbiology & Infection, 2021).
- TBM (TB meningitis): median **66 hari** menuju diagnosis (PMC, 2025).
- Kualitas fasilitas DOTS berkorelasi dengan total delay (Jogjakarta, Trop Med Int Health, 2011).

## 2. Cabang masalah (MECE, mengikuti tahap loop)

| # | Cabang | Isi | Bukti |
|---|---|---|---|
| A | **Detection gap** | ±200k kasus/tahun tak pernah masuk sistem; case finding masih pasif; kasus ringan/asimtomatik; stigma; kapasitas skrining | 1,07-1,09 jt estimasi vs ~885k ditemukan (GTR 2025; Kemenkes 2024) |
| B | **Reading/access gap** | X-ray portabel didistribusikan masif, tapi pembaca ahli langka; GeneXpert tidak merata; keputusan rujukan lambat | WHO 2021/2025: CAD direkomendasikan *justru karena* kekurangan pembaca; Kemenkes distribusi X-ray portabel |
| C | **Interpretation gap** | Skor/angka tanpa penjelasan; pasien & kader tidak paham arti; stigma | TBScreen.AI = "probability 0-100%" tanpa mekanisme penjelasan |
| D | **Treatment continuity** | Regimen 6 bulan; LTFU; beban DOTS; TPT & investigasi kontak belum optimal | Wamenkes (2026): penemuan kasus, keberhasilan pengobatan, investigasi kontak & TPT masih jadi tantangan |
| E | **System continuity/learning** | Data antar-kunjungan terfragmentasi; tak ada trajektori risiko; program tak belajar dari kasus yang lolos | SITB/monitoring manual; tidak ada mekanisme loop |

## 3. Analisis bottleneck (Theory of Constraints)

- **Inflow constraint: deteksi.** Tidak bisa mengobati yang tidak ditemukan. Setiap kasus tertunda = tambahan transmisi + risiko kematian.
- Di dalam deteksi, constraint operasional pada push nasional = **throughput pembacaan CXR di ujung rantai** (puskesmas tanpa dokter; volume naik setelah roll-out alat portabel).
- **Wedge = kompresi latensi di titik skrining & triase**, lalu sambungkan loop ke hilir (rujukan → kepatuhan → tindak lanjut).
- Cabang D & E adalah bottleneck sekunder: intervensi di sana hanya menyentuh pasien yang *sudah* tertangkap sistem.

## 4. Mengapa solusi yang ada belum cukup (terverifikasi – ini diferensiasi kita)

| Solusi | Status | Celah |
|---|---|---|
| qXR (Qure.ai), Cad4TB | Komersial, mapan, dipakai program komunitas Indonesia (dilaporkan 190 kabupaten) | Tertutup; berhenti pada prioritisasi/skor; tidak menutup loop; threshold tetap isu implementasi |
| TBScreen.AI (UGM × KONEKSI, Agu 2025) | CAD lokal pertama; ~936 citra; pilot Klaten & Mimika | Akurasi dilaporkan ~64% (swa-lapor); tanpa kalibrasi/ketidapastian; belum terbuka |
| Model riset dari dataset publik | "AUROC 0,99" | **Audit medRxiv Agu 2026**: class-conditional acquisition confounding di Montgomery/Shenzhen/Rahman/TBX11K – 0,97-1,00 pada split acak → **0,57-0,88** pada transfer LOCO → 0,569 pada kohort asing. 88,4% "normal" Rahman berasal dari satu arsip RS; semua TB dari koleksi khusus. Mode gagal klinis: scar paru → FP percaya diri (0,84) |

**Wedge statement (satu kalimat untuk juri):**
> *Bukan skor lagi – sebuah loop triase terkalibrasi yang tahu kapan dirinya ragu, tahan uji pindah-domain, berjalan di ujung rantai layanan, dan menyambung skrining → rujukan → kepatuhan → tindak lanjut.*

## 5. Cabang → intervensi (preview coverage matrix)

| Cabang | Intervensi (proyeksi dari R(x,t)) | Depth |
|---|---|---|
| A Detection | Model CXR ensemble terkalibrasi + band triase | DEEP |
| B Access | Kebijakan aksi t (target TPP WHO) + paket rujukan cetak | MEDIUM |
| C Interpretation | Grad-CAM + penjelasan bahasa manusia + sitasi guideline | MEDIUM |
| D Continuity | Kartu pasien: check-in harian → R(t) → alert eskalasi | LIGHT (demo) |
| E Learning | Ringkasan trajektori untuk klinisi/program | LIGHT (demo) |

## 6. Referensi kunci

- WHO Global TB Report 2025, §1.1 TB incidence (Indonesia 10% beban global; #2).
- WHO consolidated guidelines on TB, Module 2: Screening (2021), rekomendasi #10 (CAD menggantikan pembaca manusia, ≥15 tahun, kondisional).
- WHO policy statement: Use of CAD software for TB screening (Mei 2025) – proses TAG 2025; isu pemilihan threshold masih terbuka; WHO calibration toolkit.
- WHO Target Product Profile: sensitivitas >0,90; spesifisitas >0,70.
- Bilal A. "Auditing Class-Conditional Acquisition Confounding Across Five Open TB CXR Corpora", medRxiv 2026 – repo audit: github.com/AhmadBilal227/tb-cxr-triage-pakistan.
- Kemenkes: TOSS-TB; ~885k kasus ditemukan 2024; proyeksi ~1,08-1,09 jt kasus 2024-2026; TB ditetapkan sebagai emergency.
```

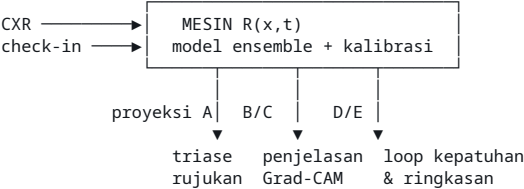
```
# Arsitektur & Scope Freeze (ADR-003)

## 1. Invarian inti: state risiko `R(x, t)`

Satu objek komputasi per pasien yang berevolusi waktu:

...
R(x, t) = {
  p_tb : probabilitas TB terkalibrasi (temperature-scaled ensemble),
  u : ketidakpastian (ensemble disagreement),
  z : peta bukti spasial (Grad-CAM),
  band : aksi { rujuk-prioritas | ragu-periksa-manusia | negatif-skinning },
  t : riwayat pembaruan (trajektori)
}
...

**Metode penyatu:** satu mesin risiko + satu kebijakan aksi. Setiap cabang pohon masalah = **proyeksi** dari `R` yang sama – bukan fitur yang hidup sendiri-sendiri.

...


```

graph LR
 CXR --> R["MESIN R(x,t)
model ensemble + kalibrasi"]
 check_in[check-in] --> R
 R --> A["proyeksi A
triase rujukan"]
 R --> BC["B/C
penjelasan
Grad-CAM"]
 R --> DE["D/E
loop kepatuhan
& ringkasan"]

```


...

## 2. Coverage matrix – SCOPE MVP (FROZEN setelah approval)

| Cabang | Proyeksi dari R | Fitur MVP | Depth | Bukti sukses |
|---|---|---|---|---|
| A Detection | p_tb + u | Upload CXR → skor + CI + band | **DEEP** | Sens@Spec, AUROC+CI, ECE, tabel transfer |
| B Access | band → aksi | Triage band + surat rujukan (print/PDF) | MEDIUM | Kesepakatan band vs guideline; 1 alur <3 menit |
| C Interpretation | z → penjelasan | Overlay Grad-CAM + teks ID + "apa yang TIDAK ditunjukkan" | MEDIUM | Paham oleh 3+ orang non-medis (user test kecil) |
| D Monitoring | R(t) update | Kartu pasien: check-in minum obat/gejala → alert naik | LIGHT (demo) | Skenario demo memburuk → eskalasi tampil |
| E Continuity | R(t) → ringkasan | Export ringkasan klinisi (print) | LIGHT (demo) | Artefak printable muncul |

**Di luar scope (bukan roadmap yang dijanjikan, hanya ide yang sengaja TIDAK dikerjakan):** audio batuk, pediatric, multi-penyakit, integrasi SITB, auth multi-user, mobile native.

## 3. Data plan

| Dataset | Isi | Peran | Catatan |
|---|---|---|---|
| TBX11K (vbookshelf/tbx11k-simplified) | 8.399 berlabel 5122: **660 active TB**; 7.600 negatif (3.800 healthy + 3.800 sick-but-no-TB); 139 latent dikeluarkan; +3.302 `unknown_*` tanpa label (tidak dipakai) | **Train/val utama** | Label dari `data.csv` (otoritatif); task sulit by design: sick-but-no-TB = negatif |
| Rahman (tawsifurrahman) – folder `TB_Chest_Radiography_Database/` | **700 TB / 3.500 normal** | **Train** | WAJIB dedup – 88,4% "normal" dari satu arsip RS; semua TB dari koleksi khusus (Bilal 2026) |
| NLM Shenzhen | **336 TB / 326 normal** (662 citra) | **External test** | Sumber resmi NLM; tidak boleh tersentuh training |
| NLM Montgomery | **58 TB / 80 normal** (138 citra) + ClinicalReadings (usia/sex) | **External test** + subgroup | idem |
| Mendeley Pakistani TB CXR V2 | kohort RS Pakistan | External tambahan (stretch) | doi:10.17632/8j2g3csprk.2 |

Imbalance train ≈ 1.360 TB : 11.100 negatif (~1:8) → pakai `--balanced` + metrik yang tidak sensitif kelas (AUROC, sens@spec) sebagai metrik utama.

**Aturan integritas (tidak bisa dinegosiasi):**
1. Dedup perseptual (pHash/embedding) + provenance audit mengikuti metode Bilal 2026 (tools + source-matched hashes tersedia publik).
2. Laporkan jumlah duplikat yang dibuang, di README dan evaluasi.
3. Split manifest dikunci sebelum training (file JSON versioned di repo).
4. Tidak ada citra dari korpus eksternal yang masuk training – dicek ulang via hash sebelum final submission.

## 4. Protokol evaluasi (senjata rigor)

| Aspek | Metrik/metode |
|---|---|
| Akurasi utama | **Sensitivitas @ spesifisitas 0,90 & 0,70** (target TPP WHO: sens >0,90, spec >0,70) + 95% CI bootstrap |
| Diskriminasi | AUROC + CI |
| Kalibrasi | ECE, reliability curve, Brier score (sebelum vs sesudah temperature scaling) |
| Ketidakpastian | Proporsi "ragu" (deferral rate) + akurasi pada subset ragu |
| Transfer/honesty | Tabel: within-corpus vs provenance-dedup LOCO vs unseen-corpus (meniru desain audit) |
| Baseline pembandingan | (1) linear probe fitur beku, (2) 1 arsitektur fine-tune, (3) ensemble + kalibrasi |
| Subgrup | Usia/jenis kelamin (tersedia di NLM); pembahasan kualitatif scar-FP |
| Limitations (wajib di PDF & video) | Retrospektif; confounding akuisisi dataset publik; tanpa data Indonesia; bukan alat diagnosis |

## 5. Stack teknis

| Layer | Pilihan | Alasan |
|---|---|---|
| ML | PyTorch + timm (EfficientNet-B0, ConvNeXt-T), scikit-learn | Cepat dilatih, terbukti, export ONNX mulus |
| Training | Colab/Kaggle GPU (T4) atau lokal bila tersedia | B0 pada ~10k citra: ~1-2 jam/arsitektur |
```

Serving	FastAPI + onnxruntime (CPU), Docker → deploy HF Spaces	Gratis, publik, stabil untuk demo
FE	Next.js + Tailwind, mobile-first, ID-first	Kecepatan build + kualitas UI (kekuatan kita)
Offline (stretch)	onnxruntime-web (WASM) + PWA	Momen demo "airplane mode tetap jalan"
Repo	mono-repo: `ml/` `api/` `web/` `docs/` `submission/`	Bersih untuk code PDF & juri

6. Narasi demo video (3 menit)

1. **Masalah** (20 dtk): angka BPS/WHO + satu kalimat wedge.
2. **Skrining** (60 dtk): kader upload CXR → band + heatmap + penjelasan + surat rujukan.
3. **Loop** (40 dtk): kartu pasien → check-in → R(t) naik → alert eskalasi → ringkasan klinisi.
4. **Rigor** (40 dtk): tabel evaluasi – termasuk **jatuhnya performa di unseen corpus** (kejujuran ini yang membedakan).
5. **Roadmap jujur** (20 dtk): selesai vs direncanakan vs tidak dikerjakan.

```
# Eksekusi & Risk Register

## 1. Timeline (asumsi: deadline aman 7 Okt; jika 13 Okt terkonfirmasi → 8-13 Okt hanya polish/stretch)

| Hari | Tanggal | Kerja | Output |
| --- | --- | --- | --- |
| D0 | 2 Okt (malam) | Freeze docs; kirim email panitia; download data (Kaggle API); init repo; setup Colab | docs frozen; data lokal; repo jalan |
| D1 | 3 Okt | Prep data: dedup pHash, split manifest, EDA; baseline linear probe; mulai fine-tune arch-1 | Manifest split + baseline terukur |
| D2 | 4 Okt | Fine-tune arch-2 (+arch-3 bila lancar); temperature scaling; ensemble; eval harness | Tabel & grafik evaluasi v1 |
| D3 | 5 Okt | API + web end-to-end; Grad-CAM; triage band; surat rujukan; layar loop; deploy HF + Vercel | Demo hidup |
| D4 | 6 Okt | Polish UI; final eval run; 1-pager PDF; code PDF; writeup Devpost; video (script+rekam); submit draft | Submission draft |
| D5 | 7 Okt | Buffer + QA + **SUBMIT** | Submitted ☒ |
| D6+ | 8-13 Okt | HANYA stretch tercatat ADR: WASM offline, korpus eksternal tambahan, efisiensi | Opsional |

## 2. Deliverables & Definition of Done

| Deliverable | DoD |
| --- | --- |
| Repo GitHub publik | README (masalah, cara run, hasil, limitation), LICENSE, `submission/` lengkap; `make`/script reproducible |
| 1-Pager PDF | Tepat 1 halaman: masalah + angka bersitasi, solusi, bukti eval, roadmap |
| Code PDF | Auto-generate dari repo (script `ml/tools/export_code_pdf.py`) |
| Demo video (~3 mnt) | Narasi EN (juri internasional), UI ID; demo fitur benar-benar jalan; seksi rigor & roadmap jujur |
| Devpost writeup | Same story: problem → wedge → eval → honesty → roadmap |

## 3. Risk register

| # | Risiko | P | Dampak | Mitigasi |
| --- | --- | --- | --- | --- |
| R1 | Deadline ganda (7 vs 13 Okt) | M | Tinggi | Email hari ini ($4); submit ≤ D5; buffer 8-13 Okt tidak dipercaya |
| R2 | GPU tidak tersedia | M | Tinggi | Colab/Kaggle T4; model kecil (B0); checkpoint tiap epoch; arch-3 opsional |
| R3 | Leakage/confounding dataset | H | Tinggi | Dedup + provenance audit + split terkunci; jadikan *bagian cerita*, bukan kejutan |
| R4 | Tanpa validator klinis | H | Sedang | Framing "triase, bukan diagnosis"; sitasi guideline resmi; limitations eksplisit |
| R5 | Scope creep | H | Tinggi | ADR gate; zero-sum; fitur tanpa baris coverage matrix = ditolak |
| R6 | Download Kaggle gagal (auth) | L | Sedang | kaggle CLI + fallback unduh manual; mirrors |
| R7 | Deployment HF Spaces mati saat judging | L | Sedang | Video demo = rekaman layar lokal; link demo sebagai bonus, bukan satu-satunya bukti |

## 4. Draft email verifikasi deadline

**To:** biocataanalysis@gmail.com
**Subject:** UnivaBio – quick check on submission deadline

> Hi UnivaBio team,
> I noticed the Rules page lists "Hackathon End: October 7" while the Devpost schedule shows submissions until October 13 (11:45pm EDT). Could you confirm which deadline applies for submissions?
> Thank you!

## 5. Kebijakan change control

1. Semua keputusan tercatat sebagai ADR di `00-charter.md` $5; perubahan = ADR baru yang menggantikan (superseding).
2. Zero-sum: tambah 1 fitur = hapus 1 fitur setara.
3. Fitur wajib punya baris di coverage matrix (`02-arsitektur.md` $2).
4. Keputusan besar (>30 menit debat) ditolak default demi momentum.

## 6. Log progres

- **2 Okt – D0 selesai:** rencana frozen (ADR-003/005/006); data terunduh (TBX11K 8.399 berlabel + Rahman 4.200 + NLM 800);
**manifest v1** (train 10.288 / val 1.815 / external 800; 338 duplikat internal + **19 kebocoran cross-corpus** dibuang); repo publik berisi kode + artefak manifest; metadata klinis NLM 789 baris (21 scar-flagged); **smoke test ML end-to-end di CPU** – 2 bug nyata ditemukan & diperbaiki (kondisi spesifikitas terbalik; exporter ONNX butuh `dynamo=False` + toleransi relatif); UI web lengkap (Skrining, Pendampingan, Ringkasan klinisi, Tentang) build hijau.
```

```
# Evidensi Data – Manifest v1 (2 Okt 2026)

## 1. Pipeline

`ml/data/prepare_data.py`: discovery berlabel → pHash 64-bit → klaster Hamming ≤3 (union-find transitif) → proteksi integritas
external → split val per-klaster (seed 42). Output: `manifest.json`, `dedup_report.json`, `train.csv`, `val.csv`, `external.csv`
(path relatif – portable lokal ≠ Colab).

## 2. Hasil dedup (angka konkret untuk submission)

| Metrik | Nilai |
| ---|---|
| Citra berlabel diproses | 13.260 |
| Tak-terklasifikasi | 3.442 (3.302 `unknown` tanpa label + 139 latent + 1 tak tercatat) |
| **Duplikat internal dibuang** | **338** |
| **Kebocoran cross-corpus dibuang** | **19** citra train terbukti near-dup citra external NLM → dihapus dari training |
| Korpus external | utuh 800 (Shenzhen 662 + Montgomery 138) |

> 19 itu penting: contoh konkret bahwa dataset publik memang saling terkontaminasi – persis mekanisme yang didokumentasikan audit
Bilal (medRxiv 2026). Kita tidak sekadar mengklaim, kita mengukur dan Membuktikan lalu membuangnya.

## 3. Split v1 (FROZEN – jangan berubah)

| Split | n | TB | NEG | Komposisi |
| ---|---|---|---|---|
| train | 10.288 | 1.123 | 9.165 | Rahman 3.552 (584 TB) + TBX11K 6.736 (539 TB) |
| val | 1.815 | 211 | 1.604 | Rahman 637 + TBX11K 1.178 |
| external | 800 | 394 | 406 | Montgomery 138 + Shenzhen 662 |

- Imbalance train ≈ **1:8,2** → training wajib `--balanced`; metrik utama AUROC & sens@spec.
- `sick_but_no_tb` (3.800 di TBX11K) dipertahankan sebagai negatif → task sengaja lebih sulit & lebih realistis untuk triase ("bukan
TB" termasuk penyakit paru lain).

## 4. Keterbatasan (jujur – akan masuk PDF/video)

1. pHash menangkap near-duplikat langsung; **reprocessing agresif multi-langkah bisa lolos** (provenance recovery sulit bahkan di
paper audit).
2. Label dari dataset publik bersifat radiografis, bukan konfirmasi bakteriologis.
3. Tidak ada data Indonesia di training/evaluasi → domain shift wajib dilaporkan apa adanya.

## 5. Cara verifikasi ulang

```bash
.venv/bin/python ml/data/prepare_data.py --raw data/raw --out ml/data/artifacts # deterministik (seed 42)
python -c "import json; m=json.load(open('ml/data/artifacts/manifest.json')); print(m['meta']['counts'])"
```

## 6. Hasil training v1 & "momen kejujuran" (2 Okt)

Model	Val AUROC (within-corpus)	Ext AUROC [95% CI]	Ext sens@spec90	Ext ECE
EfficientNet-B0	0,9999	0,638	–	–
ConvNeXt-Tiny	0,9861	0,675	–	–
**Ensemble (EB0+ConvNeXt, temp-scaled)**	–	**0,675 [0,636-0,712]**	0,266	0,212
Linear probe (fitur beku)	0,9802	**0,637 [0,597-0,671]**	0,254	0,322

**Interpretasi (jadi tulang punggung narasi submission):**

1. **Ilusi within-corpus terbukti pada data kita** – bahkan probe linear dengan fitur beku ImageNet mencapai 0,98 di val, tapi jatuh
ke ~0,64 saat pindah korpus. Persis pola yang didokumentasikan audit Bilal (2026): evaluasi split-acak membesarkan angka, deployment
membongkarnya.
2. **Over-triage eksternal:** 79,5% citra masuk band "rujuk prioritas" (ambang warisan val tidak transfer); band "negatif skrining"
masih memuat 24,7% TB → sistem **tidak boleh** dipakai rule-out tanpa konfirmasi manusia. Ini justifikasi empiris untuk pita "ragu" +
human-in-the-loop – dan mengukur secara konkret *gap threshold* yang diakui WHO masih terbuka.
3. **Kalibrasi tetap tantangan lintas-domain:** ECE eksternal 0,212-0,322 setelah temperature scaling (T=1,2-2,4). Model overkonfiden
ekstrem (logit val ~786).
4. **Subgrup adil:** AUROC eksternal F 0,669 / M 0,682; usia 0-30: 0,631 · 30-50: 0,707 · 50+: 0,737 (tidak ada disparitas besar;
celah di kelompok muda).
5. **Aksi perbaikan (ADR-007):** studi representasi SSL (DINOv2 + RAD-DINO khusus X-ray dada) – audit melaporkan probe SSL mencapai
~0,88 pada transfer terkontrol; hasil menyusul di bagian 7.
6. **Deferral berbasis ketidakpastian (ensemble disagreement):** pada data eksternal, kelompok 20% dengan disagreement tertinggi
punya TB-rate 31% vs 54% pada sisanya →  $\tau_u=0,40$  dipakai di API `model_config.json` sebagai pemicu band "ragu".

## 7. Studi representasi SSL (ADR-007) – HASIL FINAL (2 Okt)

Representasi (probe linear, fitur beku)	Val AUROC	**Eksternal AUROC [95% CI]**	sens@spec90	ECE
**RAD-DINO** (X-ray dada, SSL) – **DIPILIH**	0,9982	**0,8874 [0,862-0,909]**	**0,713**	0,247
Kombinasi DINOv2+RAD-DINO	–	0,8579 [0,831-0,883]	0,670	–
Stack RAD-DINO + supervised	–	0,8092	–	–
DINOv2 (SSL umum)	0,9987	0,7357 [0,698-0,768]	0,383	0,329
Supervised ensemble (EB0+ConvNeXt)	–	0,6751 [0,636-0,712]	0,266	0,212
Linear probe (fitur ImageNet beku)	0,9802	0,6366 [0,597-0,671]	0,254	0,322

**Keputusan final (ADR-007):**

- **Skor = probe RAD-DINO tunggal** (eksternal AUROC 0,887; sens@spec90 0,713). Catatan jujur: TPP WHO mensyaratkan sensitivitas
>0,90 – kita belum sampai, dan itu dilaporkan.
- **Ketidakpastian = std 4 anggota lintas-keluarga** (RAD-DINO, DINOv2, EB0, ConvNeXt): di eksternal, defer 25% → TB-rate grup ragu
```

32% vs 55% sisanya; $\tau_u=0,25$.

- **Ensemble/stacking** dengan supervised menurunkan performa** di domain asing (0,887-0,809) → dibuang.
- Pelajaran utama: **representasi pra-latih** pada domain (X-ray dada) mendominasi fine-tuning** untuk transfer lintas-rumah-sakit (+0,21 AUROC vs supervised; sens@spec90 2,7× lebih baik). Sejalan audit Bilal (2026): SSL 0,883 pada LOCO.
- Demo API final: kasus TB asli → p 0,98/u 0,006 → "rujuk"; kasus normal domain-shift → p 0,32/u 0,37 → "ragu".